

Rocket Slicer

Quick Start Guide

2026.06



Preface

About Rocket Slicer

Rocket Slicer is developed by FibreSeek and specifically designed for continuous-fibre 3D printing. It is compatible with the FibreSeeker 3 3D printer and is primarily used for path planning and process-parameter optimization in printing high-strength, lightweight composite materials. It helps enhance printing efficiency and structural performance.

Purpose

By reading this guide, users will learn how to install the Rocket Slicer and understand its core features.

Intended Audience

This guide is intended for users, technicians, or engineers in the 3D printing industry who use the FibreSeeker 3 3D printer.

System Requirements

Minimum

- **System**
 - Windows 10 64-bit (Version 1909 or later)
 - macOS 12 Monterey or later
- **Processor**
 - 8th Gen Intel Core i5 / AMD Ryzen 5 2000 series or newer
 - Apple M1 chip or newer
 - 64-bit quad-core CPU or above
- **Memory**
 - 8 GB RAM
- **Graphics**
 - GPU supporting DirectX 12 / Metal 2
 - Latest graphics drivers compatible with WebGPU
 - Minimum 2 GB available VRAM recommended
- **Storage**
 - At least 4 GB free disk space

- 10 GB extra space recommended for cache and project files
- **Display**
 - 1280 × 720 resolution or higher

Recommended

- **System**
 - Windows 11 64-bit / macOS 13 or later
- **Processor**
 - 10th Gen Intel Core i5/i7
 - AMD Ryzen 5/7 3000 series,
 - Apple M1 chip or newer
- **Memory**
 - 16 GB RAM
- **Graphics**
 - Discrete graphics card or modern integrated GPU with DirectX 12 / Metal support
 - 4 GB or more VRAM / Unified Memory
- **Storage**
 - SSD with more than 10 GB free space
- **Display**
 - 1920 × 1080 resolution or higher

Table of Contents

Preface	1
1. Getting Started	4
1.1 Installing Rocket Slicer on Windows	4
1.2 Installing Rocket Slicer on MacOS	5
2. Logging In to the Slicer	7
3. Connecting the Printer	9
3.1 General Connection	9
3.1.1 Connecting via Local LAN Network	9
3.1.2 Connecting via Cloud	11
3.2 Adding the Printer Manually	13
4. Feature Introduction	16
4.1 Toolbar	16
4.2 Mask Mode	18
4.3 Layup Mode	21
5. Printing Mode	23
5.1 Speedy Mode	23
5.2 Reinforced Mode	23
5.3 Fortified Mode	24
6. Slicing the Model	25

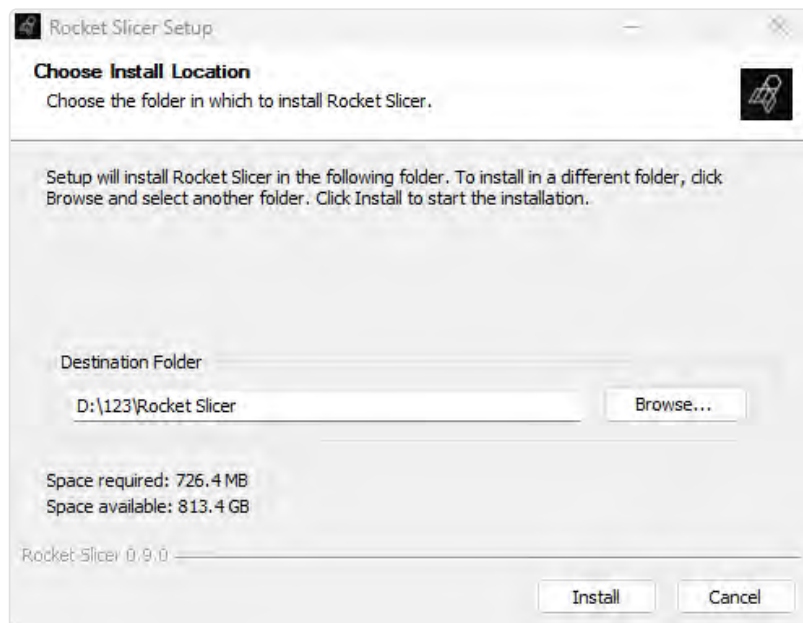
1. Getting Started

1.1 Installing Rocket Slicer on Windows

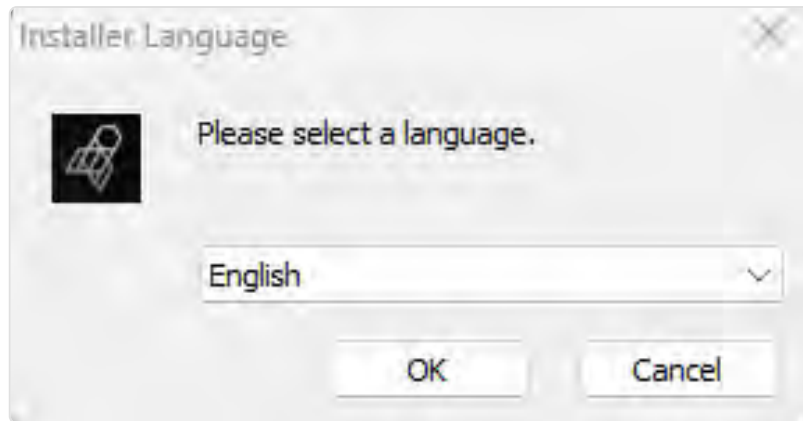
1. Click the link: [Rocket Slicer](#) to visit the official website.
2. Select **For Windows** and download the installation package.
3. Double-click the installer to run it.
4. Click **Install** and the software will be installed in the default directory.

i Note

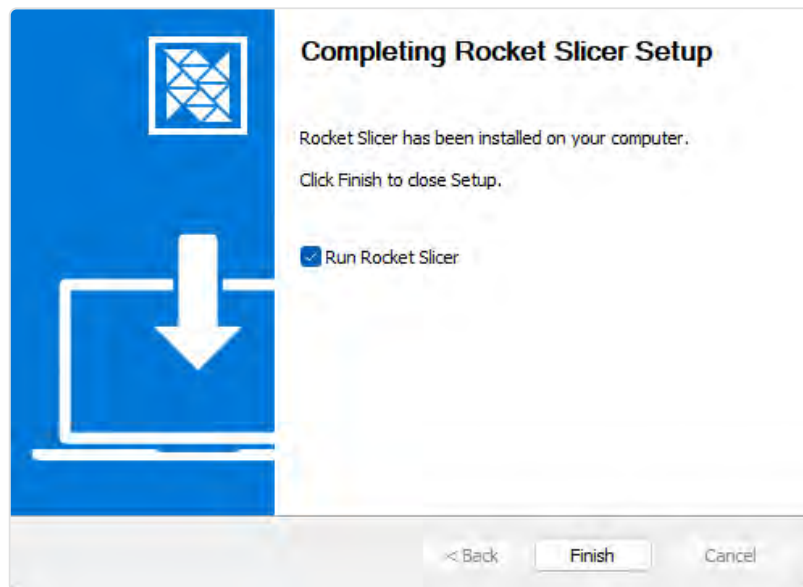
If you wish to specify an alternative installation directory, click **Browse** and select the appropriate folder.



4. Select English as the installation language.



5. After installation, click **Finish** or double-click the software icon on the desktop to launch the application.

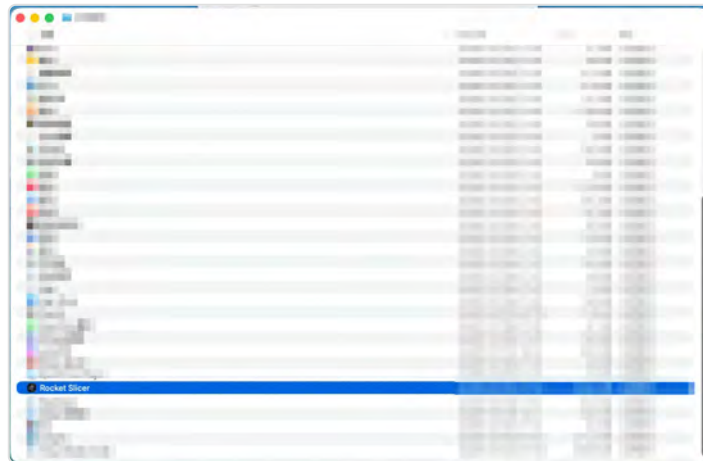


1.2 Installing Rocket Slicer on MacOS

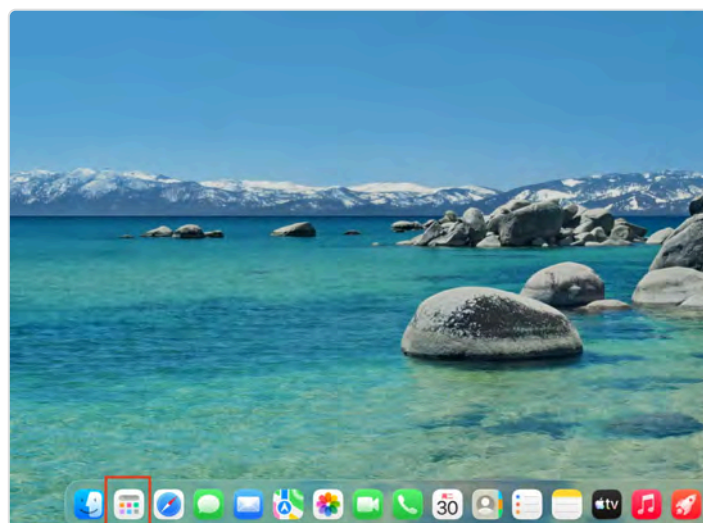
1. Click the link [Rocket Slicer](#) to visit the official website.
2. Select **For MacOS** and download the installation package.
3. Double-click the downloaded `.dmg` file, then drag the Rocket Slicer icon into the **Applications** folder.



4. Open the **Applications** folder, find Rocket Slicer, and double-click it to launch the application and enter the welcome page.



5. For future use, you can find Rocket Slicer in the **Dock** bar at the bottom of the screen or in the **Applications** folder, and double-click the icon to start the software.



2. Logging In to the Slicer

1. Double-click the software launch the program.
2. If you have not logged in, you can skip the startup animation by clicking **Skip**.

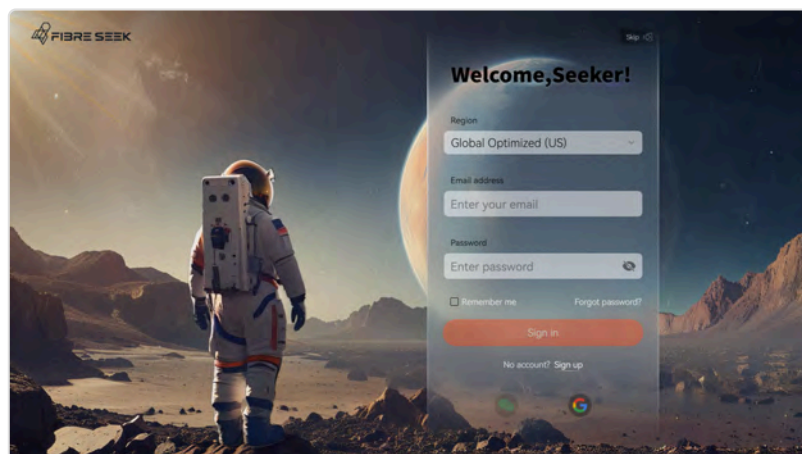


3. After skipping or finishing the animation, the system will automatically enter the login page.
4. Select **Global Optimized(US)** in the Region field.
5. Enter your email and password to log in to the software.

Note

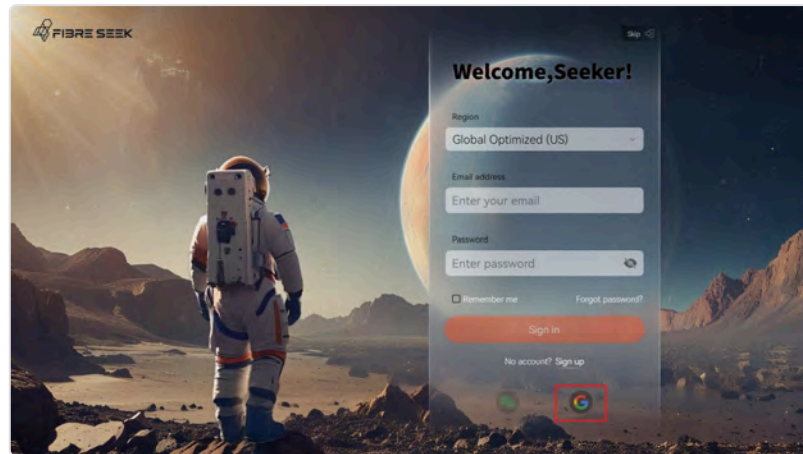
If you do not have an account, please register on the **Sign Up** page.

- Click **Sign Up** to go to the **Sign Up** page.
- Please ensure the password is at least 8 characters long, does not contain spaces or consecutive numbers or characters, and includes both uppercase and lowercase letters and numbers.
- By creating an account, you agree to the Terms of Service and Privacy Policy.

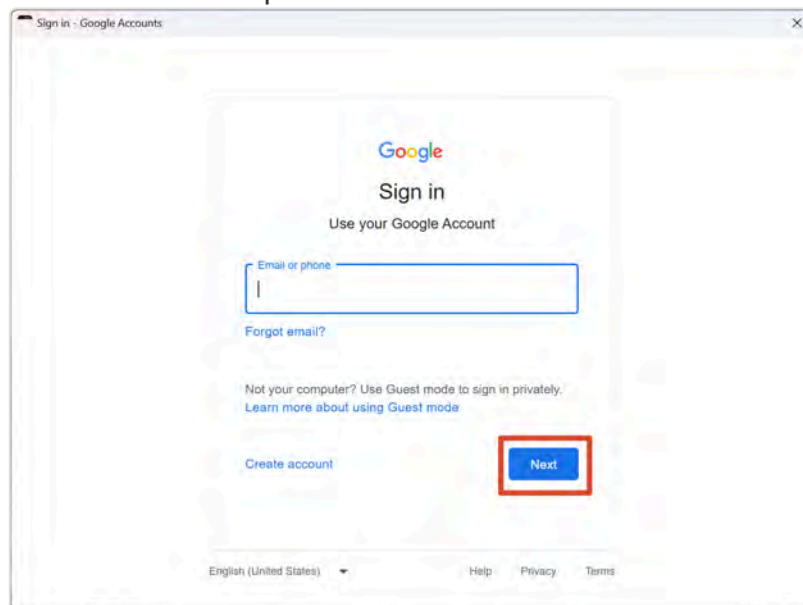


6. If you select **Remember Me**, the startup animation will no longer display on subsequent launches.
7. (Optional): You can also log in to the software with your Google account.

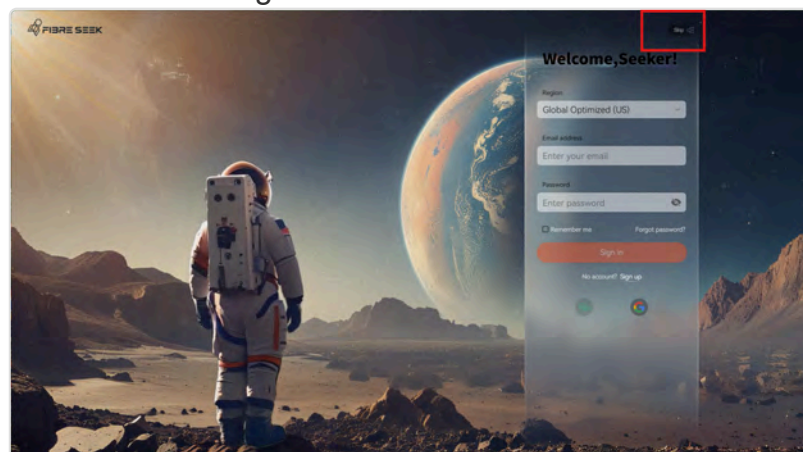
- a. Click  in the right corner.



- b. Enter your Gmail address or phone number and click **Next**.



- c. Enter your password and click **Next**.
d. Follow the prompts to finish logging in.
8. Click **Sign in**.
9. (Optional) Click **Skip** on the login page to log in as a guest. Cloud-related operations are not available in guest mode.



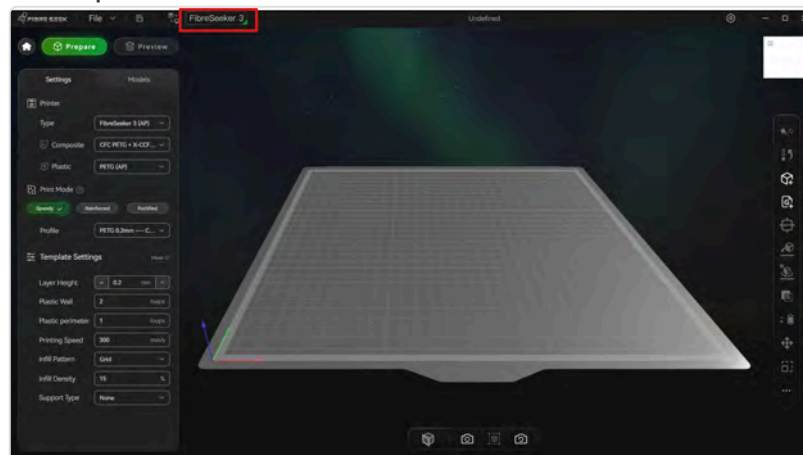
3. Connecting the Printer

There are two methods to connect the printer. Follow the steps to complete connection.

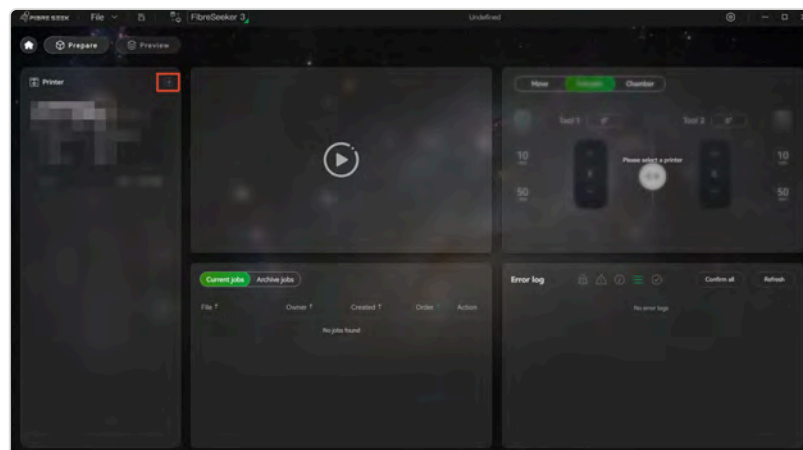
3.1 General Connection

3.1.1 Connecting via Local LAN Network

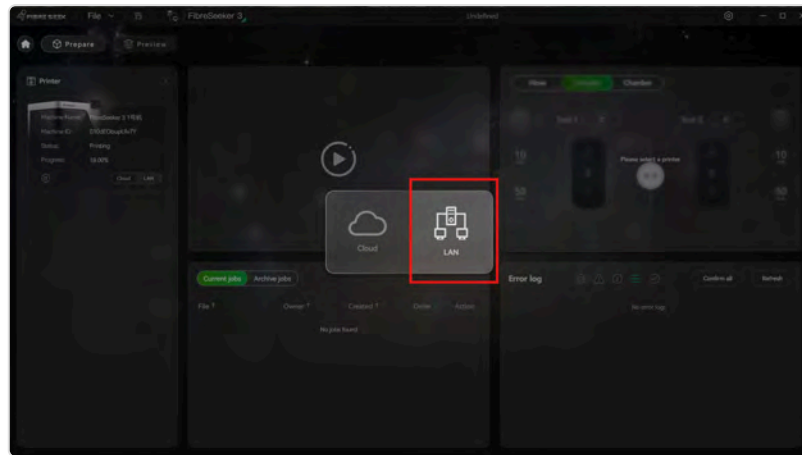
1. Click  in the **Device List** area next to  in the top toolbar and select the target printer. The default printer is FibreSeeker 3.



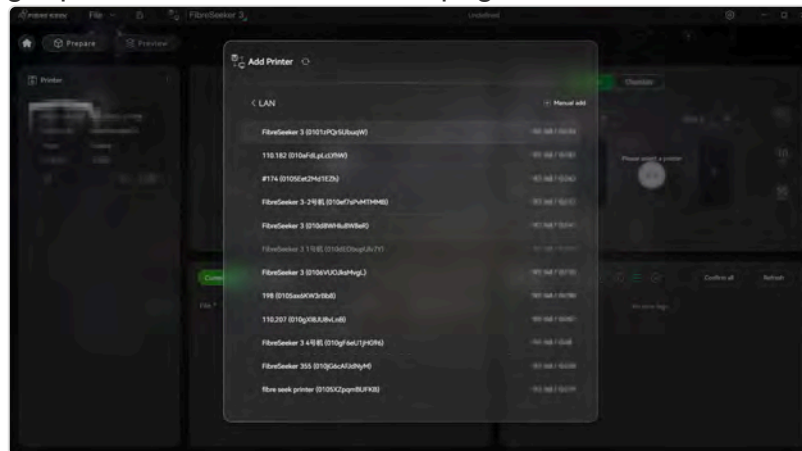
2. Click  next to the Printer area to add your printer.



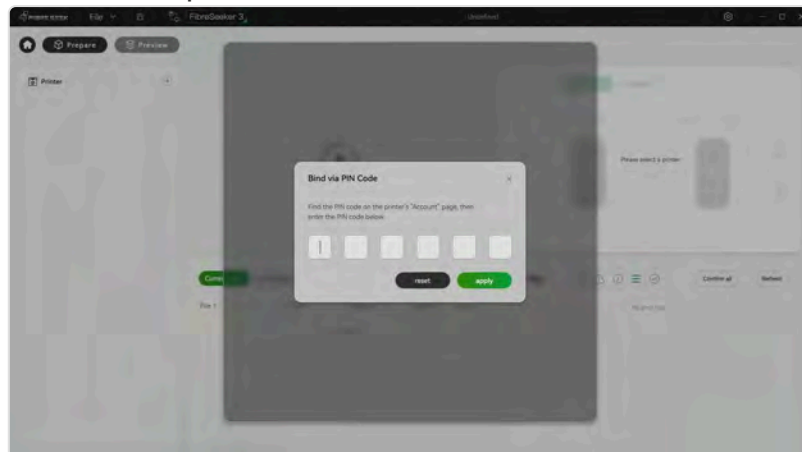
3. Select **LAN** in the pop-up window.



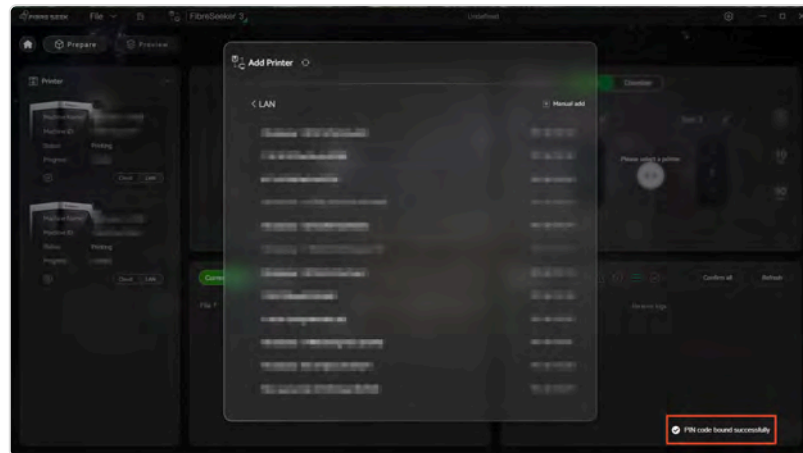
4. Select the target printer on the **Add Printer** page.



5. Enter the PIN code of the printer in the **Bind via PIN Code** window.



6. Click **Apply**.
7. After binding, a success message **PIN code bound successfully** pops up, and the printer appears in the left Printer area.

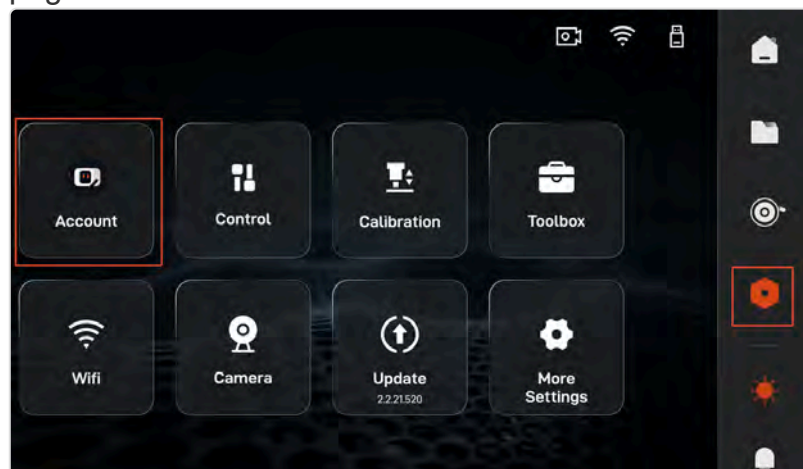


Note

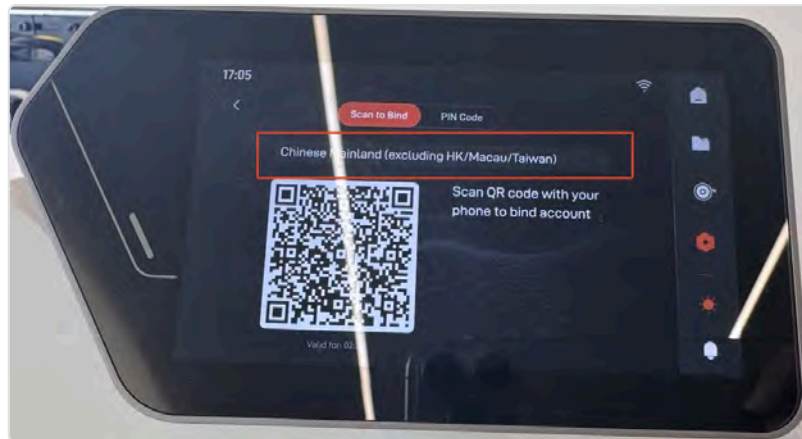
The LAN communication may be blocked by the computer system firewall or corporate network firewall. If you fail to search or bind the printer, contact your company's IT team to check firewall interception and add relevant rules to the firewall whitelist.

3.1.2 Connecting via Cloud

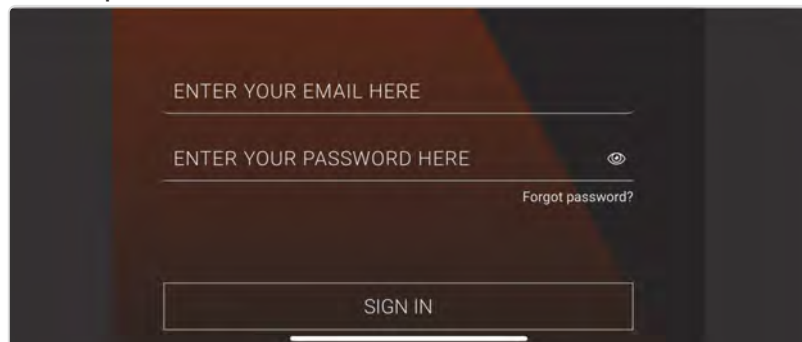
1. On the printer touchscreen, select  > **Account**, which will automatically open the Scan to Bind page.



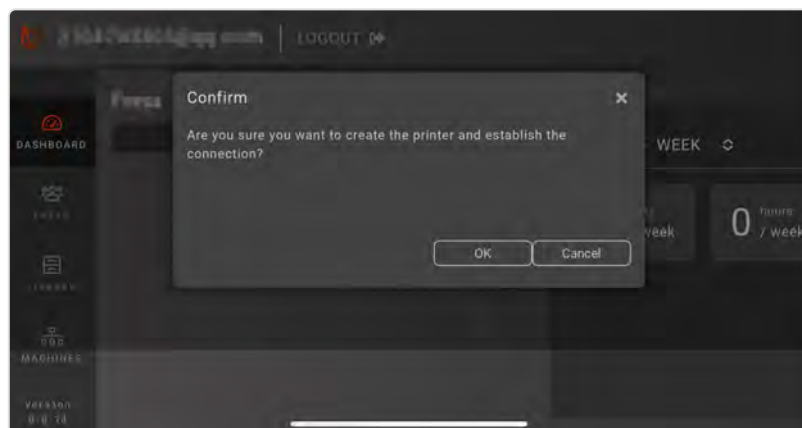
2. Select your region, then scan the QR code with your mobile phone to access the login page.



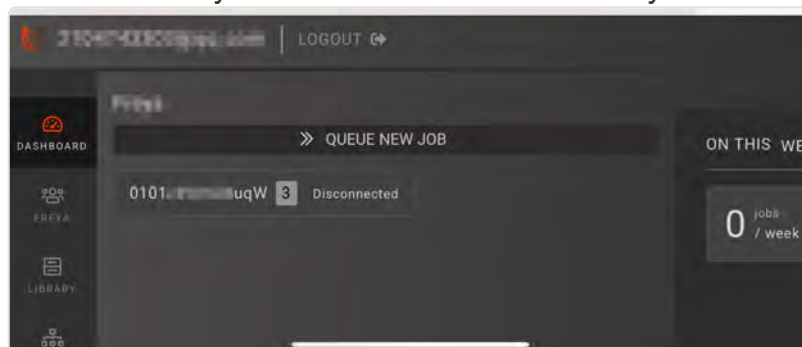
3. Enter your email and password and click **SIGN IN**.



4. Click **OK** on the phone pop-up confirmation window to finish binding the printer to your cloud account.

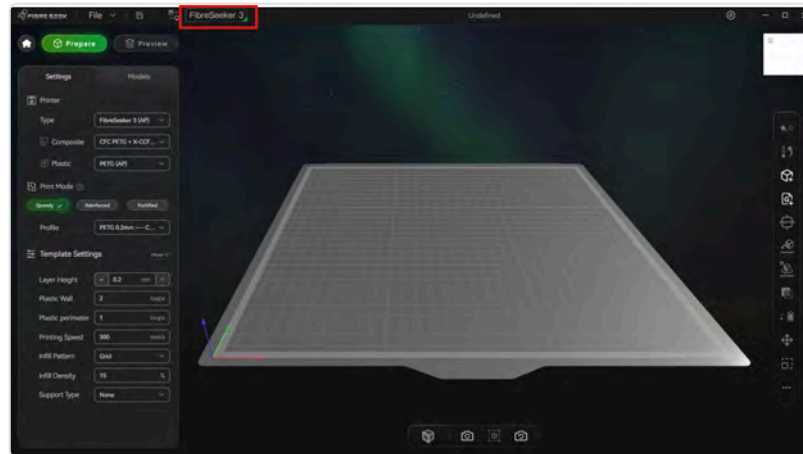


5. The printer will be bound to your cloud account successfully

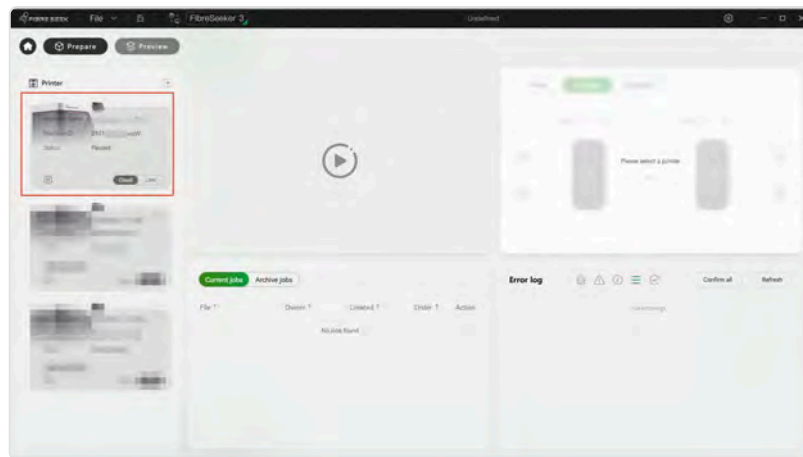


6. Launch Rocket Slicer, log in using **the identical email account** you just signed into on your mobile phone, then click **Sign in**.

- Click **FibreSeeker 3** to open the Remote Control interface.



- You can view full synchronized printer details within the Device List panel on the left sidebar.



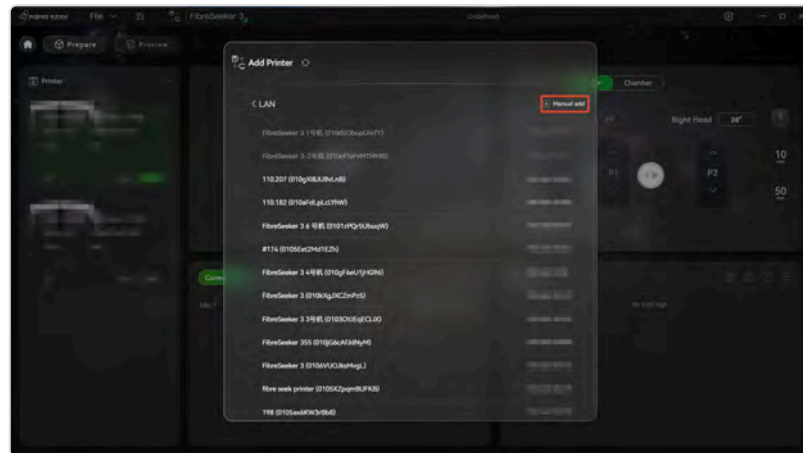
! Tip

Printer information will sync automatically between your mobile phone and Rocket Slicer desktop client, as long as you log in with the same email cloud account on both devices.

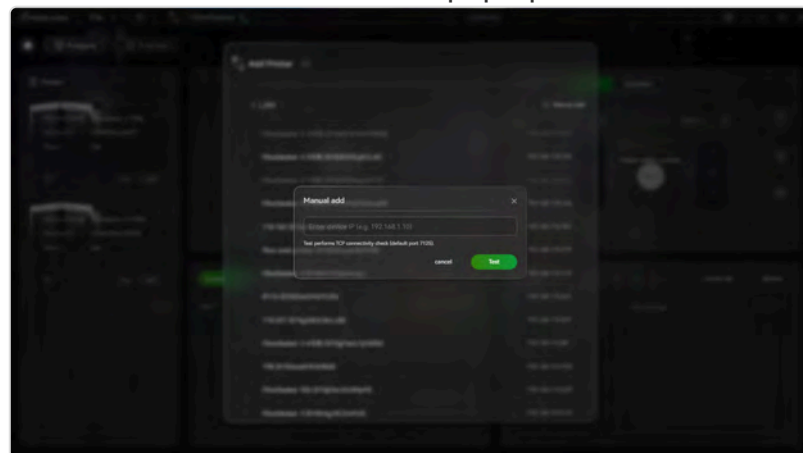
3.2 Adding the Printer Manually

If the printer is not found automatically on the **Add Printer** page, you can add it manually by entering its IP address.

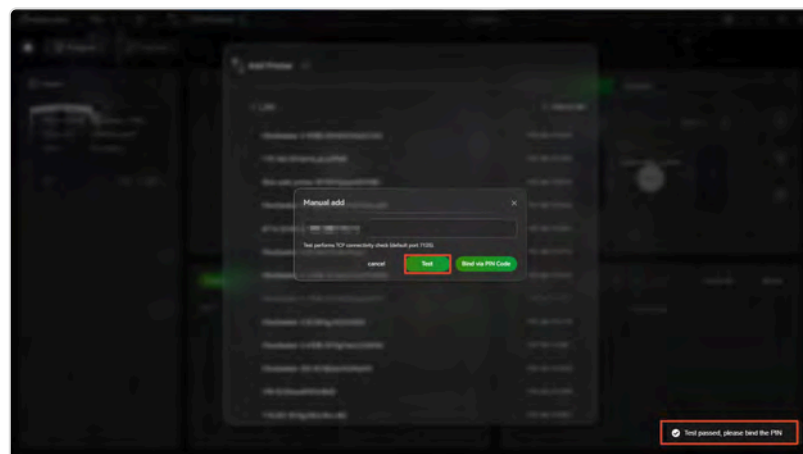
- Click **Manual add** on the **Add Printer** page.



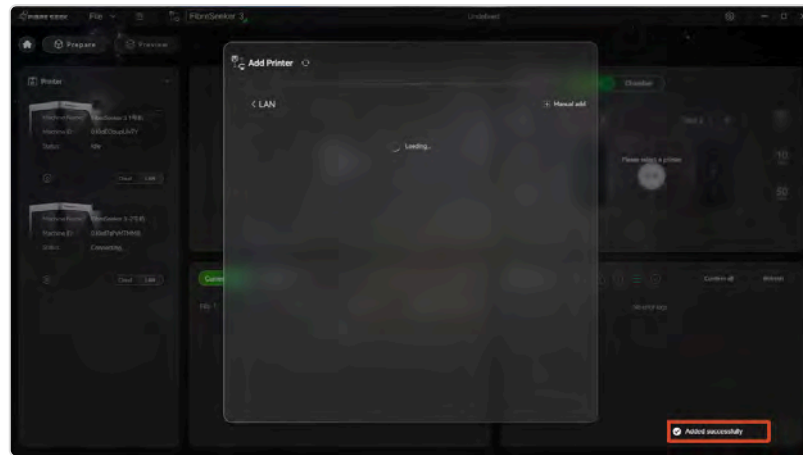
2. On the printer's home screen, tap the Wi-Fi icon at the top right to find its IP address, and enter its IP in the **Manual add** pop-up window.



3. Click **Test**, and once it succeeds, a pop-up will show "Test passed, please bind the PIN."



4. Click **Bind via PIN Code** to enter the corresponding page, and enter the PIN code of the printer.
5. Click **Apply**.
6. A pop-up confirms "Added successfully," and the printer appears in the "Printers" section on the left.



4. Feature Introduction

4.1 Toolbar

Icon	Function	Description
	Add models	Import models. Supported file formats: .stl, .step, .stp, and .3mf.
	Undo / Redo	Go back to the previous step.
	Undo to	Revert to a specified previous step.
	Select a GCode File	Select a G-code file and preview settings such as plastic/fibre path, printing time, cost, inner and outer wall infill, etc.
	Auto-layout	The software automatically identifies multiple models and arranges them according to spacing settings.
	RotatePartByFace	Select a face of the model to place it flat on the build plate.
	Auto orient	The software automatically determines the optimal orientation for the model.
	Mask Mode	Add a mask by drawing a box or cylinder over specific areas of the model.

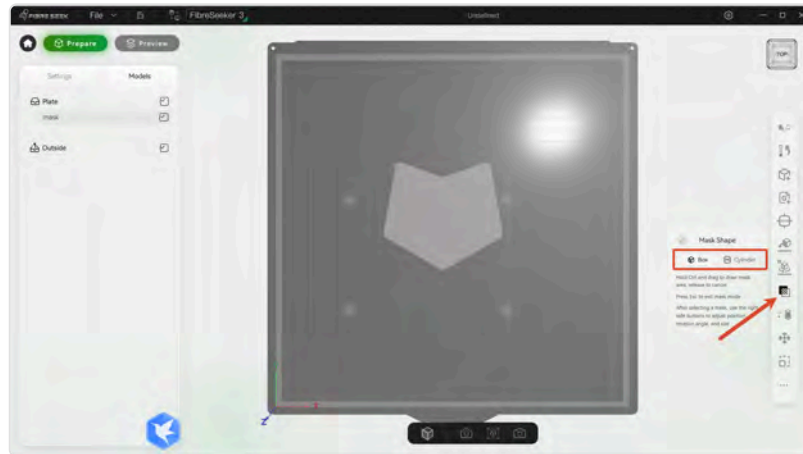
Icon	Function	Description
	Move models	Enter coordinates to control the mask position.
	Resize models	Choose manual, relative (proportional), or absolute (numeric) mode as needed and enter X/Y/Z values to scale the model.
	Rotate models	Manually rotate the model or enter X/Y/Z angle values for precise control.
	Clone models	Clone the selected model.
	Brush Mode	Drag the mouse to draw Z-seam positions, hiding inter-layer seams on non-visible surfaces or structural corners of the model to reduce post-processing difficulty and improve finished product quality.
	Layup mode	Customize the printing path layer by layer to meet different requirements for infill direction, line patterns, and more across layers.
	Measure	Click with the mouse to confirm the start and end points and quickly obtain the precise distance between two points, facilitating dimension checking and model verification.

4.2 Mask Mode

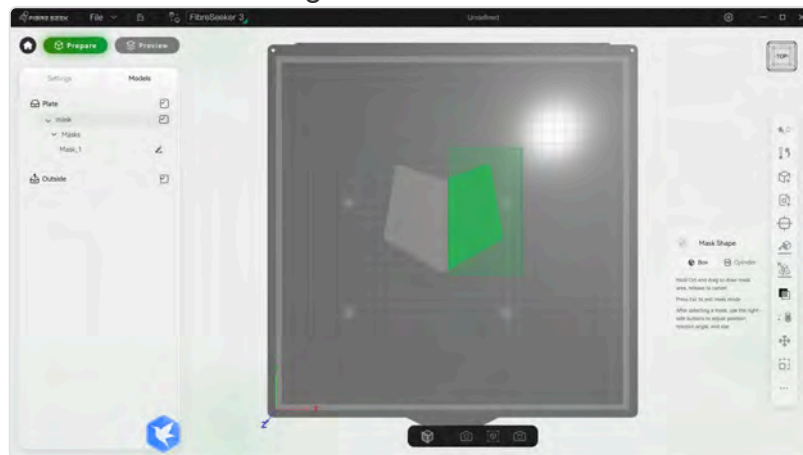
This feature allows you to add masks to specific areas of the model to precisely control the placement and quantity of fibre-reinforced materials. It also supports control over support structures: you can directly specify areas on the model where supports must be generated, or completely prohibit support generation.

Draw a mask

1. Click  to import a model in the right toolbar.
2. Choose the model and click  to select a box or cylinder as the mask shape.

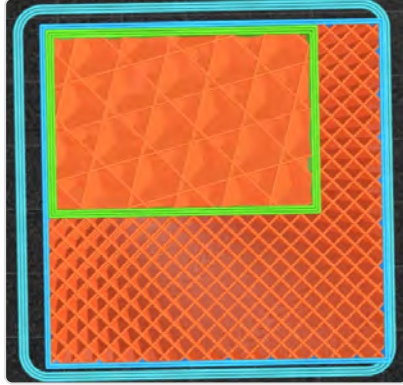


3. Press and hold the Ctrl key. Once the cursor becomes a plus sign (+), click and hold the left mouse button and drag to create the mask.

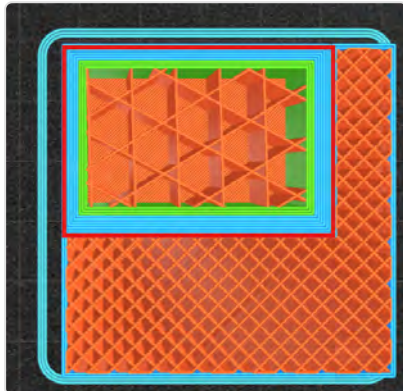


4. (Optional): When multiple masks are drawn, click  to the right of a mask to adjust its priority.
5. (Optional): Click  to access the following operations:
 - **name:** Modify the name of the mask.
 - **Mask type:**

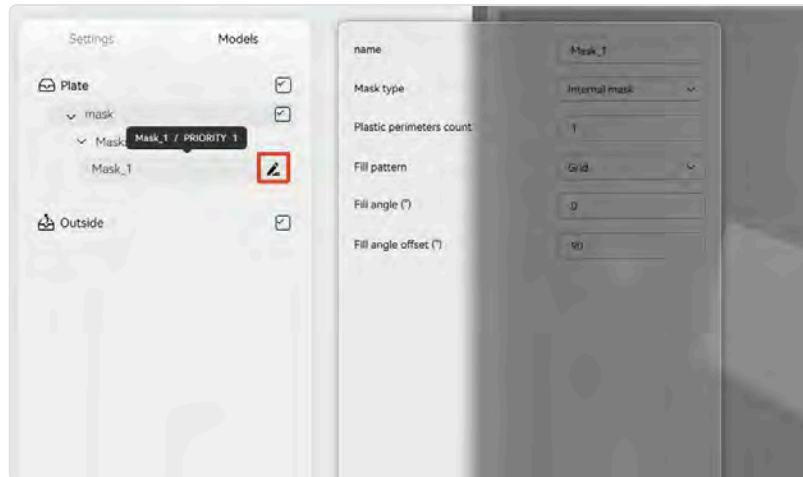
- **Internal mask:** The internal mask defines the internal structure of the part within the masked area. You can set the number of outer contour lines (shells) for this area. All regions outside the mask will be printed according to the global parameters of the profile. All internal mask bodies will share the same outer shell as the base part when printed.



- **Full mask:** The full mask creates an independent region inside the model, where print parameters (including outer wall layers, number of contour shells, and top/bottom solid layers) can differ from the base model. In Enhanced Mode, you can enable or disable reinforced outer contours and set their count, and independently set the type and density of reinforced infill.

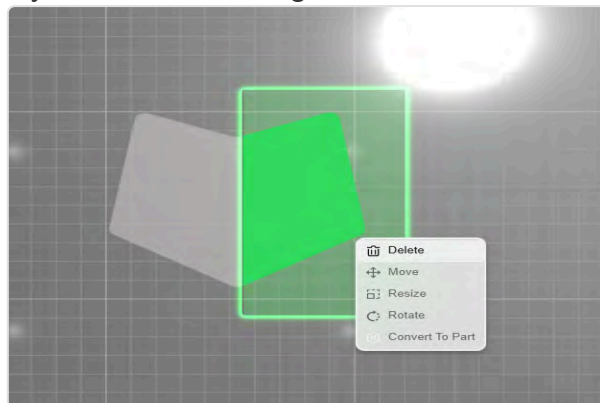


- **Support enhancer:** Forces support generation in the specified area.
- **Support blocker:** Prevents support generation in the specified area.
- **Fill pattern:** Selects the fill pattern for the mask, including grid, lines, triangles, etc. The actual patterns available are shown in the interface.
- **Fill angle:** Adjusts the angle of the fill pattern.
- **Fill angle offset:** Sets the incremental angle of the fill pattern per layer. For example, if the fill angle is 45°, and this parameter is set to 30°, the first layer path will be 45°, the next layer path will increase by 30° to 75°, and so on.



6. (Optional): Right-click the mask (or select it in the left model area and right-click) to access the following operations.

- **Rename:** Modify the name of the mask.
- **Delete:** Delete the mask.
- **Move:** Enter coordinates to control the model's position.
- **Resize:** Select the desired mode (Manual, Relative, or Absolute) and enter X/Y/Z values to scale the mask. When the Uniform option is enabled, adjusting any single dimension will automatically scale the other two proportionally.
- **Rotate:** Choose **angle** or **manual**, then enter X/Y/Z values to set the rotation.
- **Convert to Part:** Converts the target mask into a new model that can be printed as either a cylindrical or rectangular model.



7. (Optional): After converting a mask to a model, right-click it for the following operations:

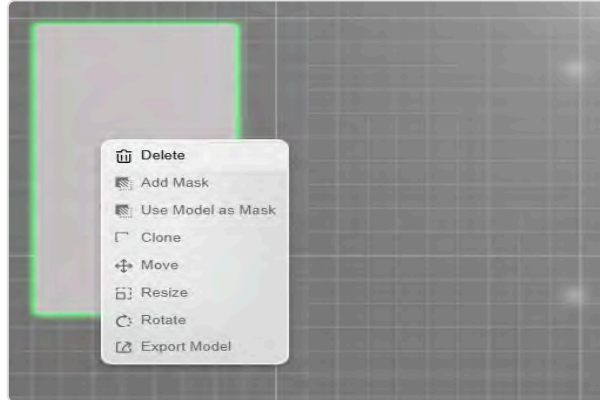
- **Delete:** Deletes the converted model.
- **Add Mask:** Adds a custom mask. Supported file formats include .stl and .obj.
- **Use Model as Mask:** Selects another model to serve as the mask for this model.



This feature requires at least 2 models.

- **Clone:** Duplicates the selected model.

- **Move:** Modify the X/Y/Z values to move the mask to the desired position.
- **Resize:** Select manual, relative, or absolute mode as needed, and modify the X/Y/Z values to scale the mask. Enable Uniform so that changing any one value automatically adjusts the others.
- **Rotate:** Select **angle** or **manual** mode, and adjust the X/Y/Z values to set the mask rotation.
- **Export Model:** Exports the model as an .stl file.

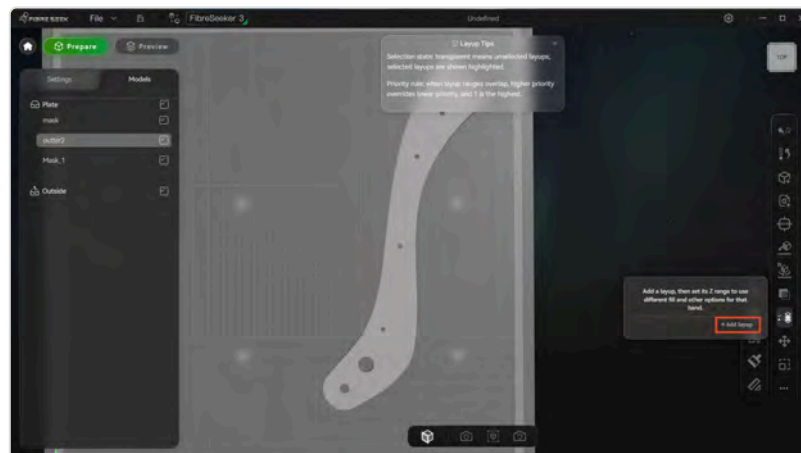


4.3 Layup Mode

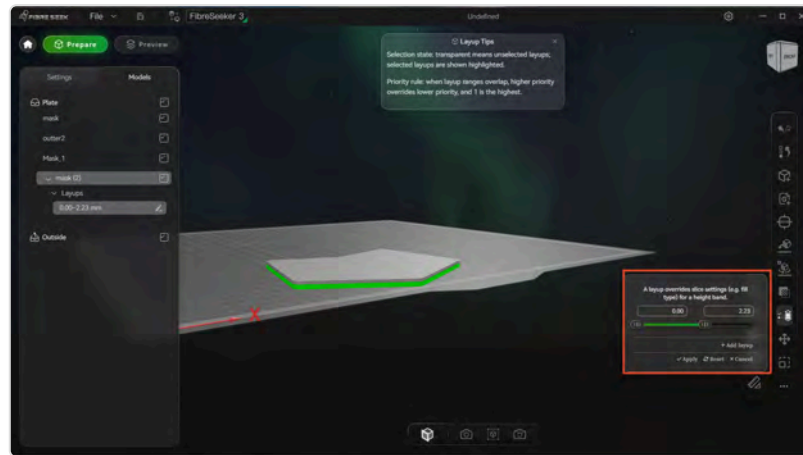
This feature allows you to customize print paths on a per-layer basis to meet different requirements for infill direction, line pattern, and more.

Steps

1. Click  in the right toolbar and select **Add layup**.



2. Enter the layer range values in the input box or drag  below the input box to adjust the range (the left side sets the minimum value and the right side sets the maximum value). In the figure below, the layer range is 0.00 — 2.23.



3. Click **Apply** to confirm.

Note

After selecting **Reinforced Mode**, you can also enable **Generate reinforced perimeters** and **Generate reinforced fill**, and adjust the parameters as needed.

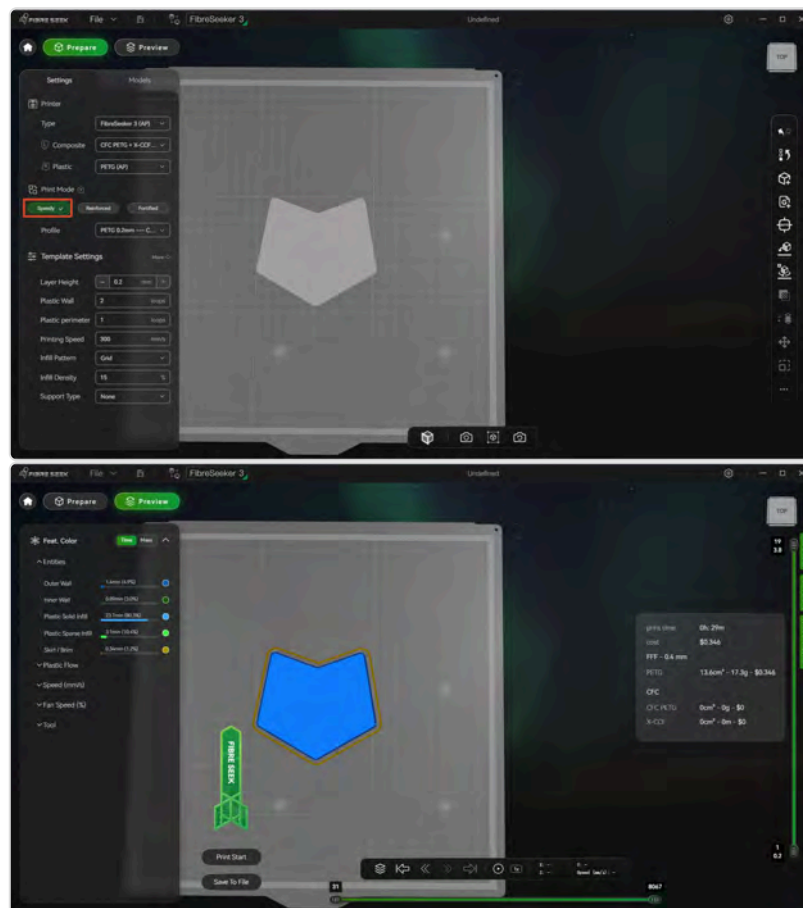
4. (Optional): Click  to the right of the layer range to access the following operations:
 - **External shell count**: Sets the number of outer shell layers for the printed model.
 - **Plastic perimeters count**: Sets the number of contour layers for the inner shell. This parameter directly affects the surface strength of the part.
 - **Fill pattern**: Selects the fill pattern, including Grid, Lines, Triangles, etc. The actual patterns available are shown in the interface.
 - **Fill angle**: Adjusts the angle of the fill pattern.
 - **Fill angle offset**: Sets the incremental angle of the fill pattern per layer. For example, if the fill angle is 45° and this parameter is set to 30°, the first layer path will be 45°, the next layer path will increase by 30° to 75°, and so on.
 - **Fill density**: Sets the density of the fill pattern.

5. Printing Mode

The FibreSeeker 3 3D printer supports three printing modes: Speedy, Reinforced, and Fortified.

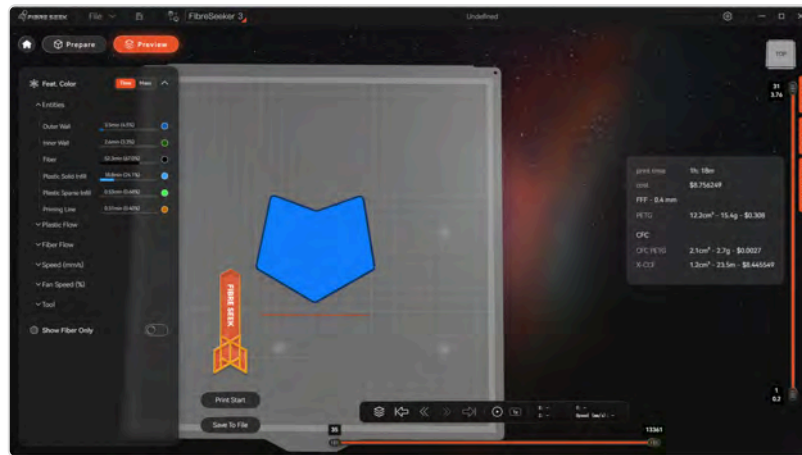
5.1 Speedy Mode

The FFF nozzle works independently, delivering faster printing speeds and higher precision, making it suitable for standard 3D printing applications.



5.2 Reinforced Mode

The CFC + FFF dual-nozzle system works in tandem to embed a fibre framework layer inside the printed part. The material ratio can be freely adjusted to achieve a precise balance between strength, weight, and cost.

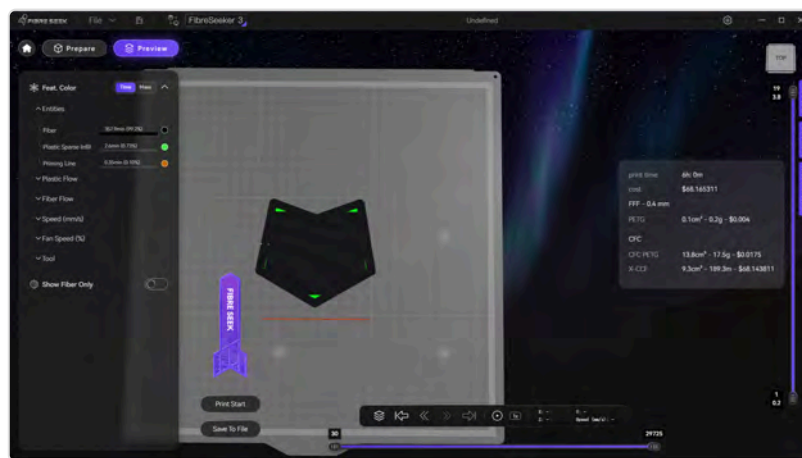


5.3 Fortified Mode

The CFC nozzle operates independently, utilizing advanced technology to produce high-strength components with a tensile strength of up to 900 MPa.

Note

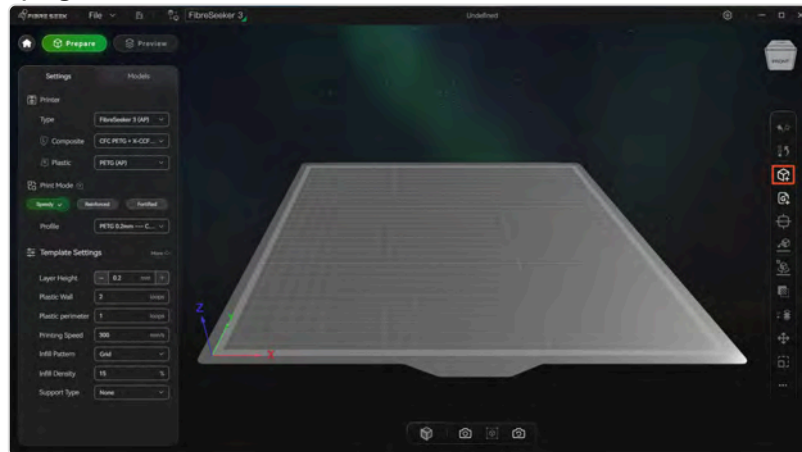
When print length or space is insufficient, plastic infill is used.



6. Slicing the Model

Within the slicing software, print parameters can be configured on the **Template Settings** area. The original parameters of built-in templates are non-editable. If you need customized template parameters, copy a built-in template first and adjust relevant print settings accordingly. For full instructions, refer to the *FibreSeeker 3 User Manual*.

1. Rocket Slicer supports two methods for importing models:
 - Click , or use the shortcut key Ctrl + O to import a slice file in the right toolbar. Supported formats include .stl, .step, .stp, and .3mf.
 - Drag and drop the model file directly onto the print plate of the Rocket Slicer preparation page.

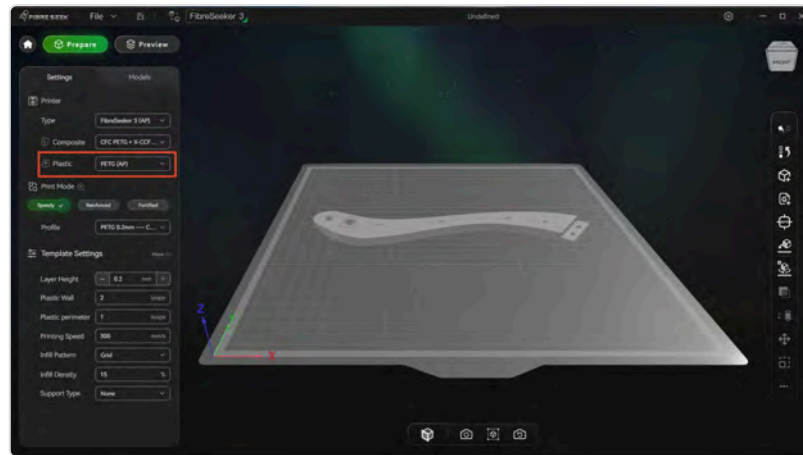


2. Click , ,  in the right toolbar to adjust the model's position, size, and orientation.

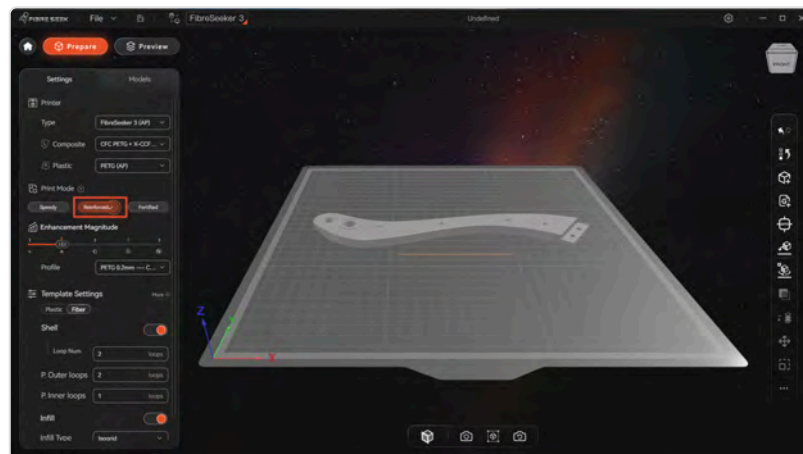
Note

When printing carbon fibre models, apply biomimetic principles to analyze the force-bearing direction and select the optimal orientation accordingly.

3. Select the target material profile (the figure below uses plastic as an example).



4. Select the desired print mode in the **Print Mode** section on the left toolbar. For more details, refer to [5. Printing Mode](#). The figure below uses Reinforced Mode as an example.



5. Click **Preview** at the top of the left toolbar to review the print parameters of the model, as shown in the figure below.



6. You can choose between the following two printing methods:
 - **Direct Print:** After connecting the printer, click **Print Start** in the lower-left corner.



- **USB Drive Transfer:** When Rocket Slicer is not connected to the printer, save the model as a G-code file on a USB drive, then insert the drive into the printer. On the printer touchscreen, tap  > **USB**, then select the target model for printing.

**Thank you for your choice! May every print
be smooth and successful, and may every one of your
creative ideas take shape perfectly.**

Disassemble/Reassemble :

Collision Block

1. Tools :

New collision block, H2.5 hex key.

2. Precautions :

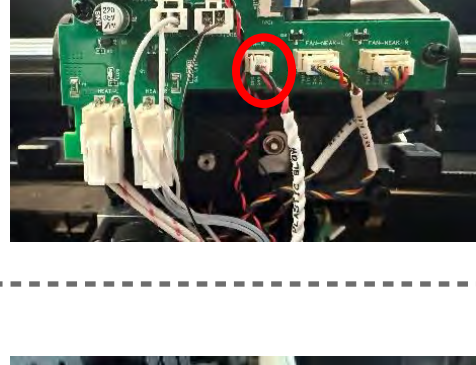
Always power off before operation. Beware of high temperature. Avoid pulling or straining cables to prevent loose connections.

1

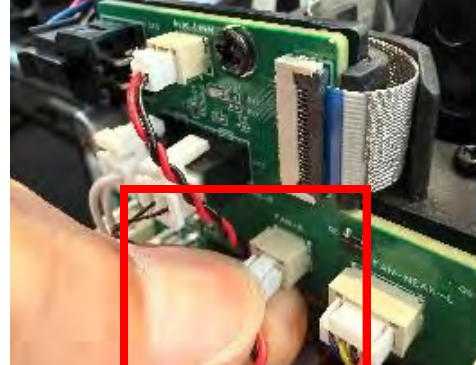


Step 1. Power Off

1. Flip the red switch on the rear of the machine to "O".

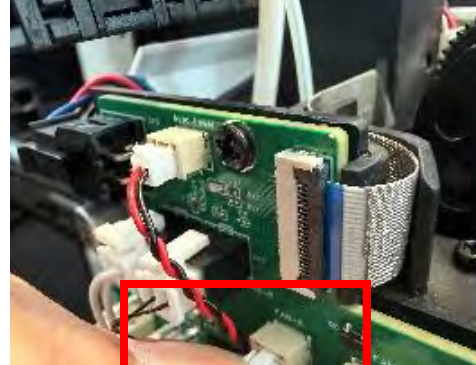


2. Pinch the lower left side of the front cover to release the latch.



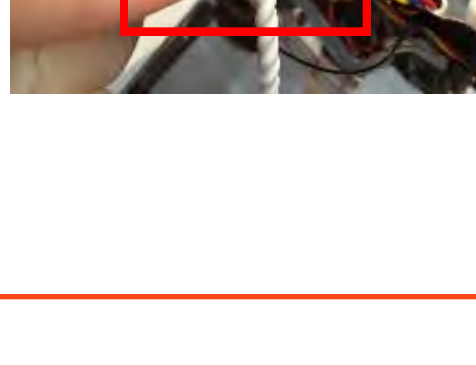
3. Pull the toolhead front cover forward to remove it.

1



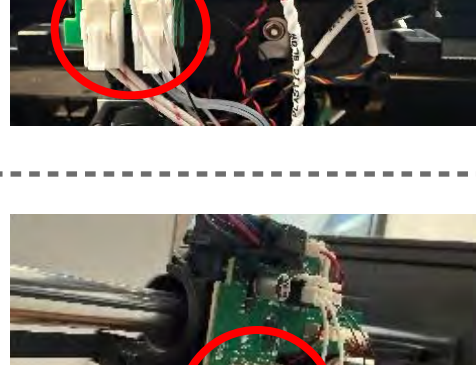
Step 2. Remove the Toolhead Front Cover

1. Pinch the latch on the fan plug at the "FAN-R" position on the mainboard.



2. Unplug the fan connector to remove the front cover.

2



Step 3. Loosen the Cable Connectors

1. Pinch the latch on the heater plug at the "HRAT-L" position on the mainboard.

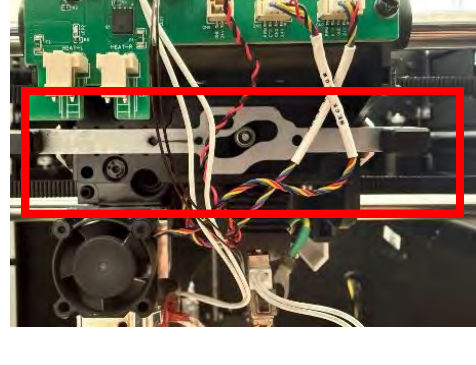


2. Pinch the latch on the heater plug at the "HRAT-R" position on the mainboard.



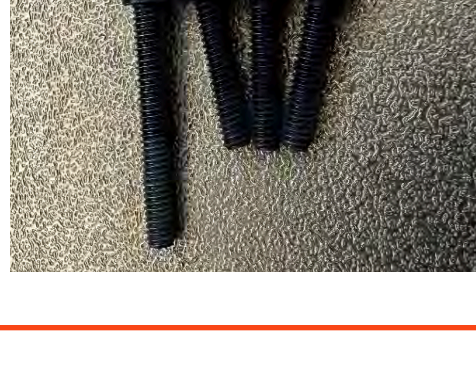
3. Unplug the heater.

3



Step 4. Remove the Collision Block Cover Fixing Screws

1. Use the H2.5 hex key.

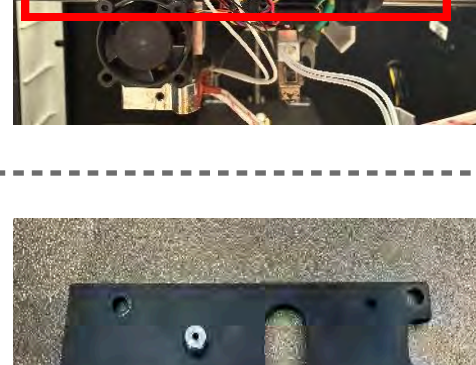


2. Remove the four screws securing the collision block cover.



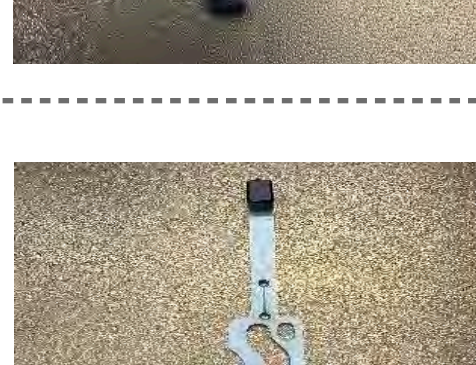
3. The top-left screw is longer than the other three.

4



Step 5. Remove the Collision Block

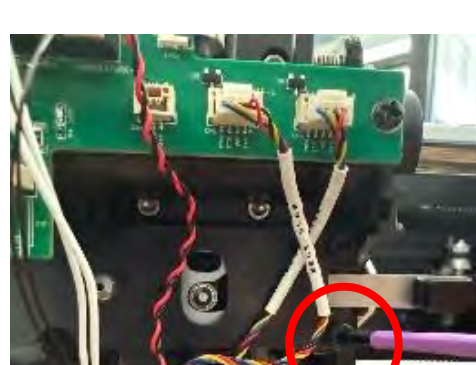
1. Pull the cover outward.



2. Grasp both ends of the collision block and pull it straight out horizontally.



5



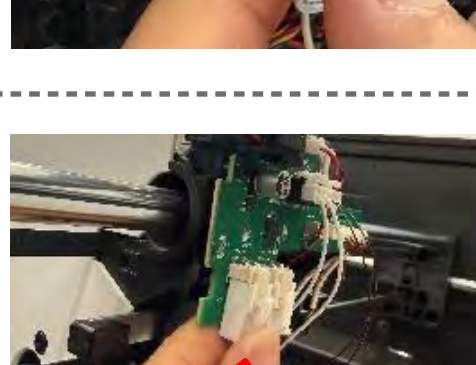
Step 6. Install the New Collision Block

1. Fit the new collision block, making sure the bearing engages correctly into the block's slide groove. Replace the cover.



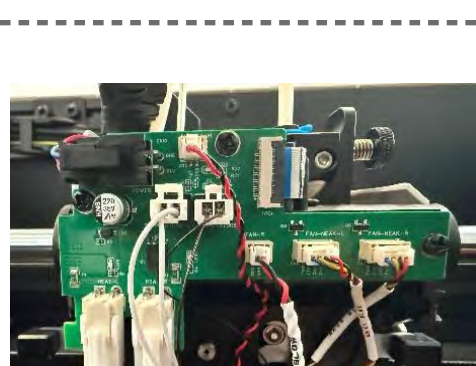
2. Use the H2.5 hex key to screw back the four screws (again, the top-left screw is the longer one). Slide the collision block to verify smooth movement.

6



Step 7. Reconnect Cables

1. Ensure the latch side of the heater cartridge connector faces the operator.



2. Plug it back into the mainboard.

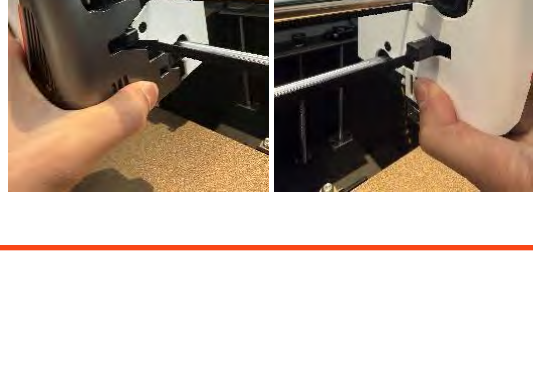


7

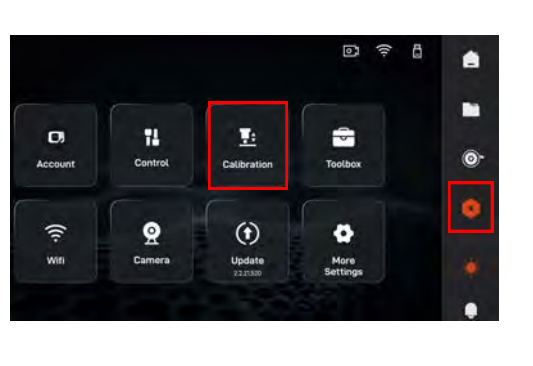


Step 8. Reinstall the Front Cover

1. Reinsert the fan plug into the corresponding port on the mainboard with the latch side facing up.

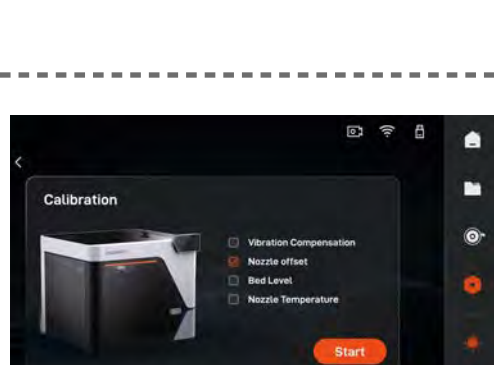


2. Pinch the lower part of the front cover, align the latch with the hole on the rear cover, and push the cover backward onto the toolhead.



3. When it clicks into place, the replacement is complete.

8



Step 9. Perform XYZ Offset

1. Power on and wait for boot. Tap the settings icon on the right side of the toolbar, and select "Calibration" → "Print Calibration".



2. Select only "Nozzle offset". Click "Start" and wait for the process to finish. The printer is ready for use.

3. Skipping this step may cause issues during printing.

9

Disassemble/Reassemble :

Fiber Air Guide

1. Tools :

Phillips screwdriver, new fiber air guide, H2 hex key

2. Precautions :

Always power off before operation. Beware of high temperature. Avoid pulling or straining cables to prevent loose connections.

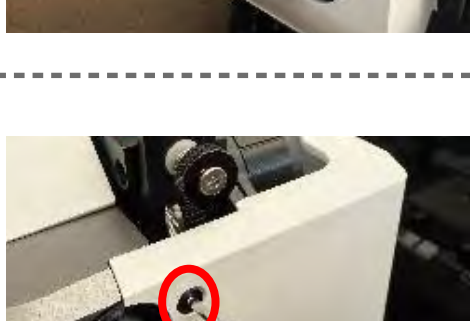
1



Step 1.
Power Off

and Remove the Fixing Screws

1. Power Off.

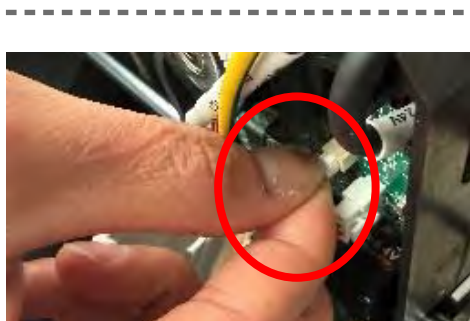


2. Use an H2 hex key.



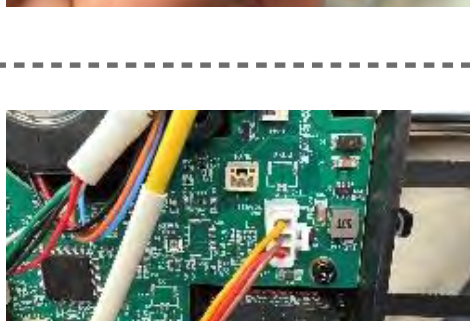
3. Remove the four fixing screws (two on the left , two on the right)

1

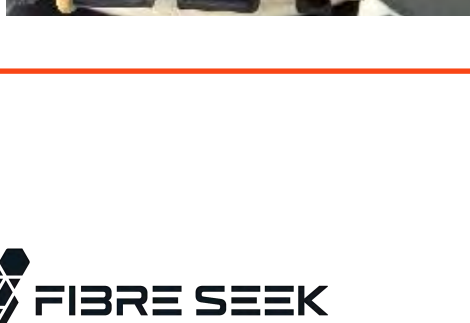


Step 2.
Loosen Cable Connectors to Remove the Rear Cover

1. Slide the toolhead rear cover horizontally backward to detach it.

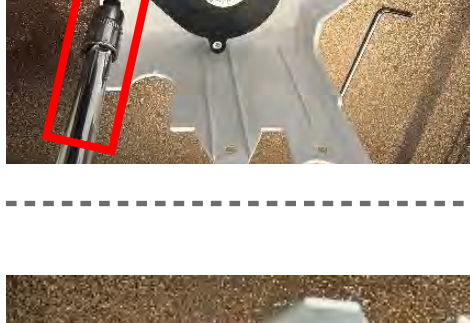


2. Press the white latch on the fan connector.



2. Remove the rear cover.

2



Step 3.
Remove the Old Air Guide

1. Use a Phillips screwdriver.



2. Remove the screw securing the fiber air guide.

3



Step 4.
Install the New Air Guide

1. Remove the old air guide.



2. Fit the new air guide so that its air inlet wraps around the fan outlet.



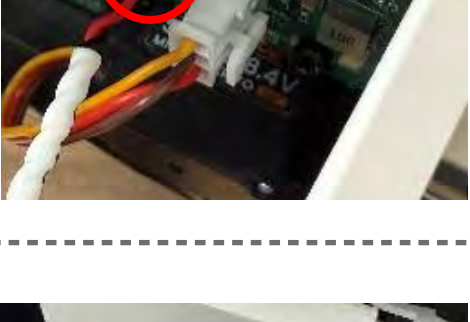
3. Secure the two fixing screws on the fan.

4



Step 5.
Reconnect Cables

1. Ensure the latch side of the fan connector faces upward.



2. Plug the fan connector back into the corresponding port on the rear mainboard.



3. Ensure the correct orientation, then reinstall the rear cover.

5



Step 6.
Reinstall the rear cover

1. Tighten the four fixing screws.



2. Tighten two screws on the left and two on the right.

6

Disassemble/Reassemble:

Fiber Hotend Heater

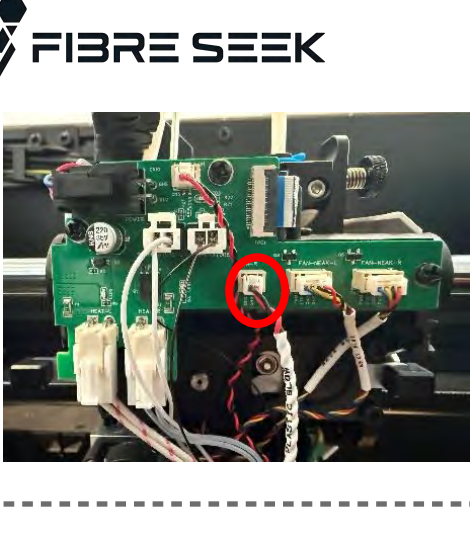
1. Tools :

New fiber hotend heater, H1 hex key, H2.5 hex key, diagonal pliers, thermal grease.

2. Precautions :

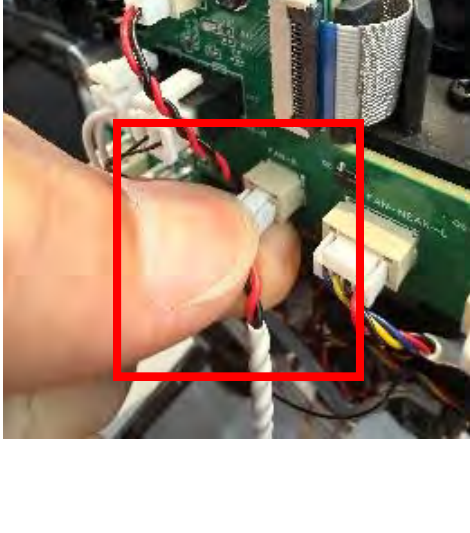
Always power off before operation. Beware of high temperature.
Avoid pulling or straining cables to prevent loose connections.

1



Step 1. Power Off

1. Flip the red switch on the rear of the machine to "O".

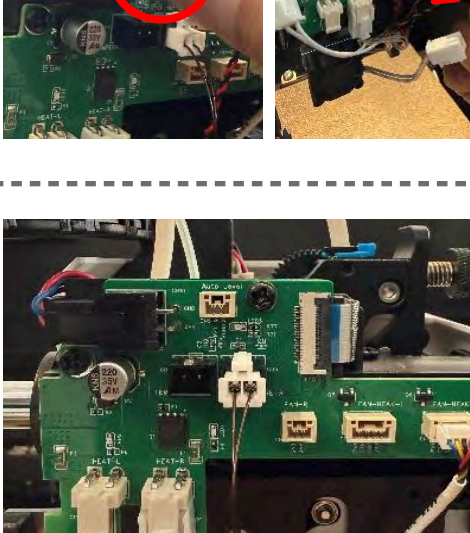


2. Pinch the lower left side of the front cover to release the latch.



3. Pull the toolhead front cover forward to remove it.

2



Step 2. Remove the Toolhead Front Cover

1. Pinch the latch on the fan plug at the "FAN-R" position on the mainboard.



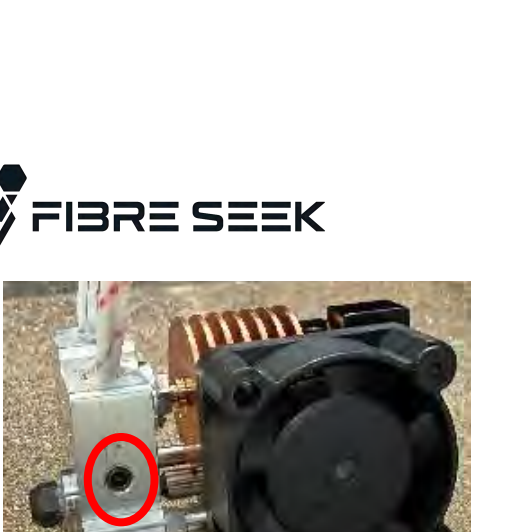
2. Unplug the fan connector to remove the front cover.

3

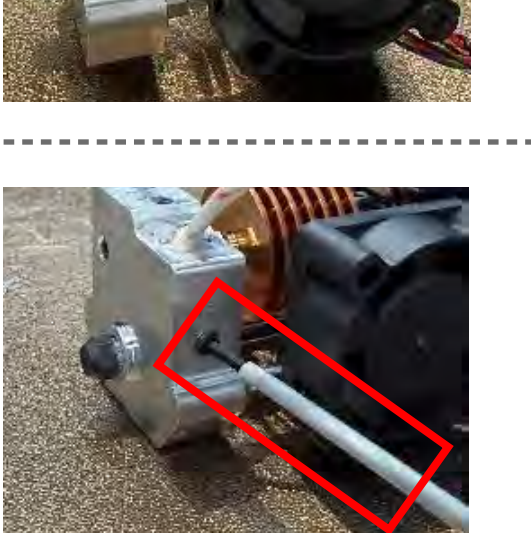


Step 3. Loosen the Cable Connectors

1. Press the latch on the connector labeled "HRAT-L" on the mainboard, then unplug the heater cartridge.



2. Press the latch on the connector labeled "TEMPERATURE-L", then unplug the temperature sensor.



3. Press the latch on the connector labeled "FAN-NEAK-L", then unplug the heat sink fan.



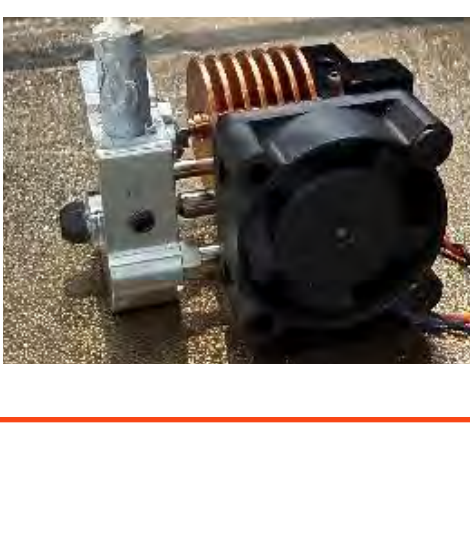
4. Press the latch on the connector labeled "Auto Level", then unplug the auto-leveling sensor.

4



Step 4. Remove the fixing screws

1. Use an H2.5 hex key.

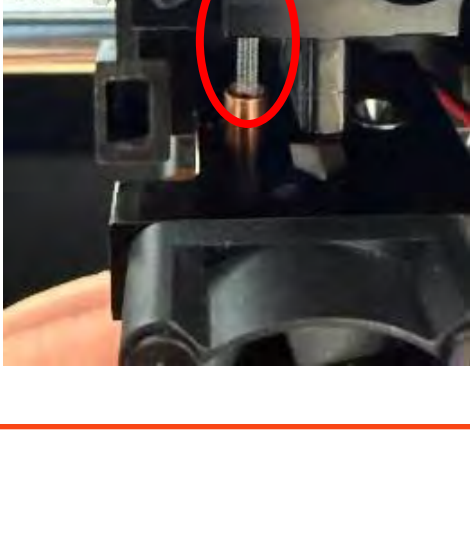


2. Remove two screws securing the left composite head hotend.



3. Remove the entire hotend assembly.

5

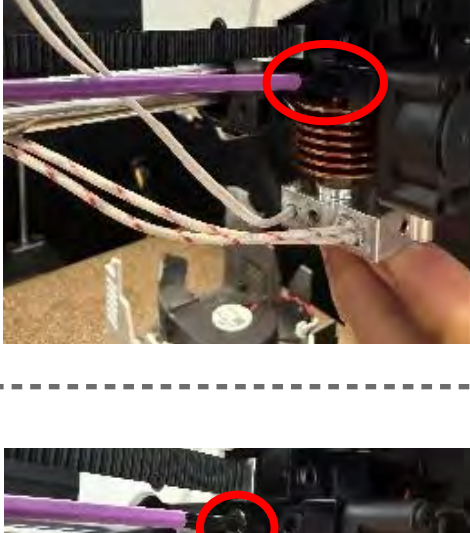


Step 5. Remove the Heater Cartridge

1. Use an H1.0 hex key to turn the set screw counterclockwise for 3-4 rotations from the bottom.

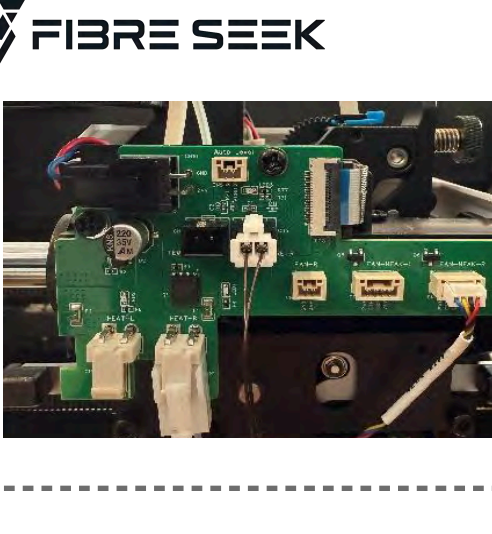
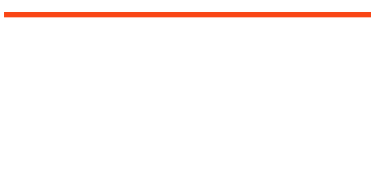


2. Loosen the heater.



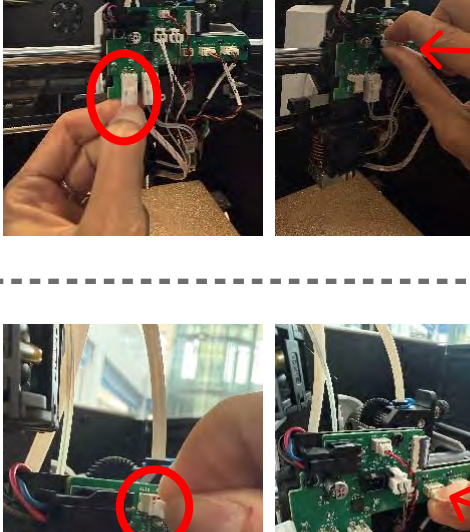
3. Gently pull the heater upward by its cable and remove it.

6



Step 6. Install the New Heater Cartridge

1. Apply a thin layer of thermal grease to the probe housing of the new fiber hotend temperature sensor. Insert the sensor vertically downward to the full depth (the entire heated section must be fully embedded in the hole).

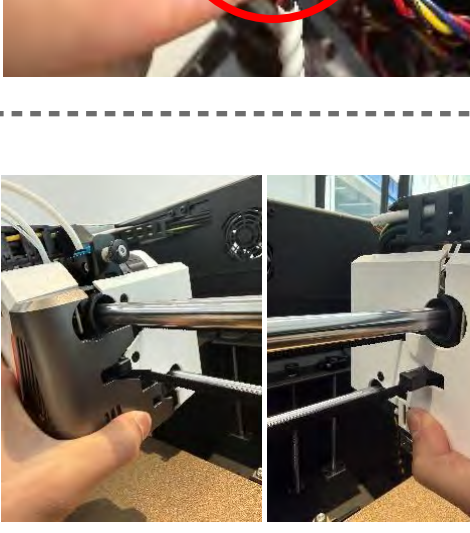
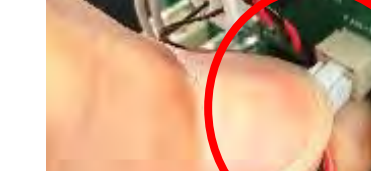


2. Use the H1.0 hex key to tighten the set screw clockwise to secure the sensor.



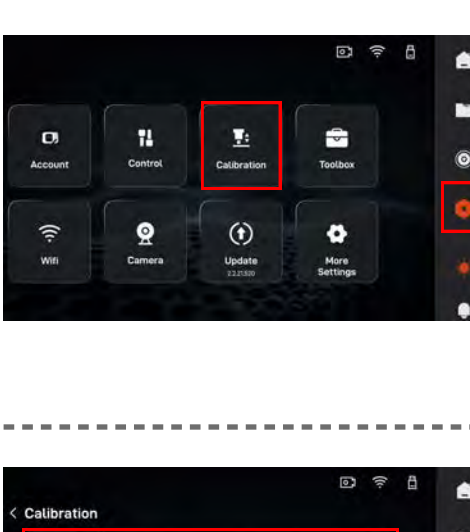
3. Do not overtighten. Excessive force will compress the probe housing and damage internal components.

7



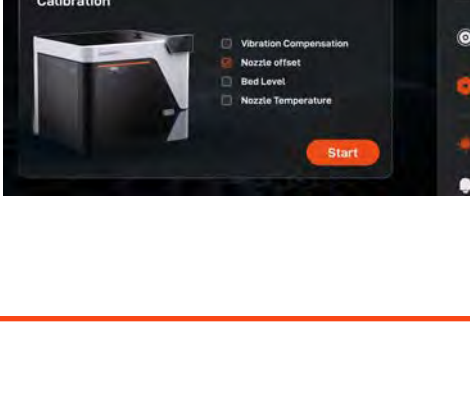
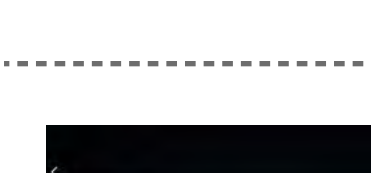
Step 7. Trim the filament inlet

1. Use side cutters to trim off about half of the excess plastic from the fiber hotend inlet. This prevents residual material in the throat tube from blocking full insertion.



2. Align the copper tube of the assembled fiber hotend with the filament path.

8



Step 8. Reinstall the hotend assembly

1. Push the hotend upward from the bottom.

2. Tighten the two fixing screws for the fiber hotend after ensuring no gaps between the two planes.

3. Keep pushing upward by hand to prevent the hotend from moving downward.

9

Step 9. Reconnect cables

1. Pinch the latch on the ceramic heater plug at the "HRAT-L" position to replug the heater.

2. Pinch the latch on temperature sensor plug at the "TEMPERATURE-L" position to replug the sensor.

3. Press the latch on the heat sink fan connector at the "FAN-NEAK-L" position to replug the fan.

4. Press the latch on the auto-leveling sensor connector at the "Auto Level" position to replug the sensor.

10

Step 10. Reinstall the Front Cover

1. Reinsert the fan plug into the corresponding port on the mainboard with the latch side facing up.

2. Pinch the lower part of the front cover, align the latch with the hole on the rear cover, and push the cover backward onto the toolhead.

3. When it clicks into place, the replacement is complete.

11

Step 11. Perform XYZ offset

1. Power on and wait for boot. Tap the settings icon on the right side of the toolbar, and select "Calibration" → "Print Calibration".

2. Select only "Nozzle offset". Click "Start" and wait for the process to finish. The printer is ready for use.

3. Skipping this step may cause issues during printing.

12

Disassemble/Reassemble : Fiber Hotend Nozzle

1. Tools :

New fiber hotend nozzle, H1 hex key, H2.5 hex key, 5 mm hex socket, diagonal pliers/cutters, thermal grease.

2. Precautions :

Always power off before operation. Beware of high temperature. Avoid pulling or straining cables to prevent loose connections.

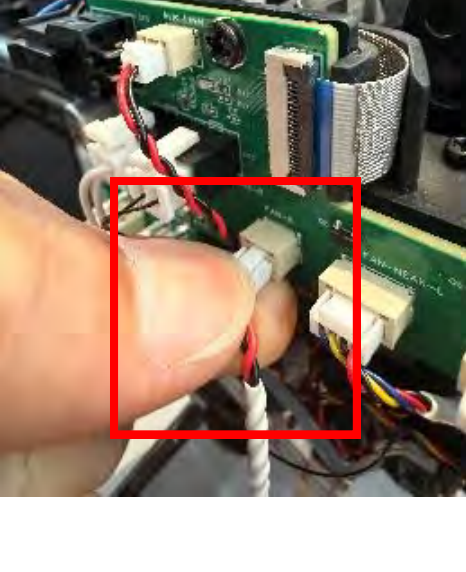
1



Step 1.

Power Off

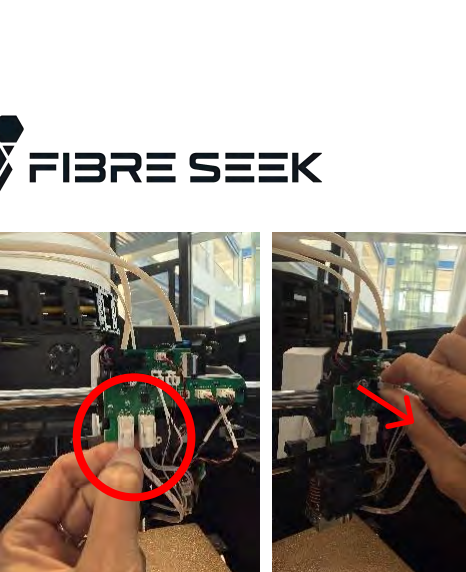
1. Flip the red switch on the rear of the machine to "O".



Step 2.

Remove the Toolhead Front Cover

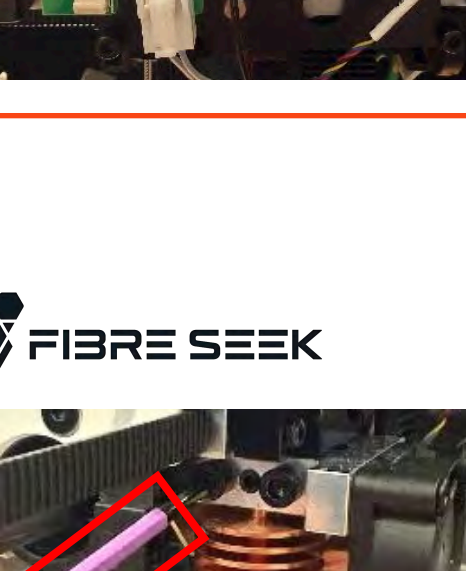
1. Pinch the latch on the fan plug at the "FAN-R" position on the mainboard.



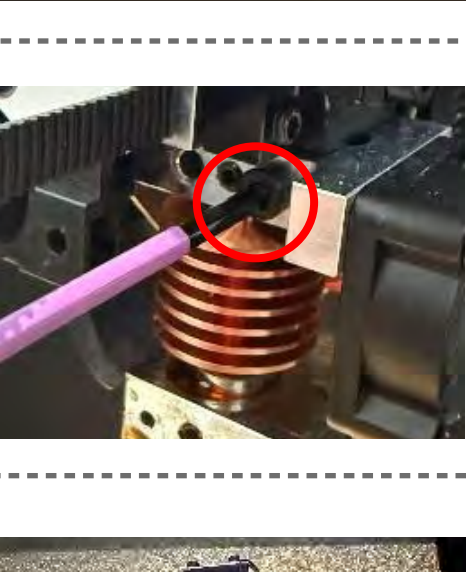
Step 3.

Loosen the Cable Connectors

1. Pinch the latch on the ceramic heater plug at the "HRAT-L" position to unplug the heater.
2. Pinch the latch on temperature sensor plug at the "TEMPERATURE-L" position to unplug the sensor.
3. Press the latch on the heat sink fan connector at the "FAN-NEAK-L" position to unplug the fan.
4. Press the latch on the auto-leveling sensor connector at the "Auto Level" position to unplug the sensor.



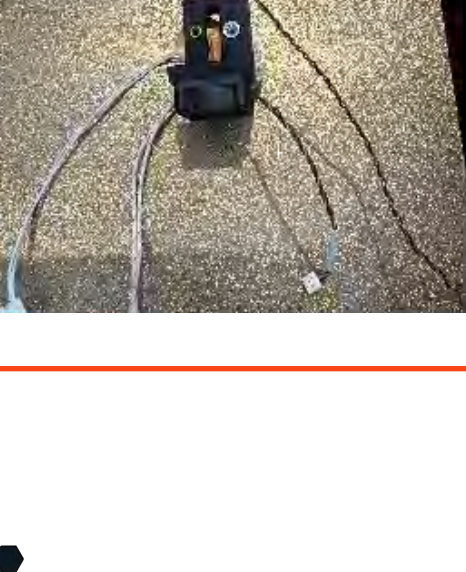
3



Step 4.

Remove the fixing screws.

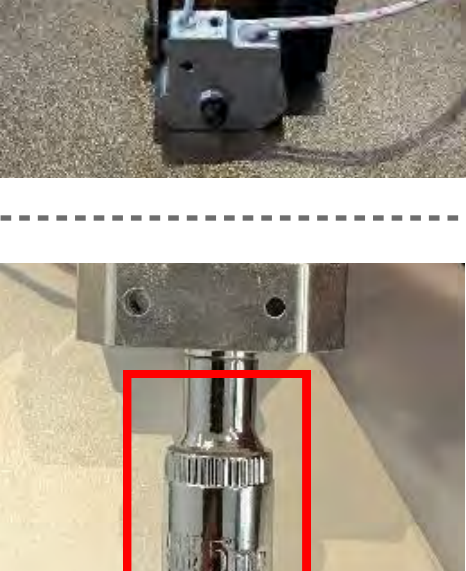
1. Use an H2.5 hex key.



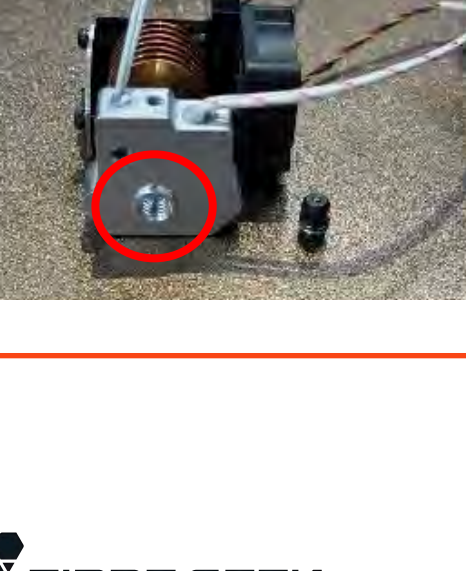
Step 5.

Remove the old nozzle

1. Use a 5 mm socket to remove the fiber hotend nozzle.
2. When the nozzle points upward, turn counterclockwise to loosen and clockwise to tighten. Reverse the direction if the nozzle points downward.
3. If it cannot be disassembled, use a heat gun to gently heat it, allowing the plastic to soften and become easier to twist loose (be careful of high temperatures!).



5



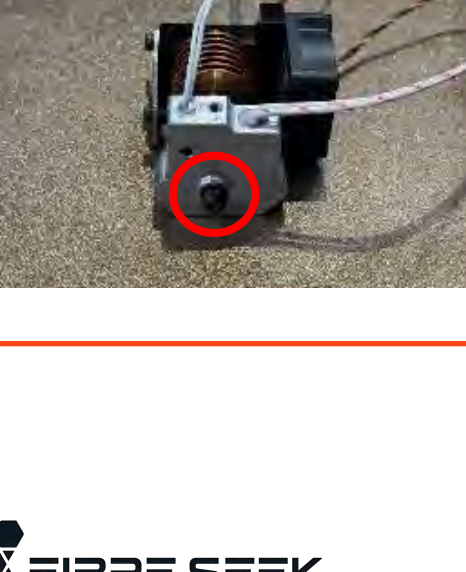
Step 6.

Prepare the new fiber hotend nozzle

1. Ensure the inside of the nozzle is completely empty.
2. Use a 5 mm socket. Turn counterclockwise to loosen and clockwise to tighten. Reverse the direction if the nozzle points downward.
3. Tighten until secure.



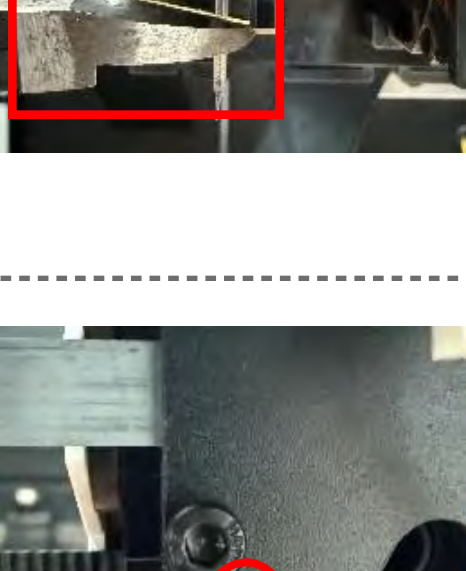
7



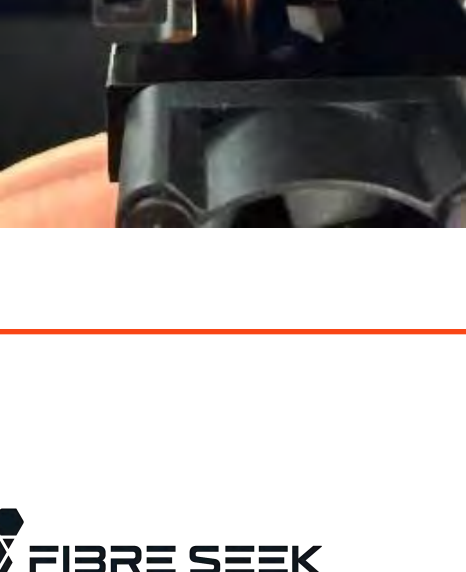
Step 7.

Trim the filament inlet

1. Use side cutters to trim off about half of the excess plastic from the fiber hotend inlet. This prevents residual material in the throat tube from blocking full insertion.
2. Align the copper tube of the assembled fiber hotend with the filament path.



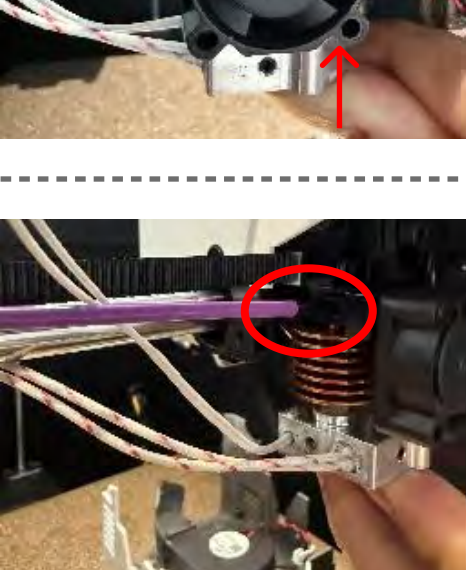
8



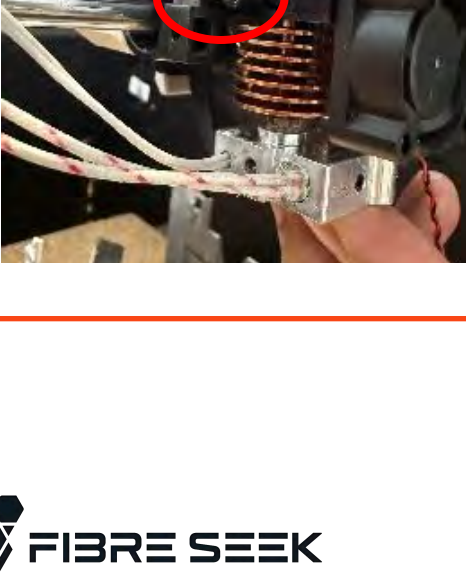
Step 8.

Reinstall the hotend assembly

1. Push the hotend upward from the bottom.
2. Tighten the two fixing screws for the fiber hotend after ensuring no gaps between the two planes.
3. Keep pushing upward by hand to prevent the hotend from moving downward.



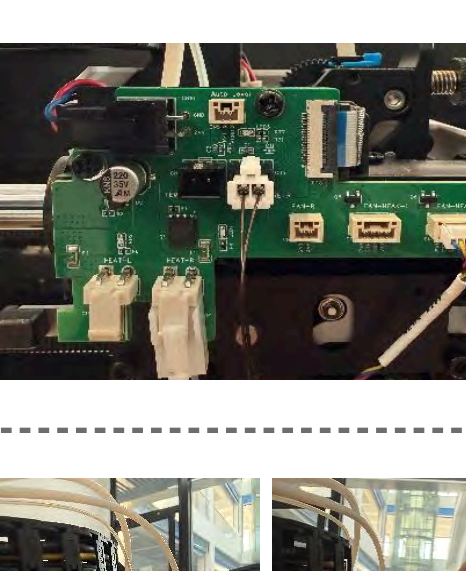
9



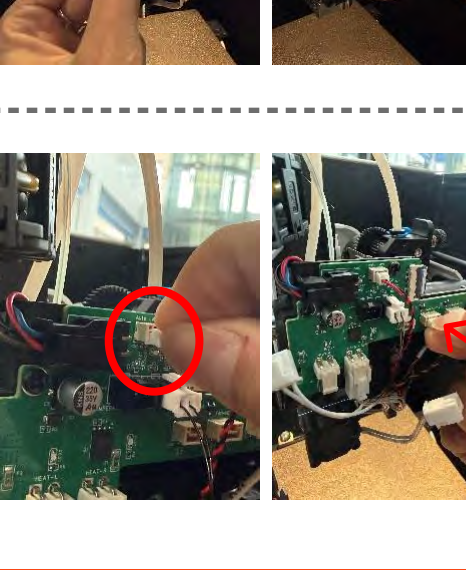
Step 9.

Reconnect cables

1. Pinch the latch on the ceramic heater plug at the "HRAT-L" position to replug the heater.
2. Pinch the latch on temperature sensor plug at the "TEMPERATURE-L" position to replug the sensor.
3. Press the latch on the heat sink fan connector at the "FAN-NEAK-L" position to replug the fan.
4. Press the latch on the auto-leveling sensor connector at the "Auto Level" position to replug the sensor.



10



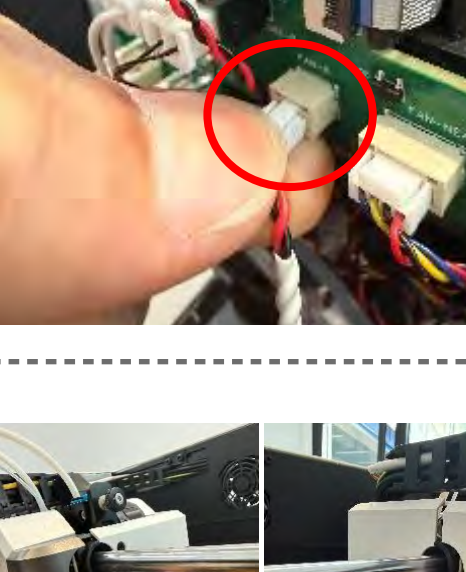
Step 10.

Reinstall the Front Cover

1. Reinsert the fan plug into the corresponding port on the mainboard with the latch side facing up.
2. Pinch the lower part of the front cover, align the latch with the hole on the rear cover, and push the cover backward onto the toolhead.
3. When it clicks into place, the replacement is complete.



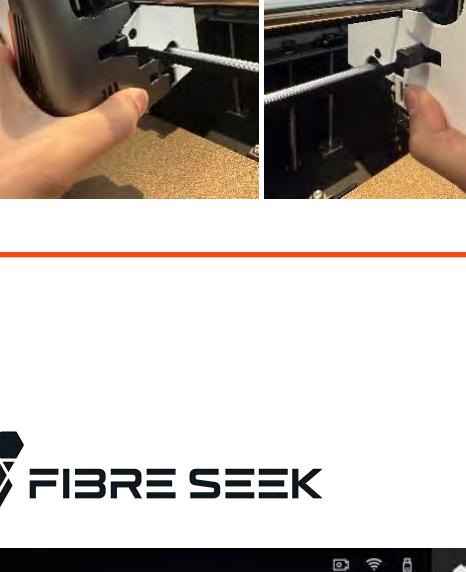
11



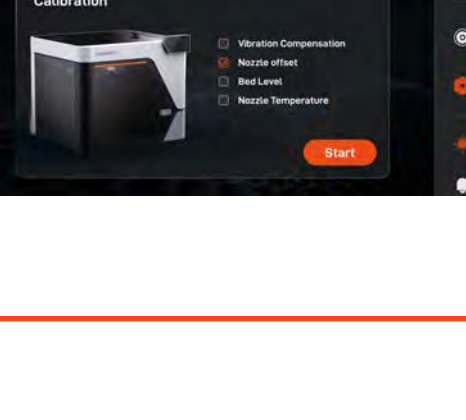
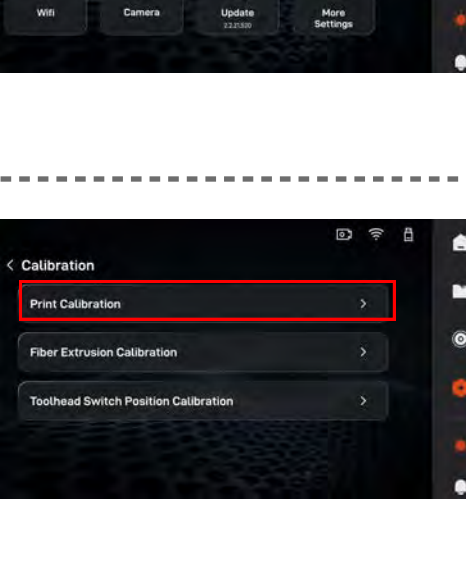
Step 11.

Perform XYZ Offset

1. Power on and wait for boot. Tap the settings icon on the right side of the toolbar, and select "Calibration" → "Print Calibration".
2. Select only "Nozzle offset". Click "Start" and wait for the process to finish. The printer is ready for use.
3. Skipping this step may cause issues during printing.



12



Disassemble/Reassemble : Fiber Hotend Temperature Sensor

1. Tools :

Fiber hotend temperature sensor, H1.5 hex key, H2.5 hex key, diagonal pliers, thermal grease.

2. Precautions :

Always power off before operation. Beware of high temperature. Avoid pulling or straining cables to prevent loose connections.

1

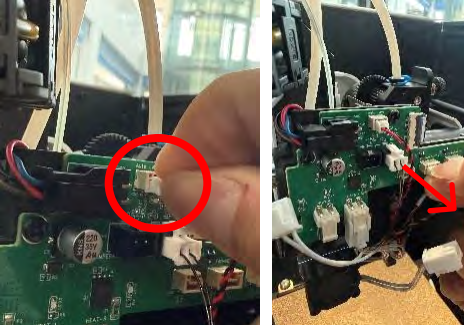


Step 1.

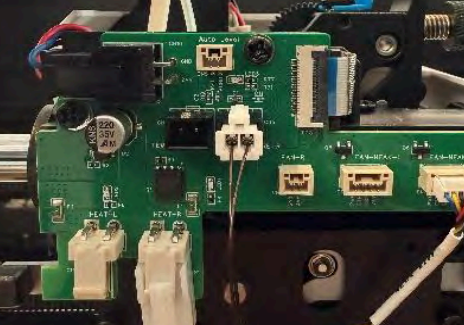
Power Off

1. Flip the red switch on the rear of the machine to "O".

2



2. Pinch the lower left side of the front cover to release the latch.



3. Pull the toolhead front cover forward to remove it.

3



Step 2.

Remove the Toolhead Front Cover

1. Pinch the latch on the fan plug at the "FAN-R" position on the mainboard.



2. Unplug the fan connector to remove the front cover.

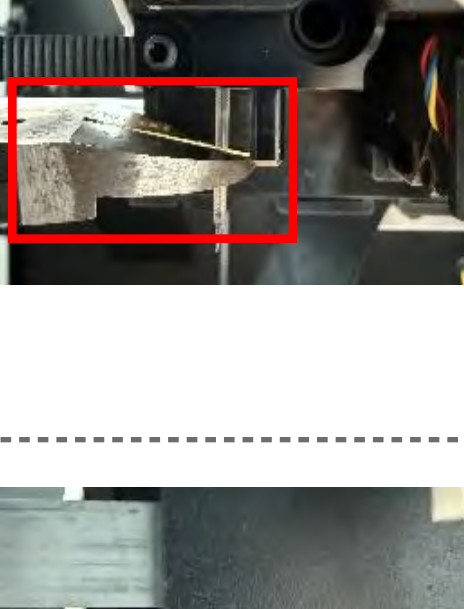
4



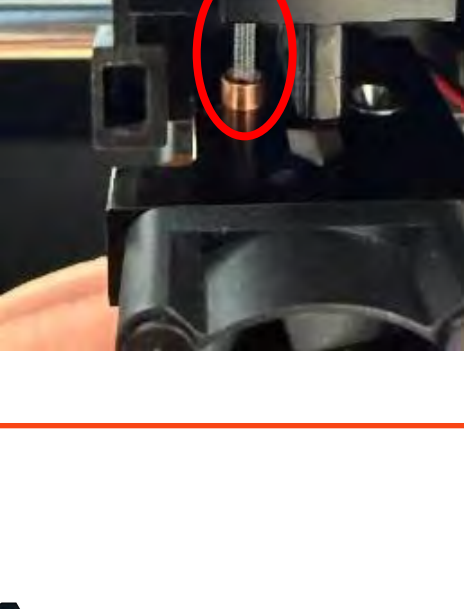
Step 3.

Loosen Cable Connectors

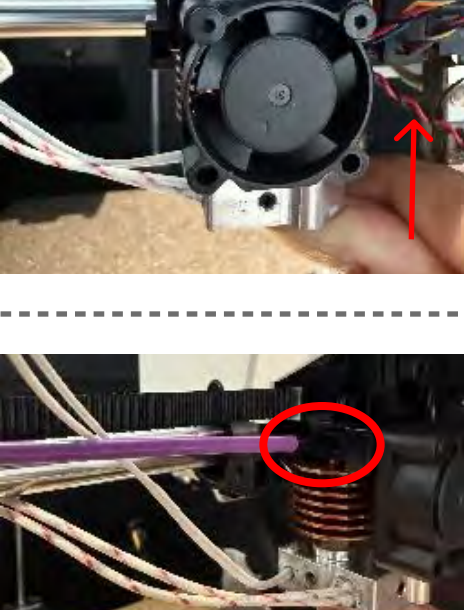
1. Pinch the latch on the ceramic heater plug at the "HRAT-L" position to unplug the heater.



2. Pinch the latch on temperature sensor plug at the "TEMPERATURE-L" position to unplug the sensor.

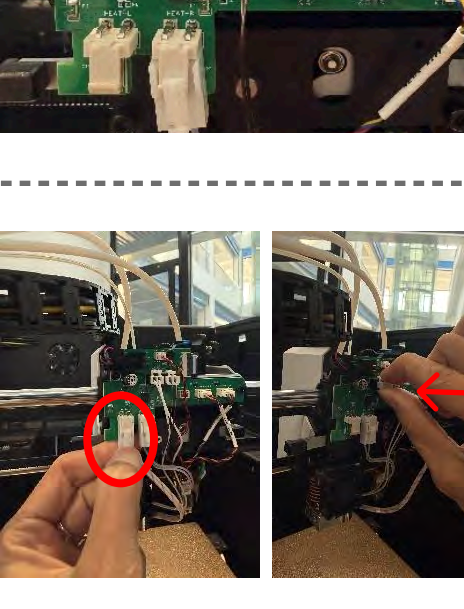


3. Press the latch on the heat sink fan connector at the "FAN-NEAK-L" position to unplug the fan.



4. Press the latch on the auto-leveling sensor connector at the "Auto Level" position to unplug the sensor.

5



Step 4.

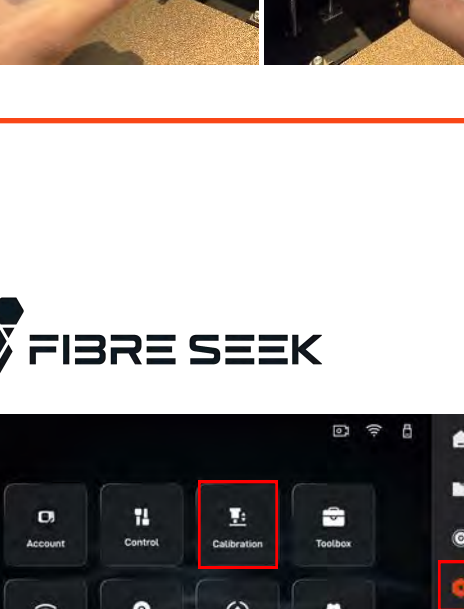
Remove the Fixing Screws

1. Use an H2.5 hex key.



2. Remove two screws securing the left composite head hotend.

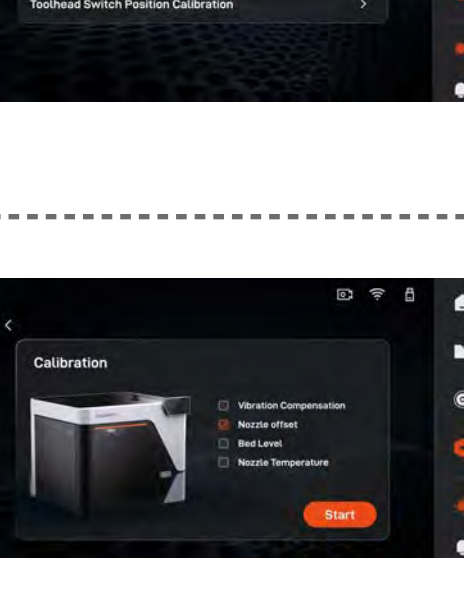
6



Step 5.

Remove the Old Temperature Sensor

1. Use the H1.0 hex key to turn the set screw counterclockwise for 3-4 rotations from the bottom.



2. Loosen the fiber hotend temperature sensor, grasp the cable from above, and slowly pull up the sensor.

3. You can remove the sensor.

7

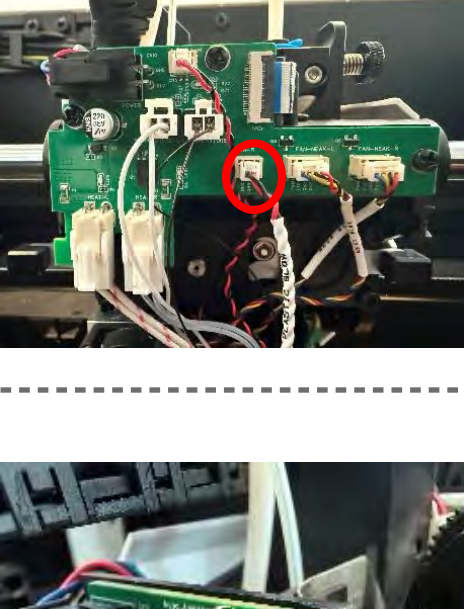
Step 6.

Install a New Temperature Sensor

1. Apply a thin layer of thermal grease to the probe housing of the new fiber hotend temperature sensor. Insert the sensor vertically downward to the full depth (the entire heated section must be fully embedded in the hole).

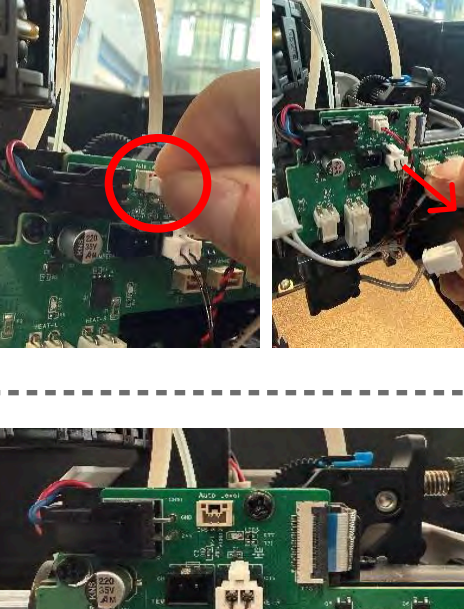


2. Use the H1.0 hex key to tighten the set screw clockwise to secure the sensor.



3. Do not overtighten. Excessive force will compress the probe housing and damage internal components.

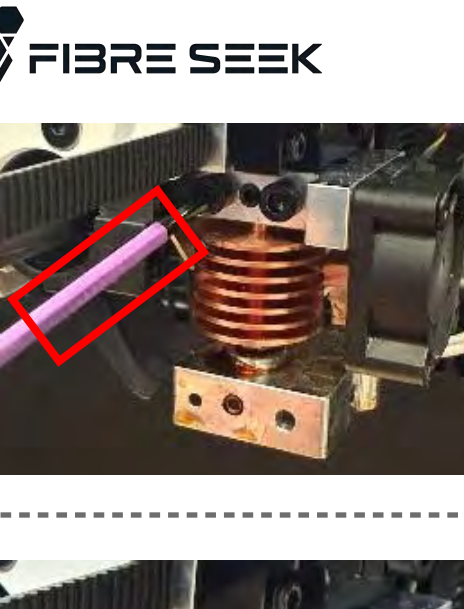
8



Step 7.

Trim the Filament Inlet

1. Use side cutters to trim off about half of the excess plastic from the fiber hotend inlet. This prevents residual material in the throat tube from blocking full insertion.



2. Align the copper tube of the assembled fiber hotend with the filament path.

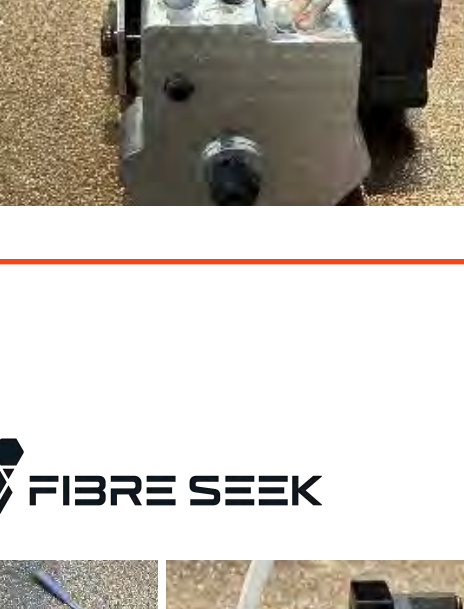
9



Step 8.

Reinstall the Hotend Assembly

1. Push the hotend upward from the bottom.

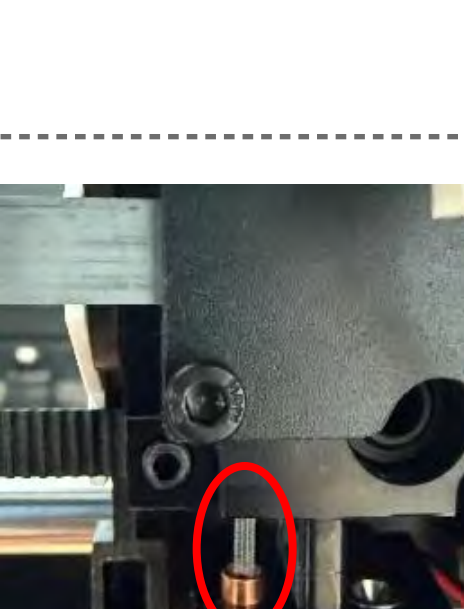


2. Tighten the two fixing screws for the fiber hotend after ensuring no gaps between the two planes.



3. Keep pushing upward by hand to prevent the hotend from moving downward.

10



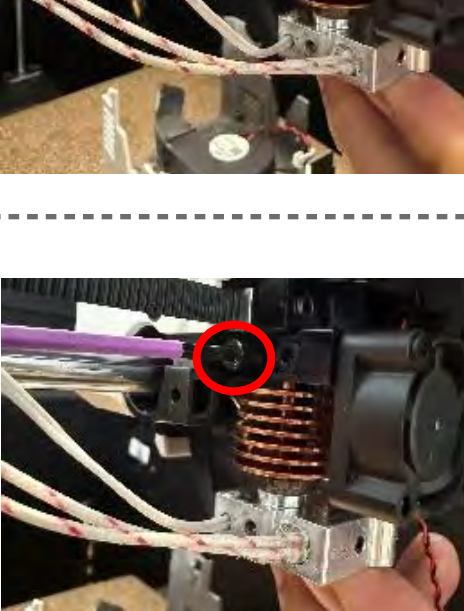
Step 9.

Reconnect Cables

1. Pinch the latch on the ceramic heater plug at the "HRAT-L" position to replug the heater.



2. Pinch the latch on temperature sensor plug at the "TEMPERATURE-L" position to replug the sensor.

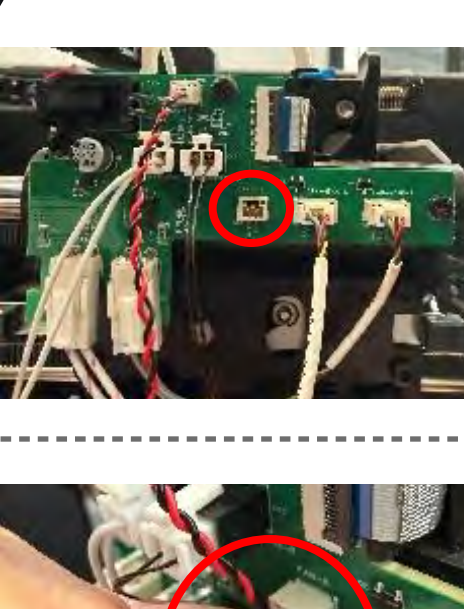


3. Press the latch on the heat sink fan connector at the "FAN-NEAK-L" position to replug the fan.



4. Press the latch on the auto-leveling sensor connector at the "Auto Level" position to replug the sensor.

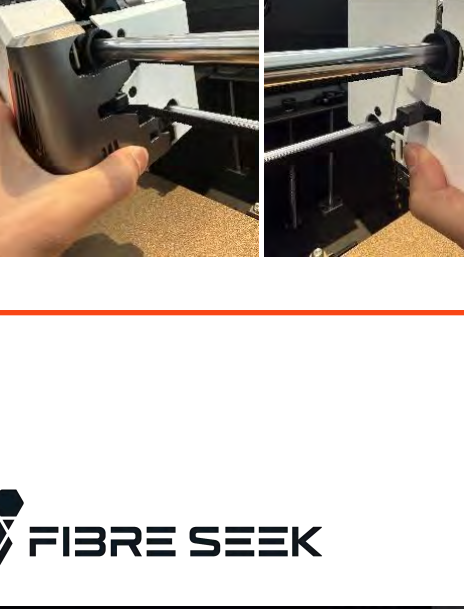
11



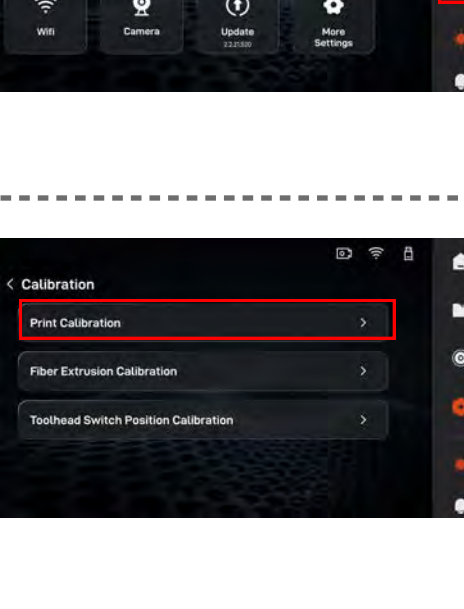
Step 10.

Reinstall the Front Cover

1. Reinsert the fan plug into the corresponding port on the mainboard with the latch side facing up.

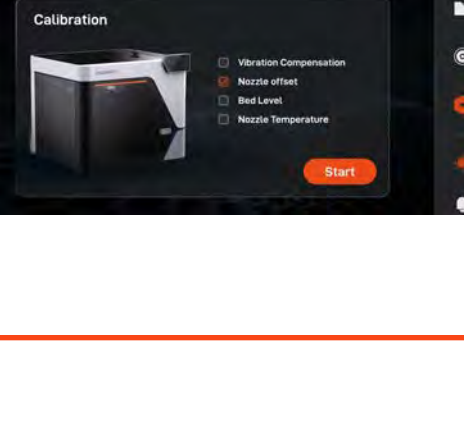


2. Pinch the lower part of the front cover, align the latch with the hole on the rear cover, and push the cover backward onto the toolhead.



3. When it clicks into place, the replacement is complete.

12



Step 11.

Perform XYZ Offset

1. Power on and wait for boot. Tap the settings icon on the right side of the toolbar, and select "Calibration" → "Print Calibration".

2. Select only "Nozzle offset". Click "Start" and wait for the process to finish. The printer is ready for use.

3. Skipping this step may cause issues during printing.

Disassemble/Reassemble :

Piezoelectric Ceramic

1. Tools :

Piezoelectric ceramic set, H2.5 hex key, H2 hex key, diagonal pliers.

2. Precautions :

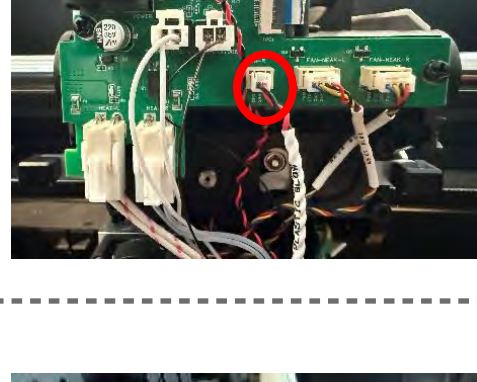
Always power off before operation. Beware of high temperature. Avoid pulling or straining cables to prevent loose connections.

1

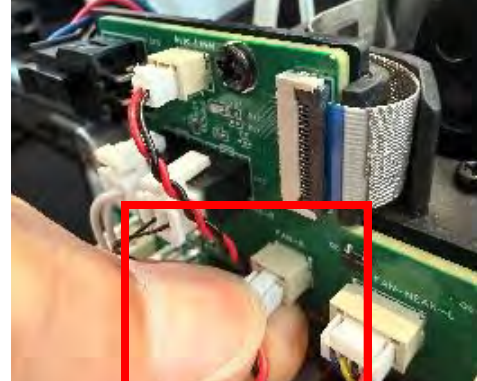
FIBRE SEEK


Step 1. Power Off

1. Flip the red switch on the rear of the machine to "O".

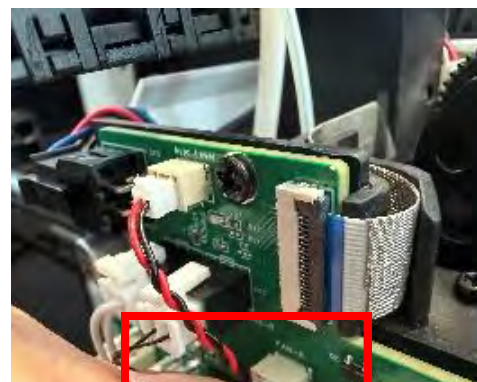


2. Pinch the lower left side of the front cover to release the latch.



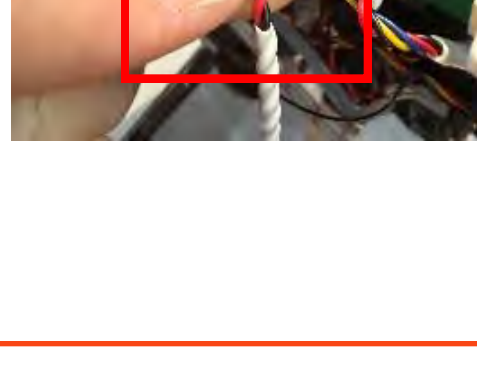
3. Pull the toolhead front cover forward to remove it.

1

FIBRE SEEK


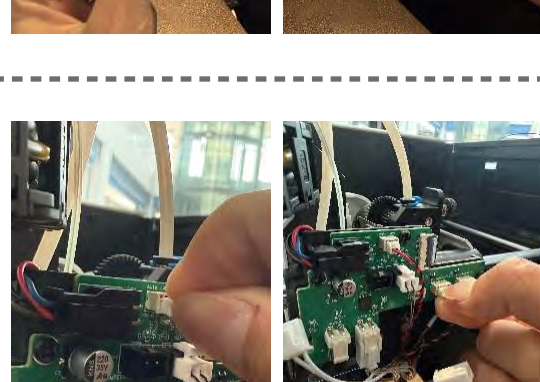
Step 2. Remove the Toolhead Front Cover

1. Pinch the latch on the fan plug at the "FAN-R" position on the mainboard.



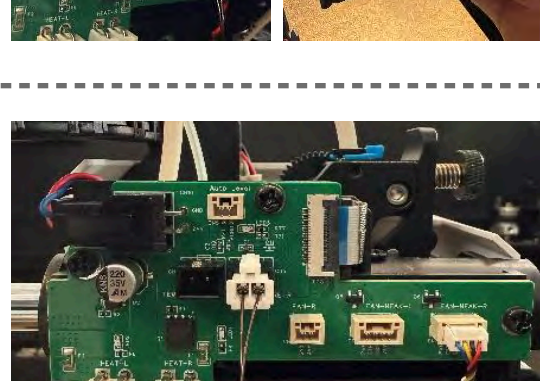
2. Unplug the fan connector to remove the front cover.

2

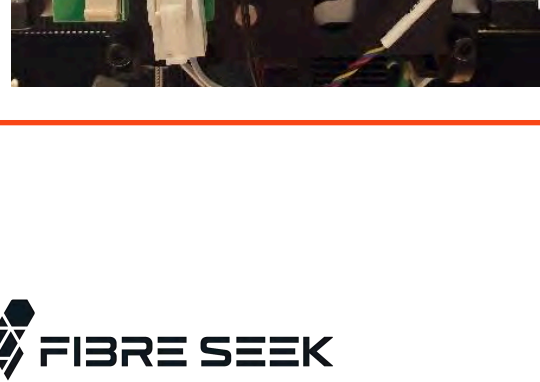
FIBRE SEEK


Step 3. Loosen the Cable Connectors

1. Pinch the latch on the heater plug at the "HRAT-L" position to unplug the heater.



2. Pinch the latch on temperature sensor plug at the "TEMPERATURE-L" position to unplug the sensor.



3. Press the latch on the heat sink fan connector at the "FAN-NEAK-L" position to unplug the fan.

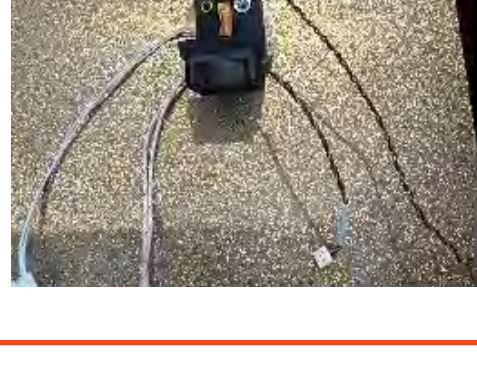
4. Press the latch on the auto-leveling sensor connector at the "Auto Level" position to unplug the sensor.

3

FIBRE SEEK


Step 4. Remove the Fixing Screws.

1. Use an H2.5 hex key.



2. Remove two screws securing the left composite head hotend.



Step 5. Replace the Piezoelectric Ceramic

1. Use an H2 hex key to remove the three fixing screws of the piezoelectric ceramic.

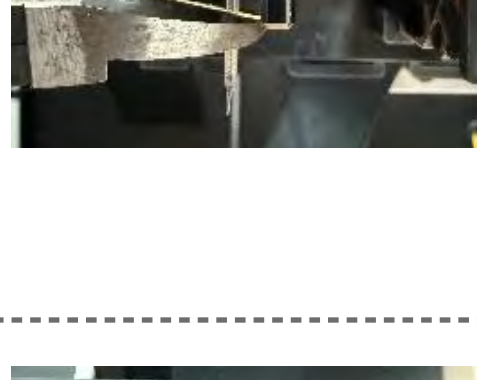


2. After removal, take out the new piezoelectric ceramic and pay attention to the installation orientation.



3. Tighten the three fixing screws to complete the replacement.

4

FIBRE SEEK


Step 6. Trim the Filament Inlet

1. Use side cutters to trim off about half of the excess plastic from the fiber hotend inlet. This prevents residual material in the throat tube from blocking full insertion.



2. Align the copper tube of the assembled fiber hotend with the filament path.

5

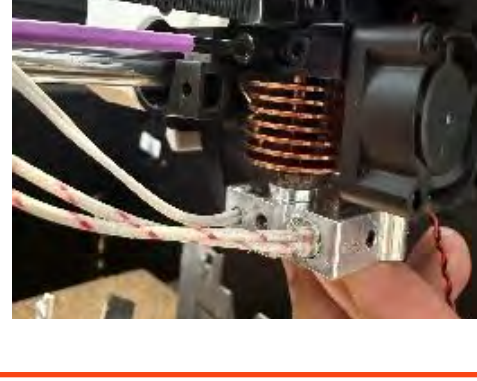
FIBRE SEEK


Step 7. Reinstall the Hotend Assembly.

1. Push the hotend upward from the bottom.

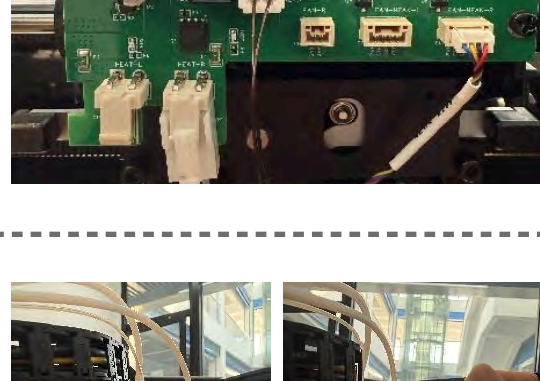


2. Tighten the two fixing screws for the fiber hotend after ensuring no gaps between the two planes.



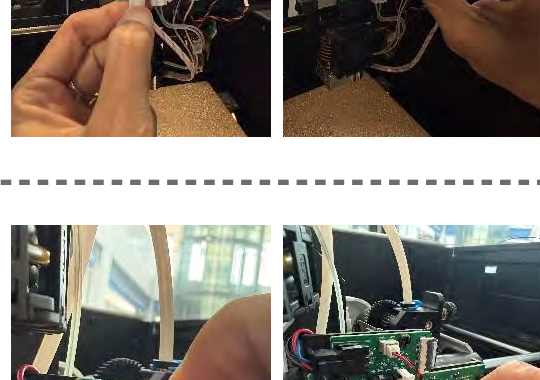
3. Keep pushing upward by hand to prevent the hotend from moving downward.

7

FIBRE SEEK


Step 8. Reconnect Cables

1. Pinch the latch on the ceramic heater plug at the "HRAT-L" position to replug the heater.

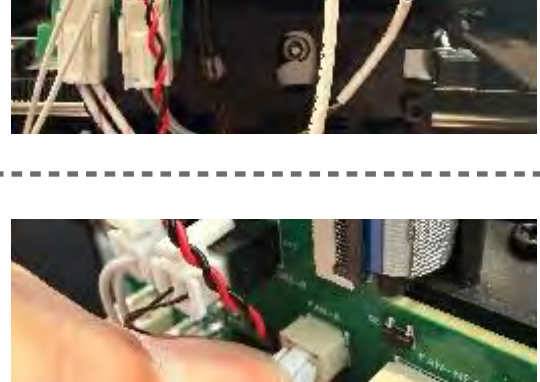


2. Pinch the latch on temperature sensor plug at the "TEMPERATURE-L" position to replug the sensor.

3. Press the latch on the heat sink fan connector at the "FAN-NEAK-L" position to replug the fan.

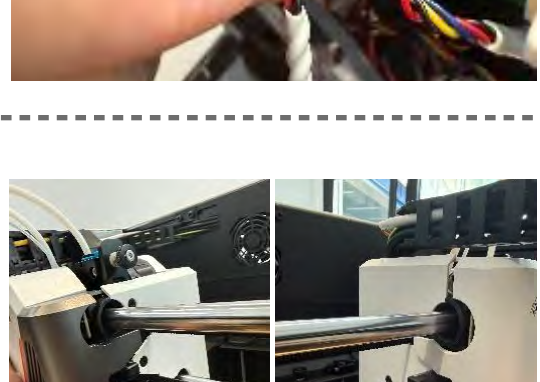
4. Press the latch on the auto-leveling sensor connector at the "Auto Level" position to replug the sensor.

8

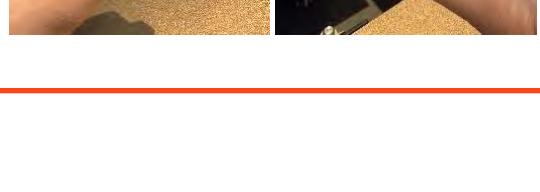
FIBRE SEEK


Step 9. Reinstall the Front Cover

1. Reinsert the fan plug into the corresponding port on the mainboard with the latch side facing up.



2. Pinch the lower part of the front cover, align the latch with the hole on the rear cover, and push the cover backward onto the toolhead.



3. When it clicks into place, the replacement is complete.

9

Disassemble/Reassemble:

Plastic Air Guide

1. Tools :

New plastic air guide, Phillips screwdriver, H3 Hex Key.

2. Precautions :

Always power off before operation. Beware of high temperature. Avoid pulling or straining cables to prevent loose connections.

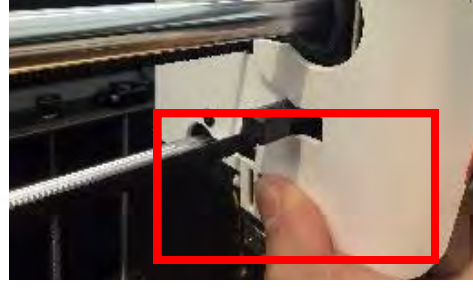
1



Step 1.

Power Off

1. Flip the red switch on the rear of the machine to "O".

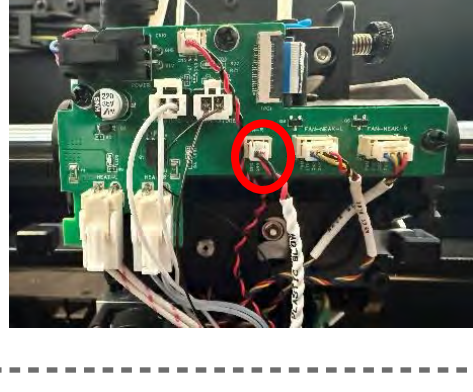


2. Pinch the lower left side of the front cover to release the latch.



3. Pull the toolhead front cover forward to remove it.

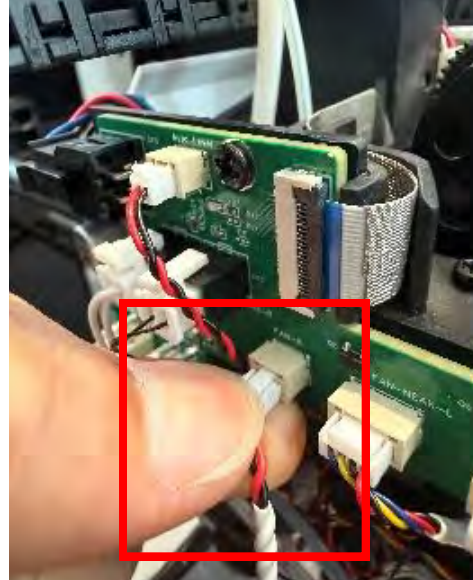
1



Step 2.

Remove the Toolhead Front Cover

1. Pinch the latch on the fan plug at the "FAN-R" position on the mainboard.



2. Unplug the fan connector to remove the front cover.

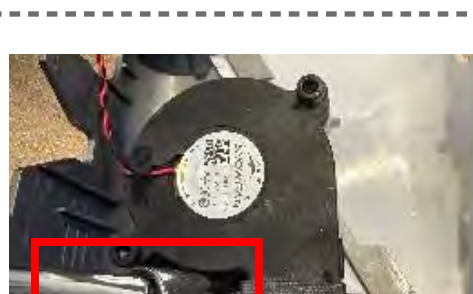
2



Step 3.

Remove the Old Air Guide

1. Use the Phillips screwdriver to remove the screw securing the plastic air guide.



2. Use the H3 hex key.



3. Remove the two screws holding the fan.

3



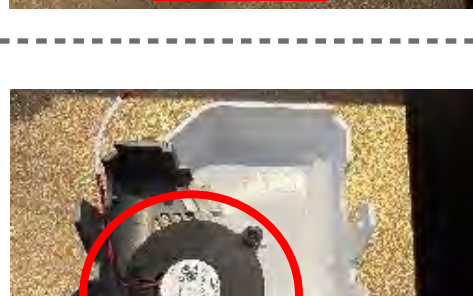
Step 4.

Install the New Air Guide

1. Align the screw holes of the new air guide with the original fixing positions, then tighten the Phillips screw.

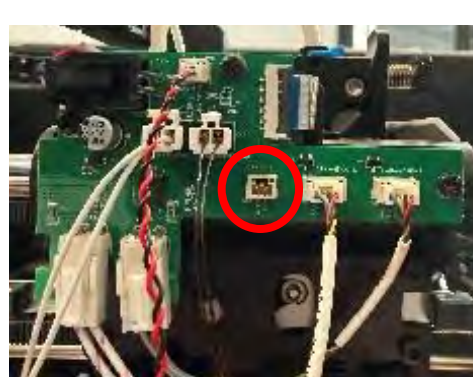


2. Insert the fan outlet into the groove of the air guide.



3. Tighten the two fan screws. Assembly is now complete.

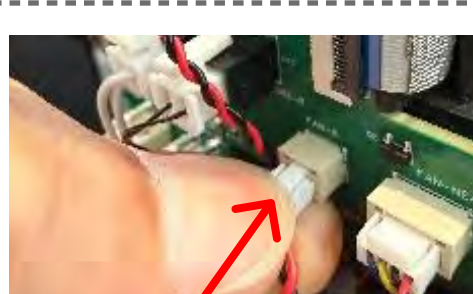
4



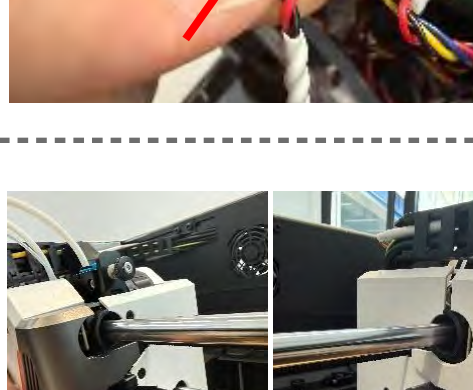
Step 5.

Reinstall the Front Cover

1. Reinsert the fan plug into the corresponding port on the mainboard with the latch side facing up.



2. Pinch the lower part of the front cover, align the latch with the hole on the rear cover, and push the cover backward onto the toolhead.



3. When it clicks into place, the replacement is complete.

5

Disassemble/Reassemble :

Plastic Hotend Ceramic Heater

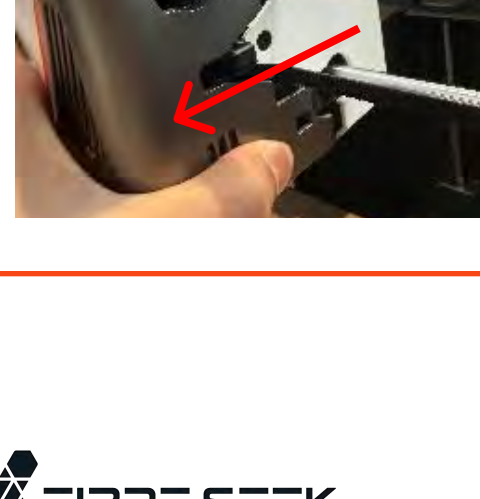
1. Tools :

Plastic hotend assembly, H1.5 & H2.5 hex keys, thermal grease.

2. Precautions :

Always power off before operation. Beware of high temperature. Avoid pulling or straining cables to prevent loose connections.

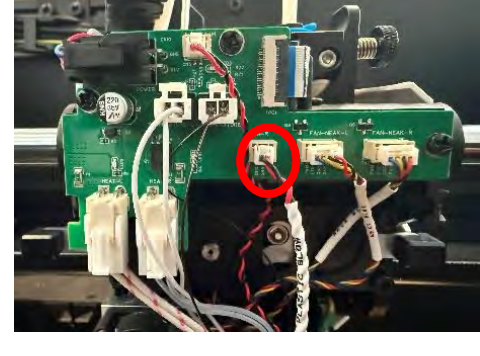
1



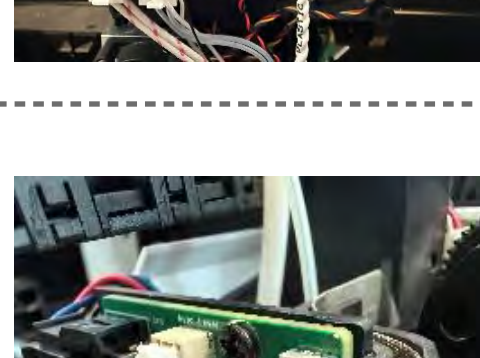
Step 1.

Power Off

1. Flip the red switch on the rear of the machine to "O".

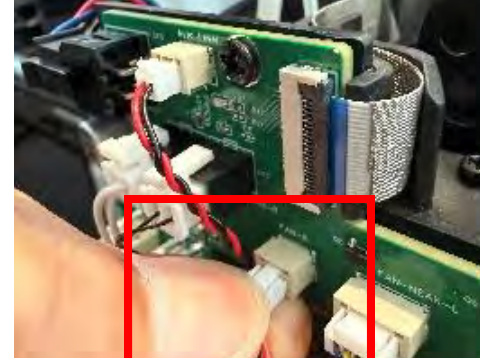


2. Pinch the lower left side of the front cover to release the latch.



3. Pull the toolhead front cover forward to remove it.

2



Step 2.

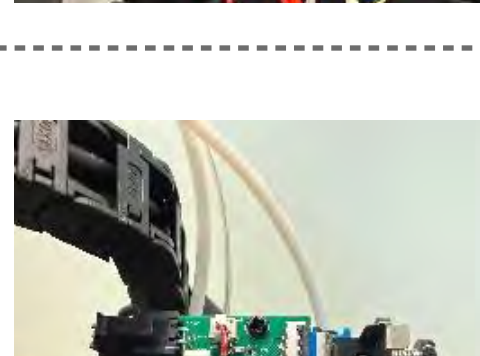
Remove the Toolhead Front Cover

1. Pinch the latch on the fan plug at the "FAN-R" position on the mainboard.



2. Unplug the fan connector to remove the front cover.

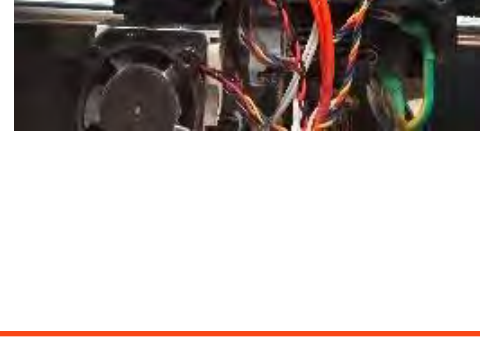
3



Step 3.

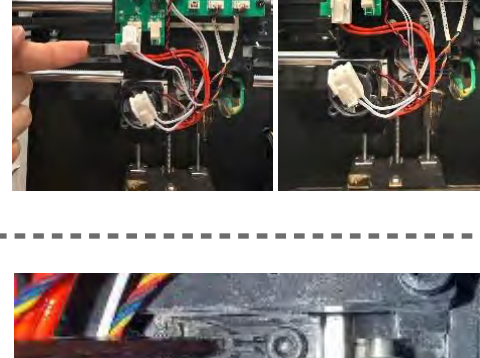
Loosen the Cable Connectors

1. Disconnect the two cables connected to the plastic hotend.



2. Pinch the latch on the ceramic heater plug at the "HRAT-R" position on the mainboard and unplug it. Pinch the latch on the temperature sensor plug at the "TEMPERATURE-R" position on the mainboard.

4



Step 4.

Remove the Hotend Fixing Screws

1. Press the collision block from the left side to switch the print head to plastic mode.

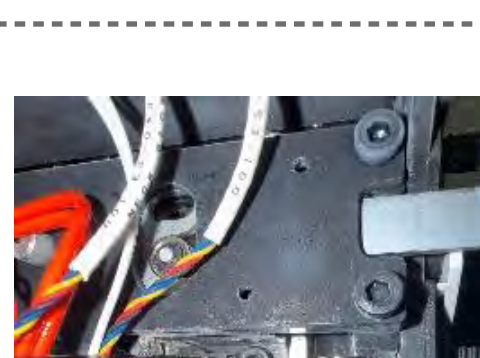


2. Remove the two front screws securing the plastic hotend with the H2.5 mm hex key.



3. Take out the entire plastic hotend assembly.

5



Step 5.

Reassemble the New Hotend

1. Prepare the new hotend assembly. Push it upward from the bottom.



2. Tighten the 2 fixing screws of the plastic hotend after ensuring no gaps between the two planes (keep pushing upward by hand to prevent the hotend from moving downward).

6



Step 6.

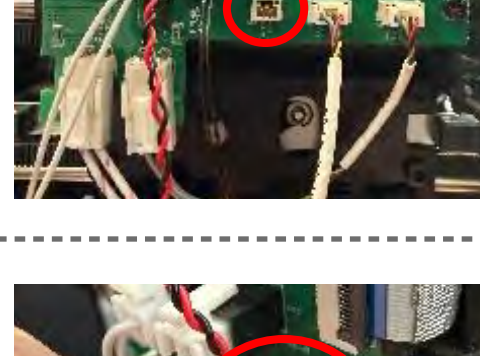
Reconnect the Cable Connectors

1. Reconnect all component cables of the plastic hotend assembly.



2. Insert the ceramic heater plug and the temperature sensor plug into the ports labeled "HRAT-R" and "TEMPERATURE-R" on the mainboard respectively.

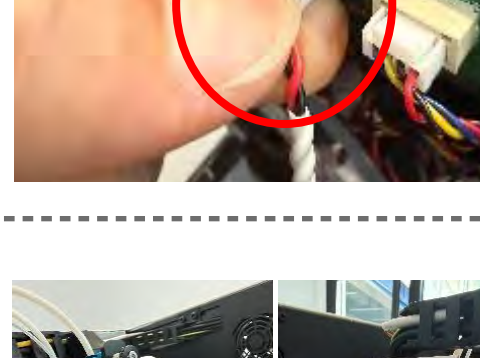
7



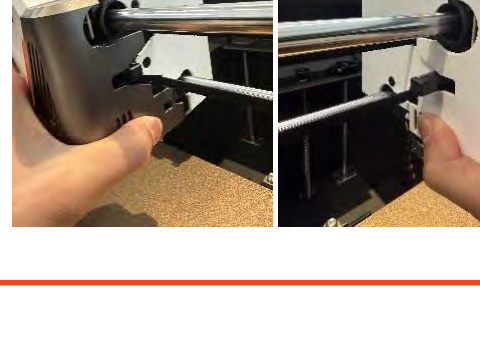
Step 7.

Reinstall the Front Cover

1. Reinsert the fan plug into the corresponding port on the mainboard with the latch side facing up.

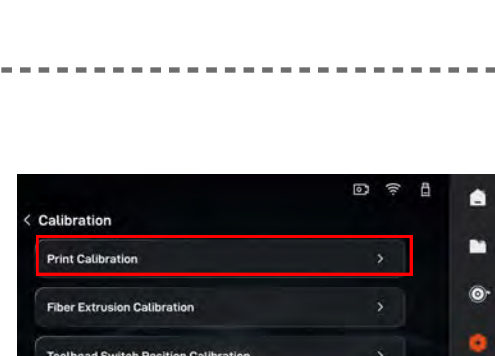


2. Pinch the lower part of the front cover, align the latch with the hole on the rear cover, and push the cover backward onto the toolhead.



3. When it clicks into place, the replacement is complete.

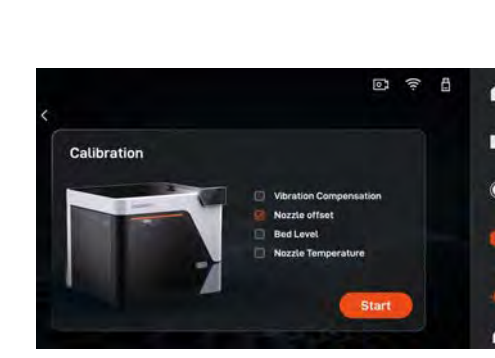
8



Step 8.

Perform XYZ Offset

1. Power on and wait for boot. Tap the settings icon on the right side of the toolbar, and select "Calibration" → "Print Calibration".



2. Select only "Nozzle offset". Click "Start" and wait for the process to finish. The printer is ready for use.



3. Skipping this step may cause issues during printing.

9

Disassemble/Reassemble :

Plastic Hotend Temperature Sensor

1. Tools:

Plastic hotend temperature sensor, H2.5 hex key.

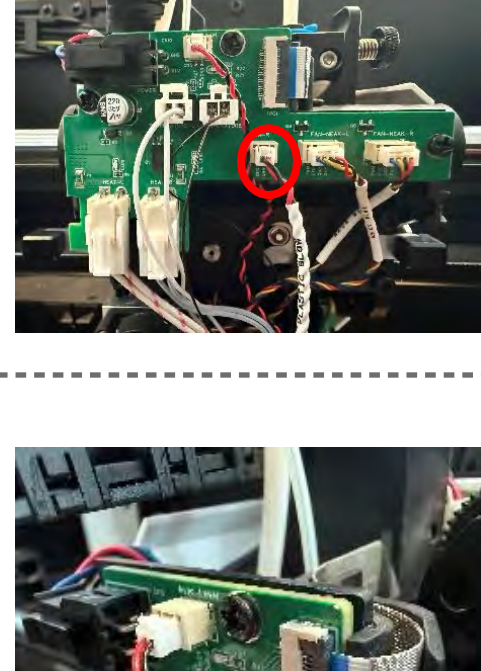
2. Precautions :

Always power off before operation. Beware of high temperature. Avoid pulling or straining cables to prevent loose connections.

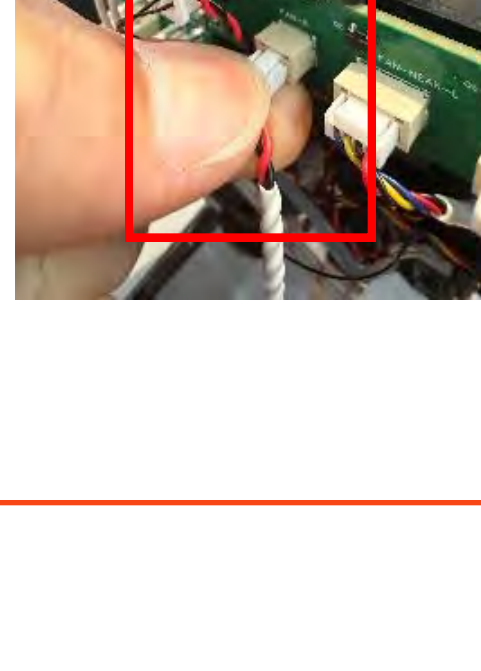
1



- Step 1.**
Power Off
1. Flip the red switch on the rear of the machine to “O”.

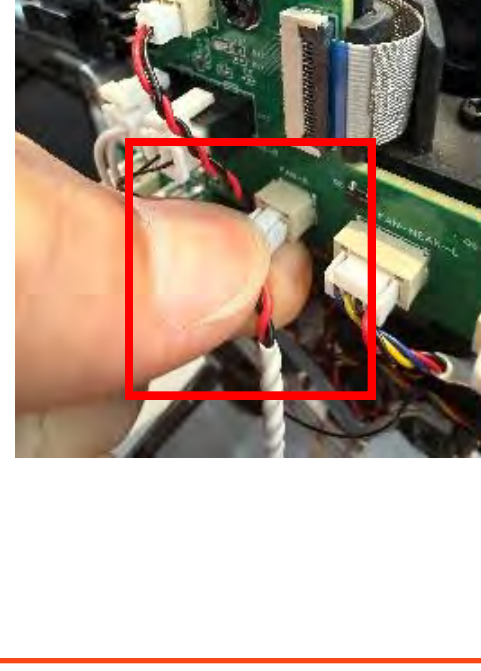


2. Pinch the lower left side of the front cover to release the latch.



3. Pull the toolhead front cover forward to remove it.

2

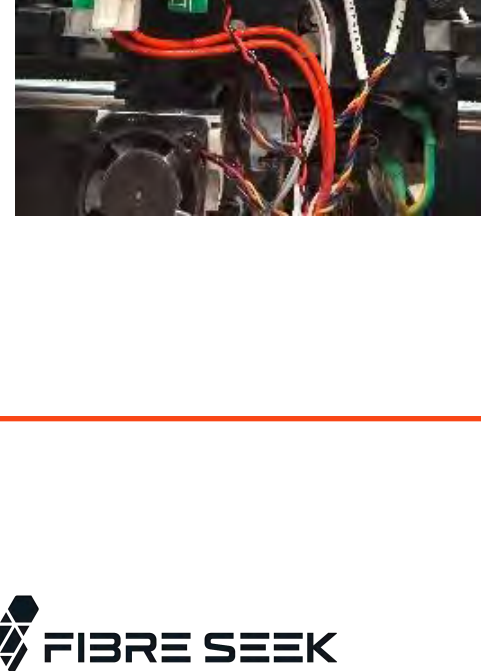


- Step 2.**
Remove the Toolhead Front Cover
1. Pinch the latch on the fan plug at the “FAN-R” position on the mainboard.

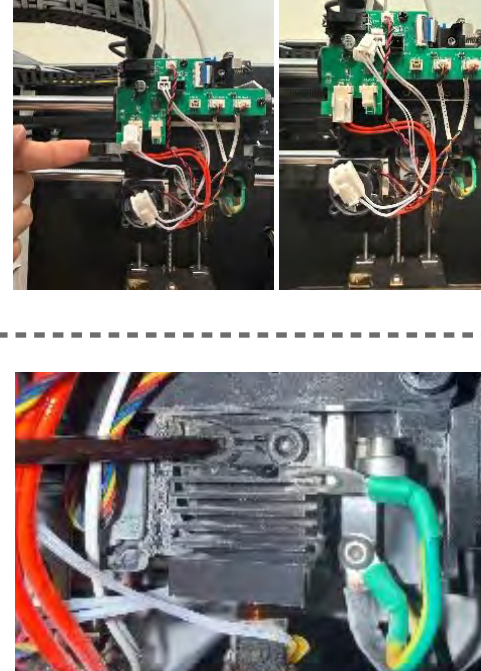


2. Unplug the fan connector to remove the front cover.

3



- Step 3.**
Loosen the Cable Connectors
1. Disconnect the two cables connected to the plastic hotend.

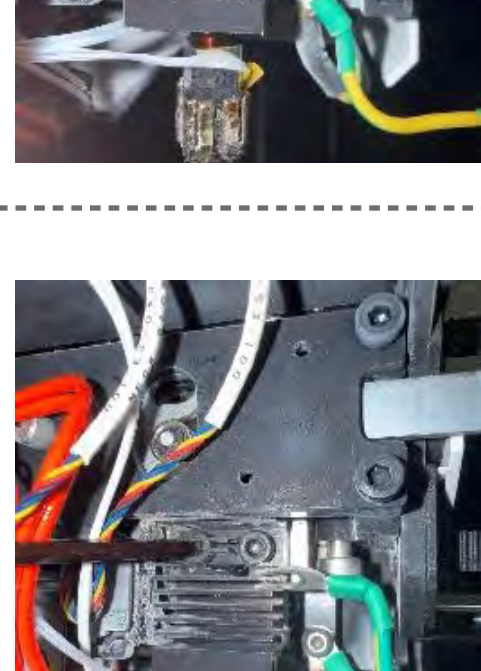


2. Pinch the latch on the ceramic heater plug at the “HRAT-R” position on the mainboard and unplug it. Pinch the latch on the temperature sensor plug at the “TEMPERATURE-R” position on the mainboard.

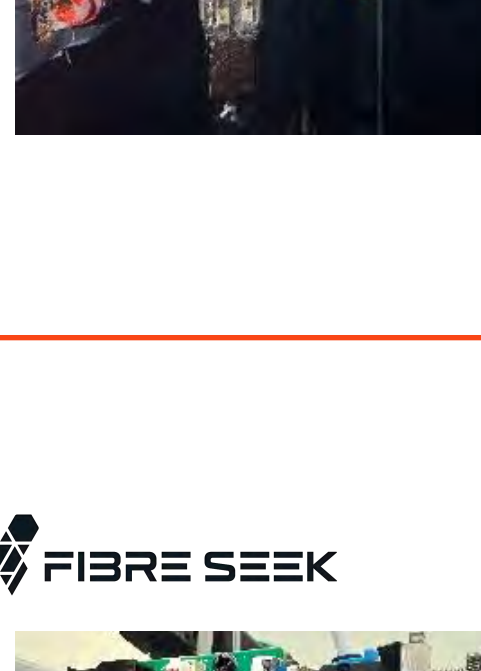
4



- Step 4.**
Remove the Hotend Fixing Screws
1. Press the collision block from the left side to switch the print head to the plastic mode.

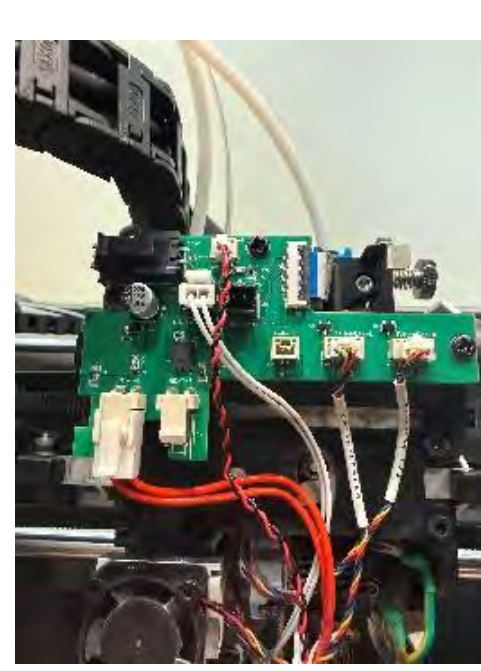


2. Remove the two front screws securing the plastic hotend with the H2.5 mm hex key.



3. Take out the entire plastic hotend assembly.

5

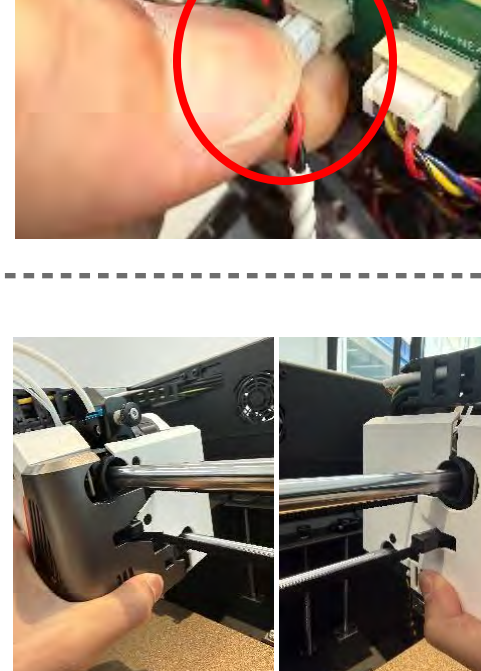


- Step 5.**
Reassemble the New Hotend
1. Prepare the new hotend assembly. Push it upward from the bottom.

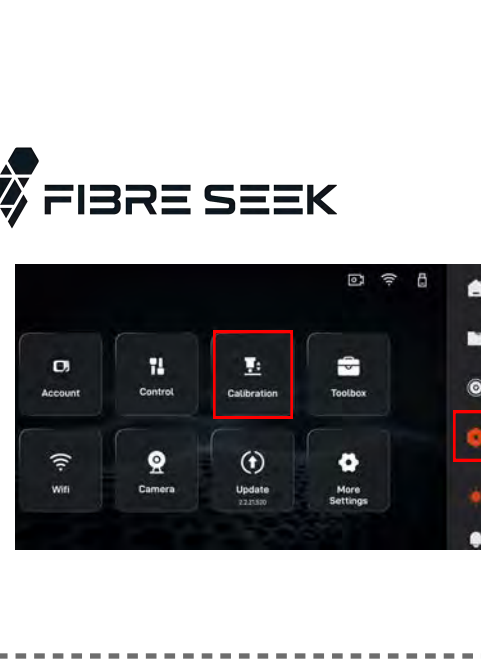


2. Tighten the two fixing screws of the plastic hotend after ensuring no gaps between the two planes (keep pushing upward by hand to prevent the hotend from moving downward).

6

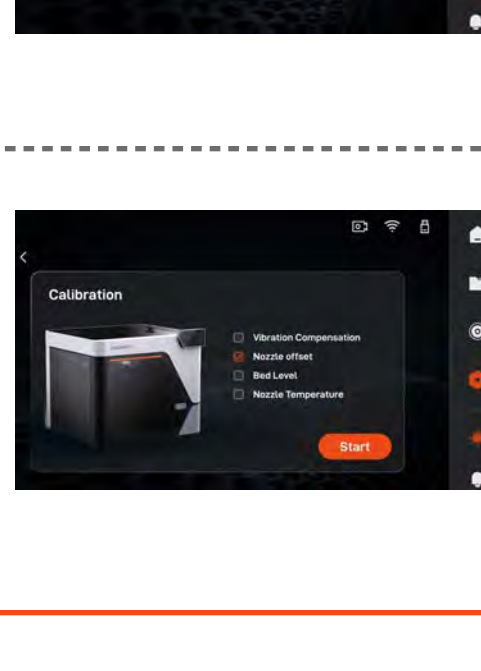


- Step 6.**
Reconnect the Cable Connectors
1. Reconnect all component cables of the plastic hotend assembly.



2. Insert the ceramic heater plug and the temperature sensor plug into the ports labeled “HRAT-R” and “TEMPERATURE-R” on the mainboard respectively.

7



- Step 7.**
Reinstall the Front Cover
1. Reinsert the fan plug into the corresponding port on the mainboard with the latch side facing up.

2. Pinch the lower part of the front cover, align the latch with the hole on the rear cover, and push the cover backward onto the toolhead.

3. When it clicks into place, the replacement is complete.

8

- Step 8.**
Perform XYZ Offset
1. Power on and wait for boot. Tap the settings icon on the right side of the toolbar, and select “Calibration” → “Print Calibration”.

2. Select only “Nozzle offset”. Click “Start” and wait for the process to finish. The printer is ready for use.

3. Skipping this step may cause issues during printing.

9

Disassemble/Reassemble :

Toolhead Fan (Fiber Nozzle)

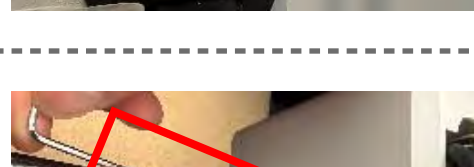
1. Tools :

New fan, Phillips screwdriver, H1.5 hex key

2. Precautions :

Always power off before operation. Beware of high temperature. Avoid pulling or straining cables to prevent loose connections.

1

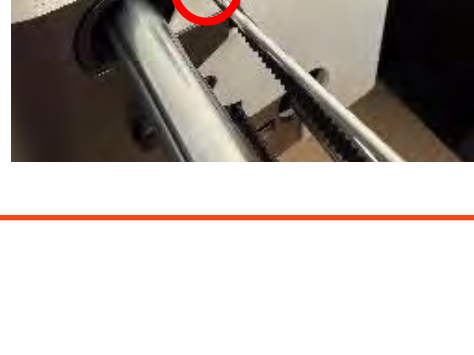


Step 1. Power Off and Remove the Fixing Screws

1. Power Off.

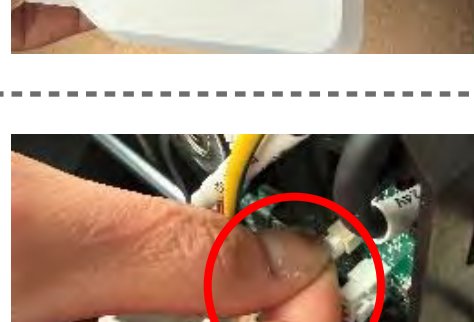


2. Use an H2 hex key.



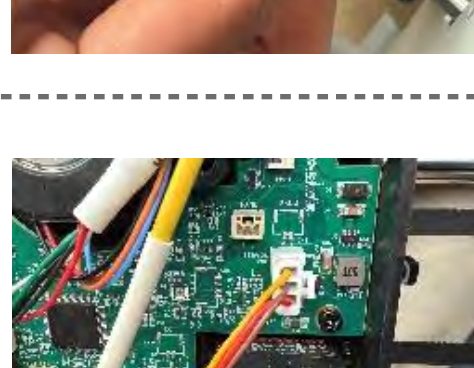
3. Remove the four fixing screws on the rear cover (two on the left, two on the right).

1



Step 2. Loosen the Cable Connectors to Remove the Rear Cover

1. Slide the toolhead rear cover horizontally backward to detach it.



2. Press the white latch on the fan connector.



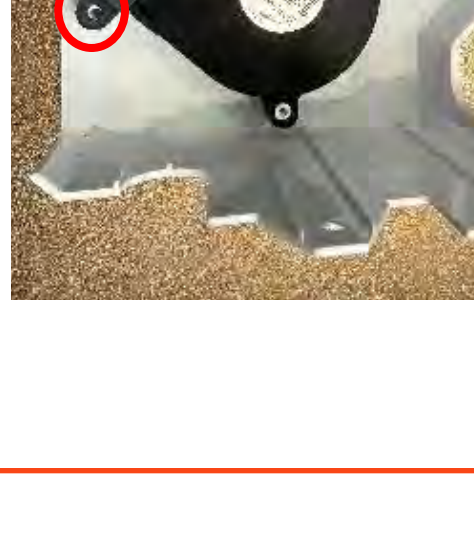
3. Remove the rear cover.

2



Step 3. Remove the Old Air Guide

1. Use a Phillips screwdriver.



2. Remove the screw securing the fiber air guide.

3



Step 4. Install the New Air Guide

1. Remove the air guide and pull the old fan upward.



2. Take the new fan and install it in the original position.



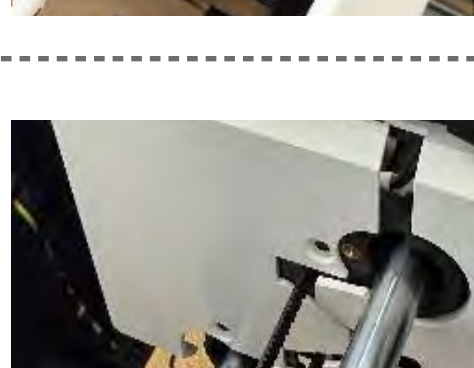
3. Tighten the screw on the air guide. The air inlet of the air guide should completely wrap around the fan's air outlet.

4



Step 5. Reconnect Cables

1. Ensure the latch side of the fan connector faces upward.



2. Plug the fan connector back into the corresponding port on the rear mainboard.



3. Ensure the direction and reinstall the rear cover.

5



Step 6. Reinstall the Rear Cover

1. Tighten the four fixing screws.



2. Tighten two screws on the left and two on the right.

6

Disassemble/Reassemble:

Toolhead Fan (Plastic Nozzle)

1. Tools :

New Fan, Phillips Screwdriver, H3 Hex Key.

2. Precautions :

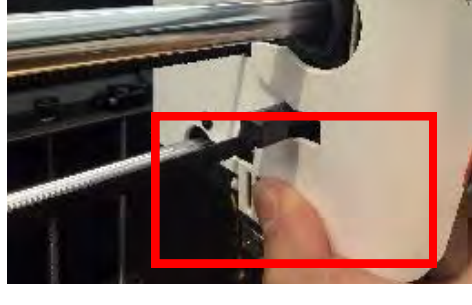
Always power off before operation. Beware of high temperature. Avoid pulling or straining cables to prevent loose connections.

1



Step 1. Power Off

1. Flip the red switch on the rear of the machine to "O".

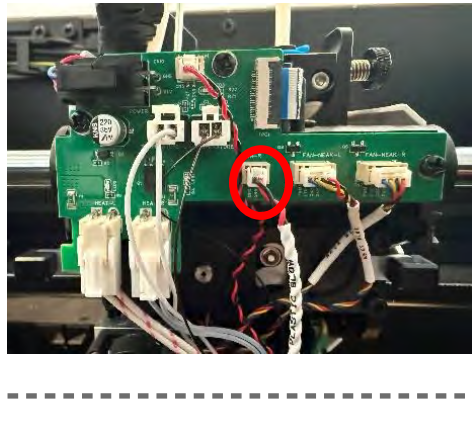


2. Pinch the lower left side of the front cover to release the latch.



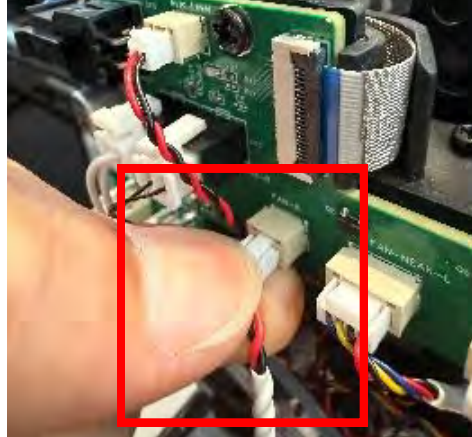
3. Pull the toolhead front cover forward to remove it.

1



Step 2. Remove the Toolhead Front Cover

1. Pinch the latch on the fan plug at the "FAN-R" position on the mainboard.



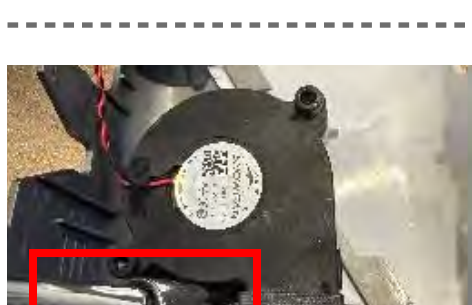
2. Unplug the fan connector to remove the front cover.

2



Step 3. Remove the Old Air Guide

1. Use the Phillips screwdriver to remove the screw securing the plastic air guide.

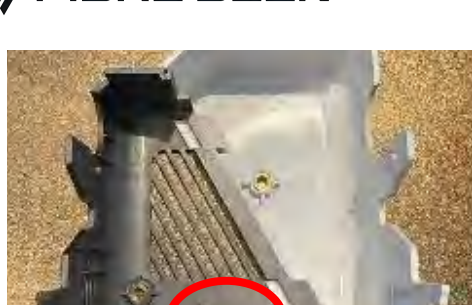


2. Use the H3 hex key.



3. Remove the two screws holding the fan.

3



Step 4. Install the new air guide

1. Align the screw holes of the new air guide with the original fixing positions, then tighten the Phillips screw.

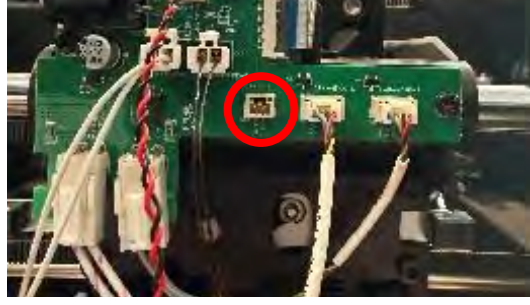


2. Insert the fan outlet into the groove of the air guide.



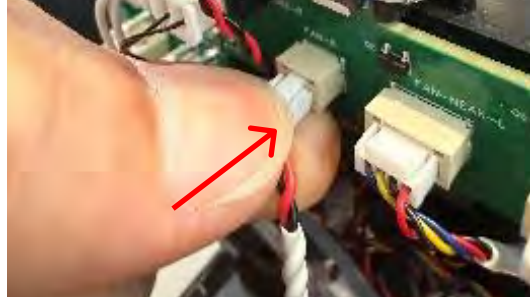
3. Tighten the two fan screws. Assembly is now complete.

4



Step 5. Reinstall the Front Cover

1. Reinsert the fan plug into the corresponding port on the mainboard with the latch side facing up.



2. Pinch the lower part of the front cover, align the latch with the hole on the rear cover, and push the cover backward onto the toolhead.



3. When it clicks into place, the replacement is complete.

5

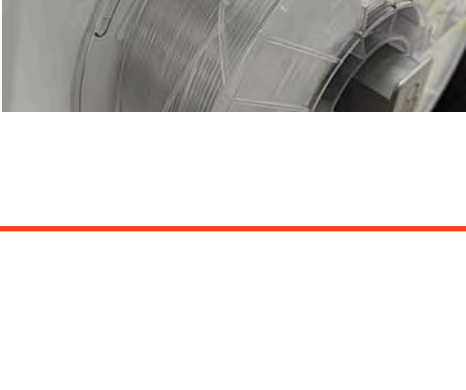
Composite Plastic Hotend Unclogging

This SOP describes how to clear clogs in the composite plastic hotend of FibreSeeker 3. Its purpose is to safely and effectively resolve nozzle clogging issues.

1. Tools

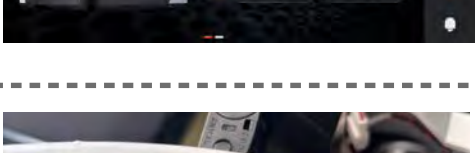
H3, H2.5, and H1.5 hex keys, an unclogging pin tool, and heat-resistant gloves.

2. Precautions:



The troubleshooting process requires heating. Wear heat-resistant gloves when working near the print hotend to prevent burns.

1



Step 1

Heat the Left Nozzle

Heat the left nozzle to the material's printing temperature (lower temperatures may block extrusion).



Step 2

Disconnect Tube and Check

Disconnect the PTFE tube. Take another piece of filament and firmly push it into the hotend. If filament comes out of the nozzle, the path is clear, and you should check the extruder for faults. A faulty extruder typically makes loud, abnormal noises during operation. If this occurs, replace the extruder immediately.

2

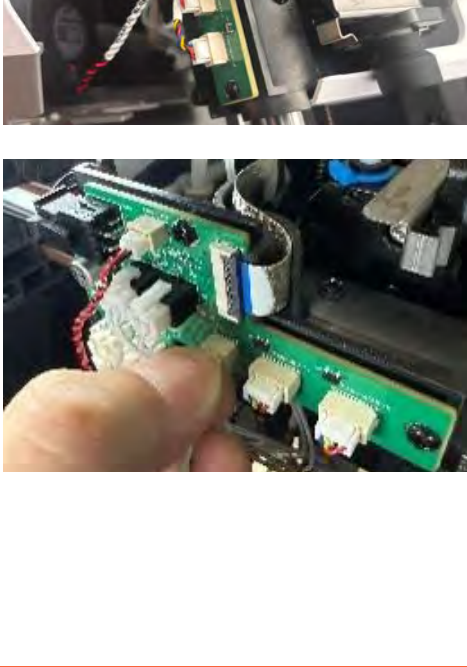


Step 3

Remove the Toolhead Front Cover

Step 3.1

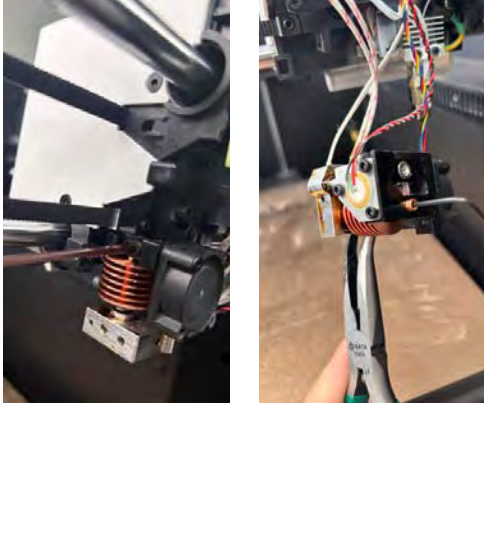
Power off the printer. Press the bottom clips on both sides of the front cover simultaneously, then pull the cover forward to remove it.



Step 3.2

Pinch the latch on the fan at the "FAN-R" position on the mainboard and unplug it to remove the front cover of the toolhead.

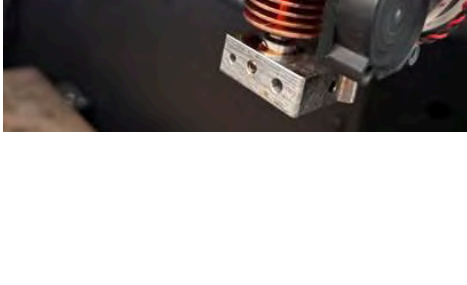
3



Step 4

Remove and Clear the Composite Hotend

If the nozzle is partially clogged, use an H2.5 hex key to remove the two fixing screws and remove the composite hotend assembly. Then, use a slender hex key or a similar tool to force through the nozzle to clear the blockage. Once filament can exit the nozzle, reassemble the hotend and extrude several times to flush out remaining debris. If the clog cannot be cleared, replace the composite hotend assembly with a new one.

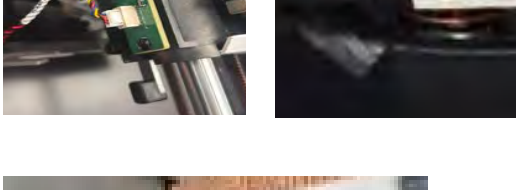


Step 5

Reinstall the Composite Hotend

Install a new hotend or reinstall the composite hotend assembly.

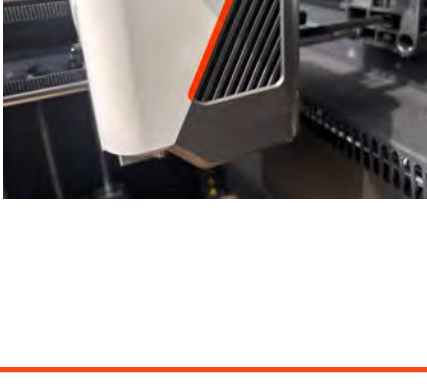
4



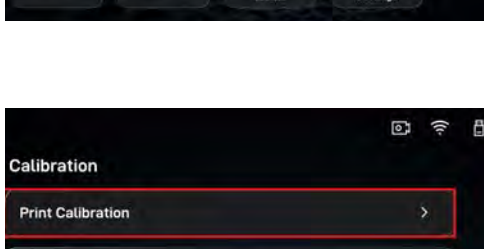
Step 6

Reinstall the Toolhead Front Cover

Plug the fan cable of the front cover back into the corresponding port on the mainboard. Make sure the two bottom clips align with their holes and snap into place.



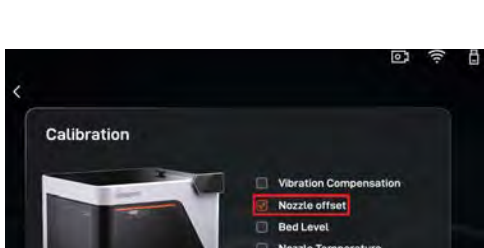
5



Step 7

Calibrate

After installation, manually calibrate the nozzle offset. Select "Nozzle offset" only.



6

Fiber Extruder Rubber Roller Replacement

This SOP describes how to replace the rubber rollers on the fiber extruder of FibreSeeker 3 safely and effectively.

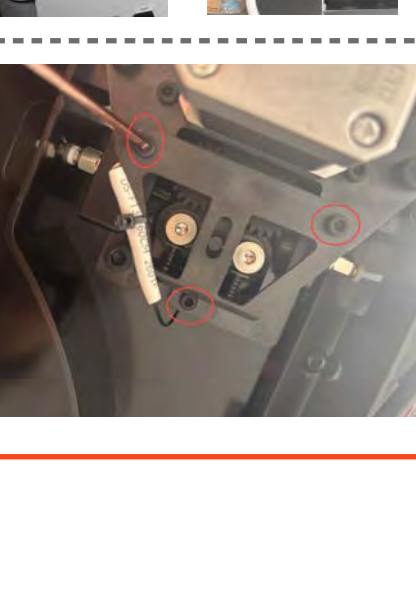
1. Tools:

H2.5 and H3 hex keys

2. Precautions:

None

1



Step 1

Click the third icon in the right toolbar as shown in the picture. On the page that opens, select “Fiber” > “Unload” to retract the fiber from the PTFE tube.

Step 2

Remove the Side Panel of the Extruder

Use an H3 hex key to remove the two screws connecting the back panel to the side panel, and the four top fixing screws. Then take off the side panel.

Step 3

Disassemble

Use an H2.5 hex key to loosen the 3 fixing screws and remove the extruder.

2



Step 4

Press the driven pulley to remove the planetary gear.

Step 5

Use an H3 hex key to remove the adjustment screws.

Step 6

Push out the bearing and remove the rubber covering. Replace the rubber coverings on the four rubber wheels. At the same time, check whether the rubber covering and the bearing are installed on the same horizontal plane.

Step 7

Simply remove the bearing and connecting rod to replace the rubber on the driving wheel.

Step 8

Install the rubber wheel, and use the H3 hex key to tighten the adjustment screw slightly.

4



Step 9

Press the driven pulley, reinstall the planetary gear, and manually rotate the gear to check if the rubber rollers wobble. Slight wobble is acceptable.

Step 10

First check if the rubber roller is not fully seated. After adjusting, check that the driven pulley adjustment screw is tightened.

Step 11

Reinstall the Extruder

5



Step 12

Test fiber extrusion.

Step 13.1

Click the settings icon on the right side of the toolbar, then select “Calibration” > “Fiber Extrusion Calibration”.

8



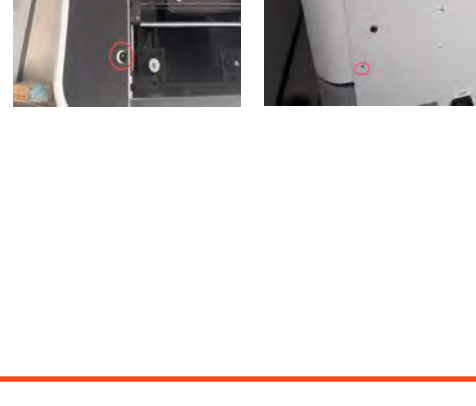
Step 13.2

Follow the on-screen instructions. Measure the length of the second segment, which should be 100 mm ± 0.5 mm. (Do not measure the first segment, which is inaccurate.)

Step 13.3

Enter the measured length in the Fiber Length field.

7



Step 14

Install the Side and Back Panels

Install the 4 fixing screws on top of the side panel. Do not install the two screws that secure the side panel to the back panel yet.

6

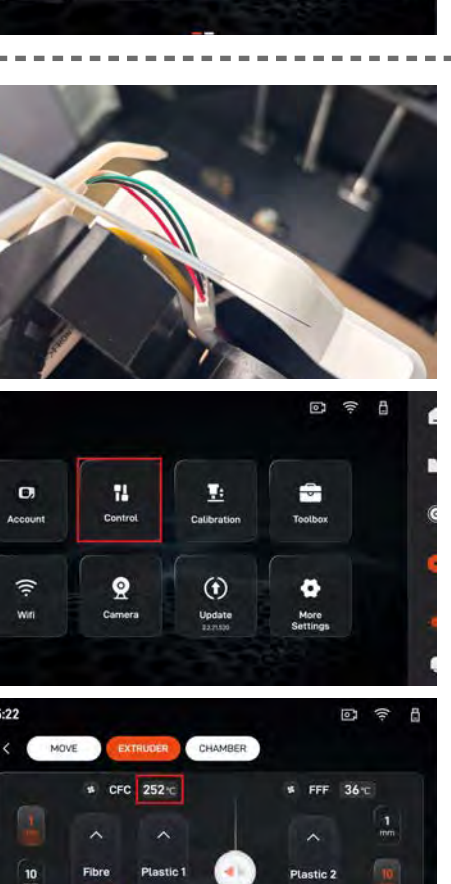
Fiber Unclogging in Composite Hotend

This SOP describes how to clear fiber clogs in the composite hotend in FibreSeeker 3, aiming to resolve clogging safely and effectively.

1. Tools:

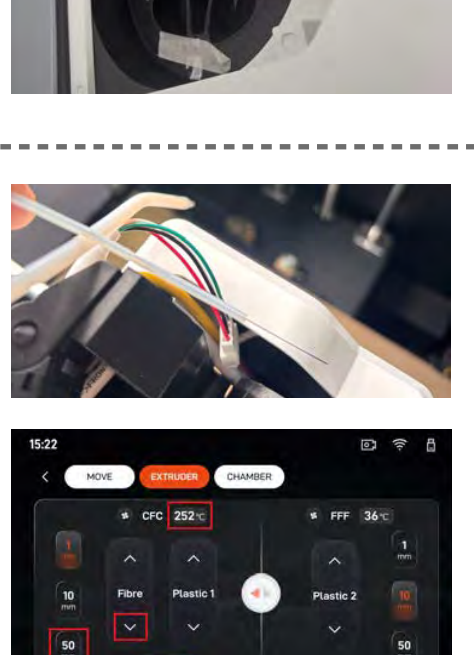
H3, H2.5, H1.5 hex keys, Phillips screwdriver, unclogging pin tool, heat-resistant gloves.

2. Precautions:



Troubleshooting requires heating. Wear heat-resistant gloves when working near the print hotend to prevent burns. Before troubleshooting, check the fiber spool for breaks or a rough, gritty feel.

1



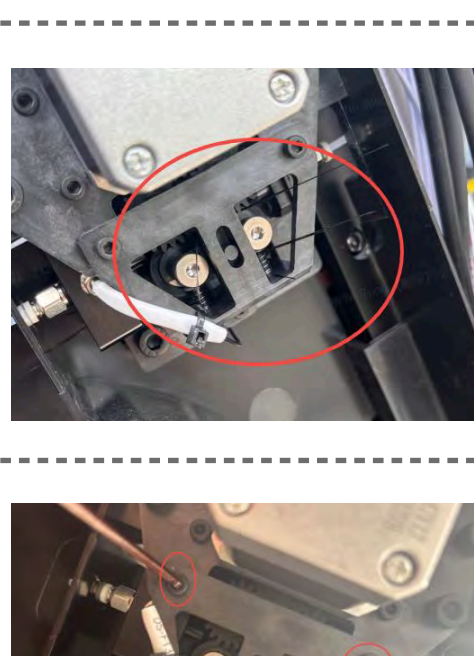
Step 1 Heat the Left Nozzle

Heat the left nozzle to the material's printing temperature. Lower temperatures may block extrusion.



Step 2.1 Check Extrusion by Removing the PTFE Tube

Remove the PTFE tube. Observe whether the fiber moves.

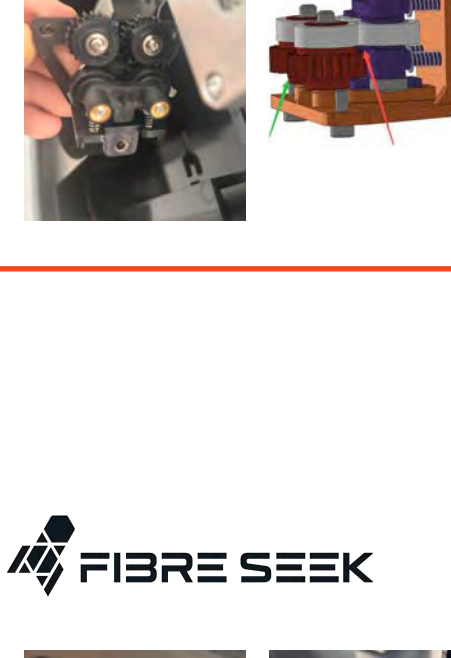


Tap the settings icon on the right side of the toolbar, then select "Control" – "Extruder". Set the temperature to approximately 250°C and the length to 50mm, then tap the down arrow to extrude.



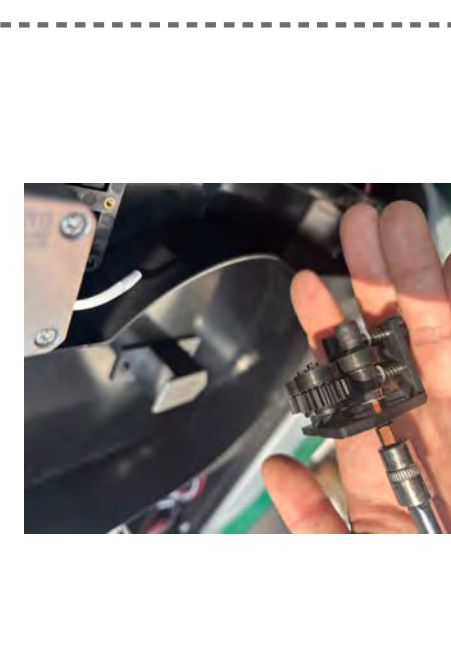
If fiber comes out of the PTFE tube at the print hotend, skip to Step 11. If it cannot be pulled out, proceed to Step 2.3.

2



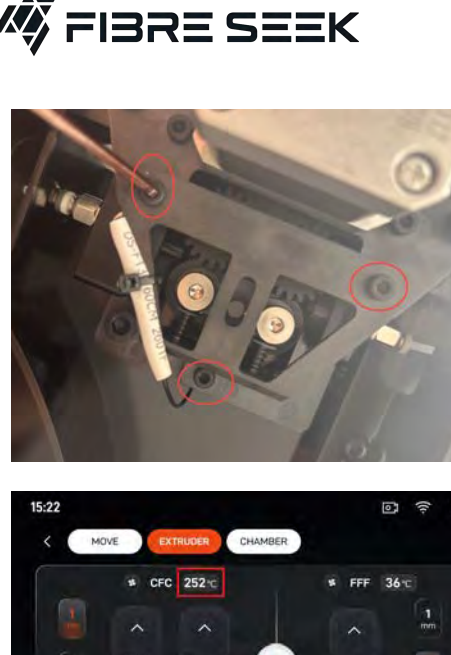
Step 2.2 Remove the Fiber

If the fiber does not move, try pulling it out by hand or with tweezers. If it comes out (the fiber broke in the middle), skip to Step 2.4. If it cannot be pulled out, proceed to Step 2.3.



Step 2.3 Observe the Spool

If the fiber cannot be pulled out, remove the fiber compartment cover. Select 50mm and press extrude twice. Watch whether the spool rotates. If it does not rotate, go to Step 3. If it rotates, go to Step 2.4.



Step 2.4 Extrude and Observe

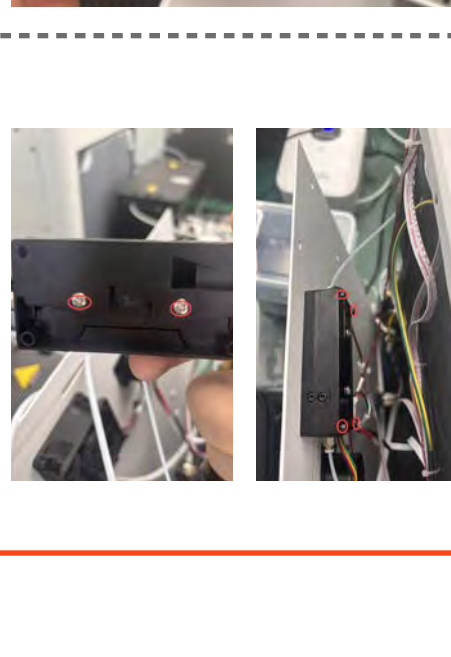
If the spool rotates, extrude 10–12 times until fiber appears inside the PTFE tube within the chamber. Then reinsert the tube and extrude to see if fiber exits the nozzle. If fiber comes out, skip to Step 10. If the spool rotates but no extrusion occurs and you hear fiber cracking, go to Step 6.

3



Step 3 Remove the Side Panel of the Extruder

If the spool does not rotate, inspect the extruder. Use an H3 hex key to remove the 2 screws connecting the rear and side panels and the 4 top fixing screws, then take off the side panel of the extruder.



Step 4.1 Inspect

The fiber extruder may have fiber debris (generally visible as fibers sticking out without disassembly).



Step 4.2 Disassemble

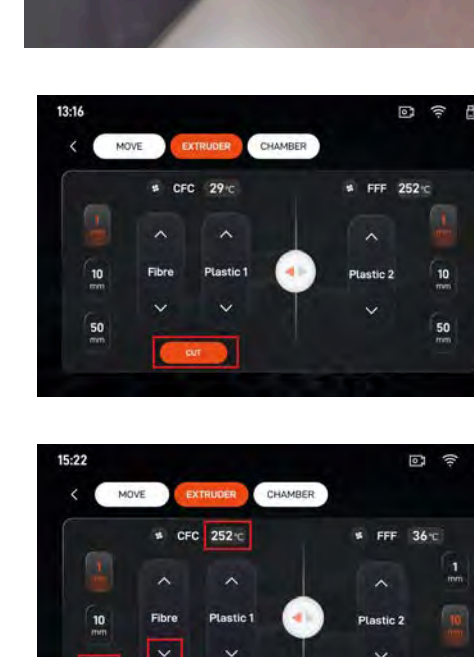
Use an H2.5 hex key to loosen 3 fixing screws and remove the extruder.

4



Step 4.3 Clean

Remove all fiber debris from the extruder and the fiber path.



Step 4.4 Check

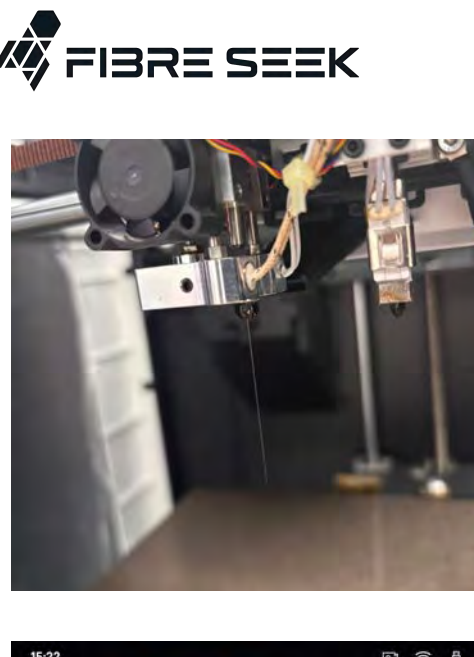
Check the rubber wheel for wear. If severely worn, replace it.



Step 4.5 Check Extruder Gear Offset

Check whether the extruder gear is offset. While rotating it, observe its movement. (Turn the part indicated by the green arrow and check if the rubber wheel indicated by the red arrow is misaligned.)

5



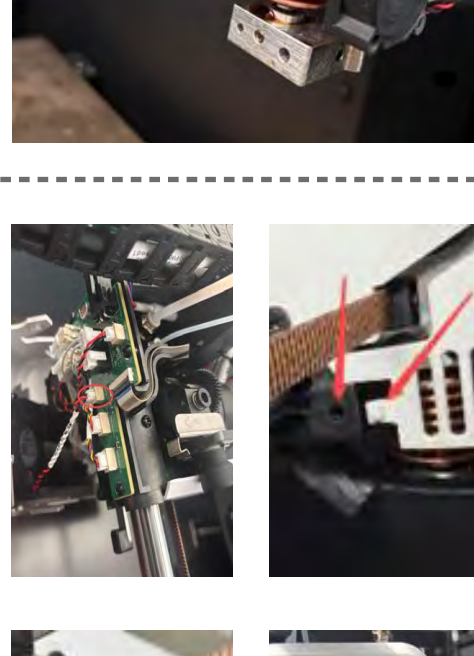
Step 4.6 Adjust

If offset occurs, press down with a wrench and release immediately to let it spring back.

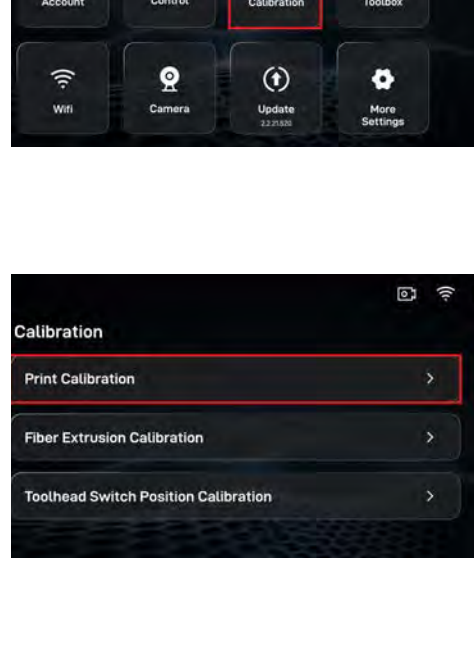


Step 4.7 Check Adjustment Screws

Use an H3 hex key to inspect the 2 adjustment screws and lightly tighten them by hand. (Tightening the screws increases the pressure on the rubber wheel. Both too tight and too loose will prevent the filament from feeding.)

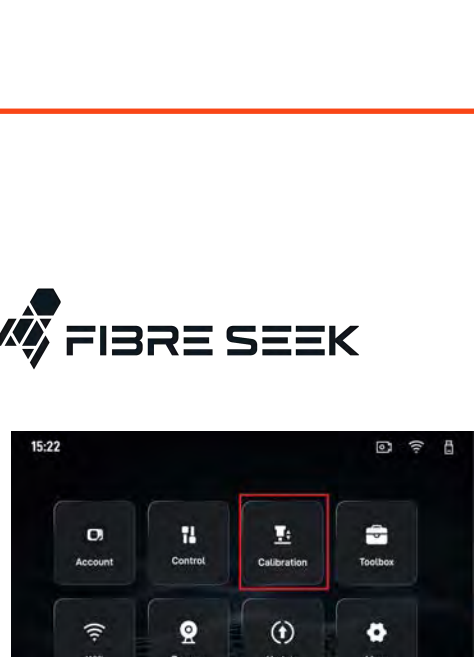


6

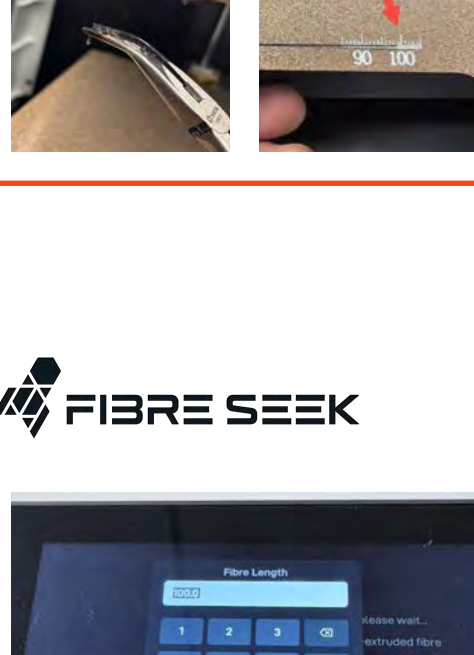


Step 5 Reinstall and Test

Reinstall the extruder, reload the filament, and test whether the extruder drives the filament correctly. Continue until filament comes out of the extruder. (After reinstallation, skip to Step 18.)

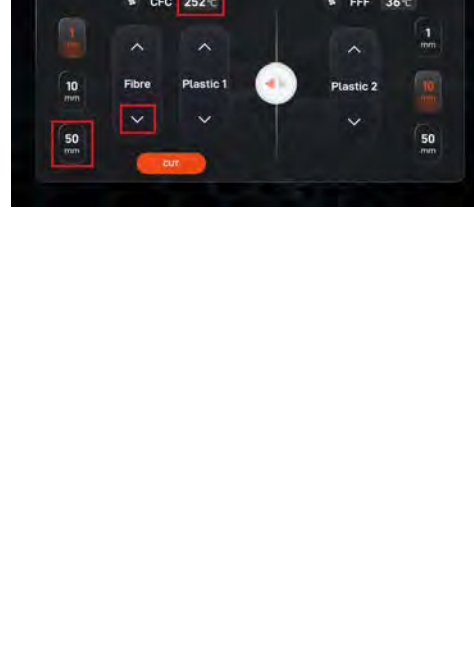


7



Step 6 Remove the Back Panel of the Extruder

The spool rotates with snapping sounds during extrusion. If filament has been extruded more than 12 times without appearing in the PTFE tube, you should also remove the back panel.



Step 6.1

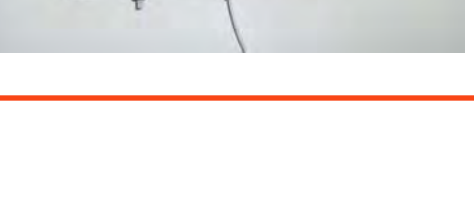
Push the sensor protection cover upward from the bottom to loosen it, then remove it.



Step 6.2

Use an H2.5 hex key to remove the 11 fixing screws on the back cover. Note: The 3 screws at the very top are longer than the others.

8



Step 7 Remove the Tension Sensor

Use an H2.5 hex key to remove the 4 fixing screws securing the tension sensor, then take off the sensor and disconnect the PTFE tube. Then use a Phillips screwdriver to remove the 2 fastening screws.

Step 8 Inspect

Inspect the inside of the tension sensor for excessive fiber debris. (Internal problems usually cause snapping sounds during extrusion.)

Step 9 Reinstall, Test, and Secure

After cleaning, reinstall the sensor. Secure the 2 fastening screws first. Click "Extrude" and check if the filament can pass through the tension sensor. If it can, tighten the 4 fixing screws. (After installation, skip to Step 18.) Then reattach the PTFE tube. Extrude several times. If the filament feeds out from the tension sensor normally, proceed to Step 11.

9

Step 10 Remove the Toolhead Front Cover

Power off the printer. Press the bottom clips on both sides of the front cover simultaneously, then pull the cover forward to remove it.

Step 10.2

Pinch the latch on the fan at the "FAN-R" position on the mainboard and unplug it to remove the front cover of the toolhead.

10

Step 11 Clear Fiber

If the clog is not severe, click "Cut", then use tweezers to pull out any fiber from the guide tube, along with any fiber that comes out with it.

Try extruding 50mm twice and check if fiber comes out of the nozzle. If it does, click "Cut" and continue testing. If still clogged, proceed to the next Step.

11

Step 12 Remove the Composite Hotend

Remove the two fixing screws with an H2.5 hex key and take off the composite hotend.

Step 13 Inspect and Clean

Inspect the fiber path in guide tube and cutter areas for clogs and clean it thoroughly.

Step 14 Clear the Hotend

Pull out the clogged fiber. Clean the pathway again with an unclogging pin tool, then insert a piece of fiber to ensure it moves through smoothly.

12

Step 15 Verify

Try to extrude 50 mm two to three times. Check if filament comes out of the nozzle. If it does, cut the filament and proceed to the next step.

Step 16 Reinstall Assembly

Once cleaned, reinstall the composite hotend assembly in the correct order.

Step 17 Reinstall the Toolhead Front Cover

Plug the fan cable of the front cover back into the corresponding port on the mainboard. Make sure the two bottom clips align with their holes and snap into place.

14

Step 18 Calibrate

After removing the composite hotend, manually calibrate the nozzle offset. (Only this calibration is needed.)

Step 18.1 Check Fiber Length

Click the settings icon on the right side of the toolbar, then select "Calibration" – "Fiber Extrusion Calibration".

Step 19.2

Follow the on-screen instructions. Measure the length of the second segment, which should be 100mm ± 0.5 mm. (Do not measure the first segment, which is inaccurate.)

Step 19.3

Enter the measured length in the Fiber Length field.

Step 19.4

Then repeat Step 19.1–19.3 twice, ensuring the length from the second extrusion is within 100 mm ± 0.5 mm each time. You may also compare two 100 mm extrusions (the nozzle must be heated above 180°C). If the lengths differ significantly, go back and check Step 4.4.

17

Step 20 Install the Side and Back Panels

Install the 4 fixing screws on top of the side panel. Do not install the two screws that secure the side panel to the back panel yet.

Step 20.2

Install the 3 longer screws to loosely hold the back cover in place and do not fully tighten. Then install the remaining 8 screws. Align the back cover gaps before tightening all screws evenly.

Step 20.3

Align the clips on the left and right ends of the sensor cover with the slots in the sheet metal. Once aligned, press the cover downward from the top until it is firmly seated.

18

Heatbed Leveling

This SOP describes the procedure for correcting an uneven heatbed on the FibreSeeker 3 3D printer.

1. Tools:

A4 paper, H2 hex wrench

2. Precautions:

The heatbed is to be heated and will reach high temperatures. Take care to avoid burns.

1



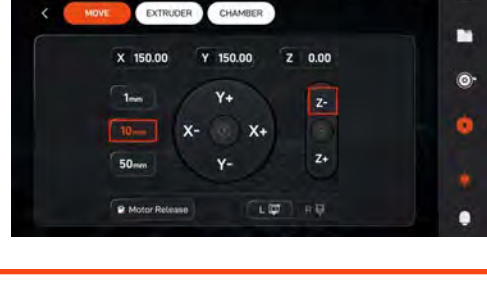
Steps Preparation

1. Preheat the heatbed to normal printing temperature (PLA: 55 °C, PETG: 75 °C). Since the bed expands with heat, leveling it cold will be inaccurate.



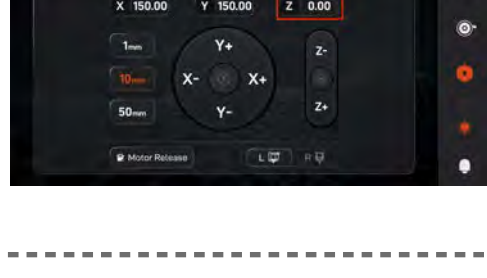
2. Clean the heatbed: remove residual filament, tape bumps, and dust to ensure a flat, even contact area.

3. Click the button circled in a red box (as shown in the picture) to move the nozzle to the home position.



4. Select 10mm and click Z-.

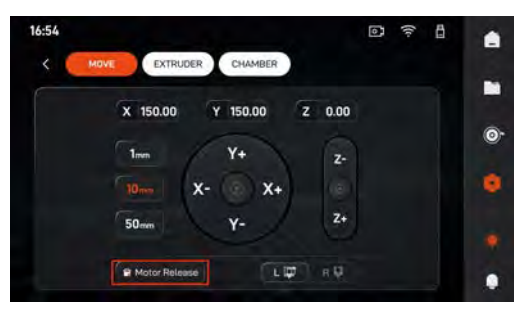
2



5. The Z axis will show 0.



6. Slightly loosen the seven retaining screws on the underside of the heatbed. Do not remove them.



7. Tap Motor Release.

3



8. Move the printhead to each corner of the heatbed, and adjust the corresponding leveling screw at that corner.



9. Insert a sheet of A4 paper between the left nozzle and the heatbed. Slide it back and forth. You should feel a light, even drag, not too tight and not too loose. Repeat this check 2–3 times at each corner.



10. Firmly tighten the seven retaining screws after sliding.

4

Single-point adjustment method

1. Move the nozzle to one corner and place the paper between the nozzle and the heatbed.

2. Turn the screw underneath that corner:

- Turn down (tighten) → heatbed rises, nozzle moves closer, paper drags more.
- Turn up (loosen) → heatbed lowers, gap increases, paper slides out easily.

3. Feeling: when sliding the paper, you should sense a light, even resistance. Neither tight nor loose.

5

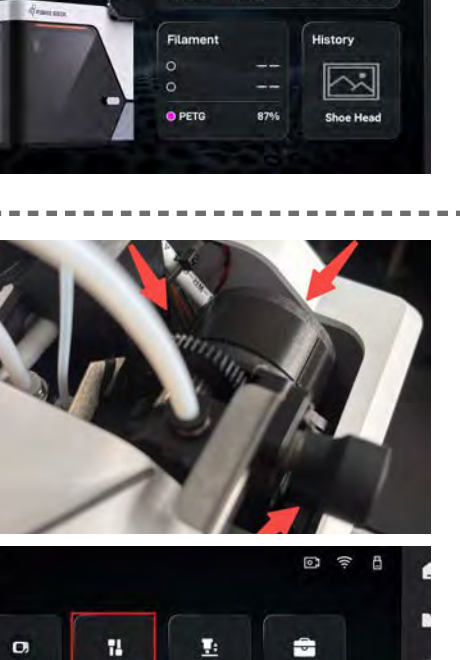
Plastic Hotend Unclogging

This SOP outlines how to clean plastic hotend clogs in the FibreSeeker 3, aiming to safely and effectively resolve clogging.

1. Tools:

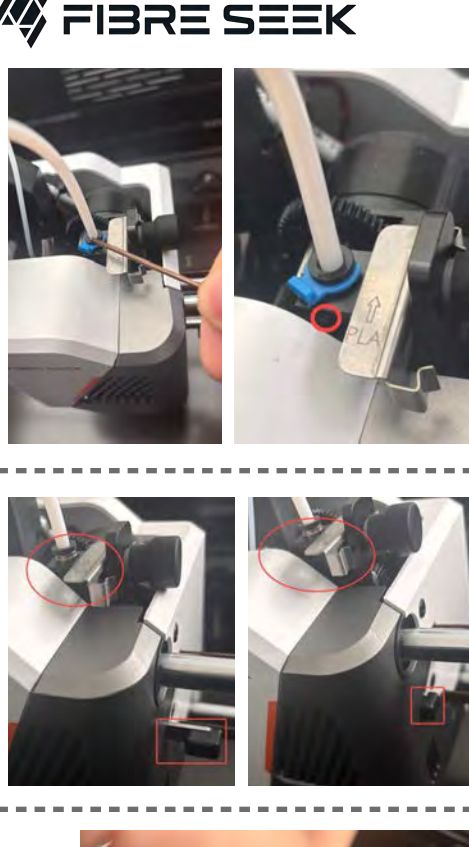
H3, H2.5, and H1.5 hex keys, an unclogging pin tool, and heat-resistant gloves.

2. Precautions:



Wear heat-resistant gloves to avoid burn accidents when you clean the nozzle. Before troubleshooting, check whether the filament is tangled.

1



Step 1

Heat the Nozzle

Heat the right nozzle to the temperature specified for the corresponding material, for example, PETG at 250°C, then try extruding again. Lower temperatures may block extrusion.

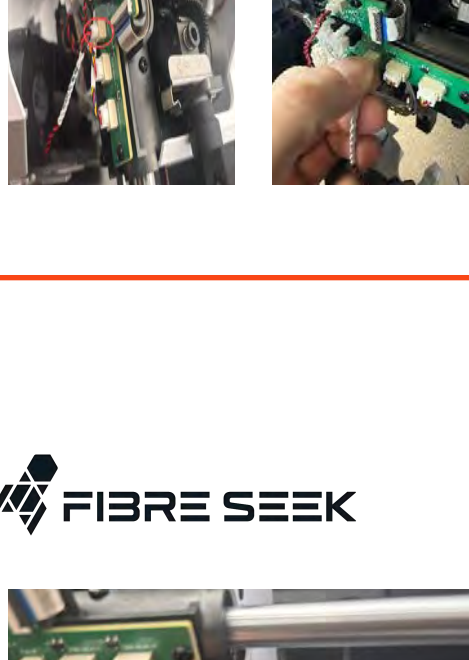


Step 2

Check Rocker Arm Screws, Extrude, and Inspect Gears

Tighten the rocker arm fastening screws and try extruding again. Tap the settings icon on the right side of the toolbar, then select "Control" – "Extruder". Set the temperature to approximately 250°C and the length to 50mm, then tap the down arrow to extrude. Observe whether the extruder gear is turning. If it turns properly, proceed to the next steps. If it does not turn, check whether the motor is operating. If the motor is not running, contact customer support for assistance.

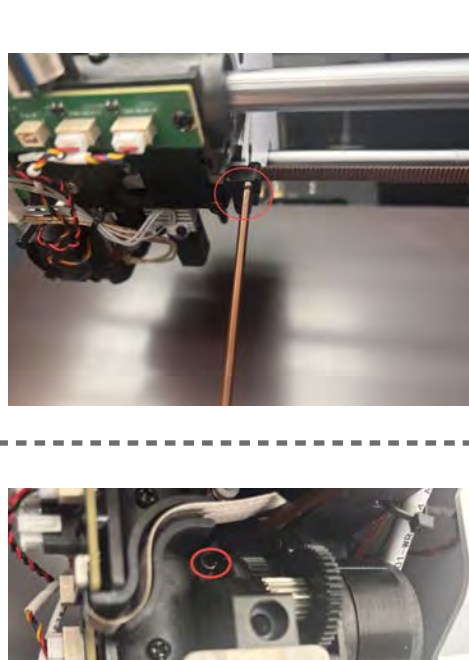
2



Step 3

Remove the PTFE Tube

If material is still not extruding, manually remove the PTFE tube: Remove the blue retaining clip. (Press the black fitting connector all the way down and pull the tube out forcefully.) If the tube cannot be removed, use an H1.5 hex key to loosen the set screws and detach the fitting connector.



Step 4

Switch the Nozzle

Push the nozzle to the left to switch it. This will cause the plastic hotend section to move upward. (You can observe changes in the set screw position in the circled area.)

Step 5

Check and Verify Filament

Manually pull out the filament left at the extruder (shown without the tube fitting). If the filament can be pulled out with minimal force, the extruder gear is worn. To verify again, use a smooth filament and reinsert it into the extruder. You should feel gear friction; the filament should be insertable with slight force. Ensure the rocker arm fastening screws are tightened. Refer to the Extruder Replacement Guide for further instructions.

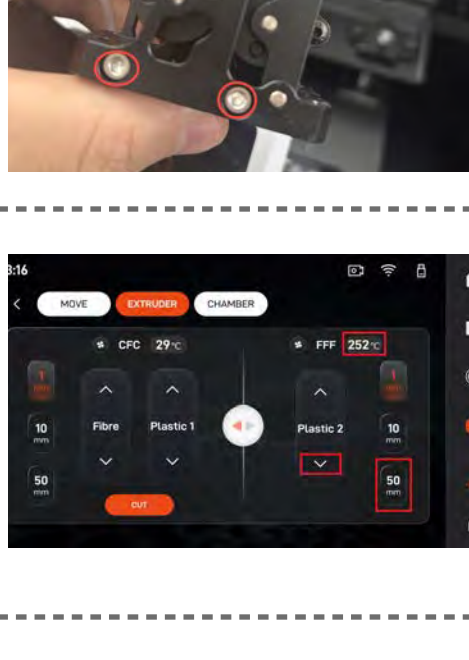
3



Step 6

Upload the Filament

If the filament cannot be pulled out with slight force, click Unload Filament. If the material still cannot eject, dismantle the extruder for inspection. If the filament can be successfully unloaded, skip the extruder dismantling step and refer to Step 7.2 Nozzle Clogging.



Step 7.1

Remove the Extruder

Step 7.1.1

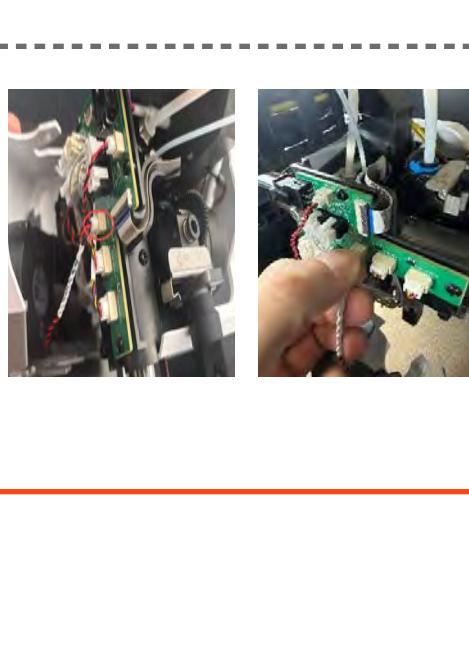
Power off the printer. Press the bottom clips on both sides of the front cover simultaneously, then pull the cover forward to remove it.



Step 7.1.2

Pinch the latch on the fan at the "FAN-R" position on the mainboard and unplug it to remove the front cover of the toolhead.

4



Step 7.1.3

Push leftward by hand to switch the nozzle. This will cause the plastic hotend section to move upward. (Skipping this step may prevent the extruder from being removed.)



Step 7.1.4

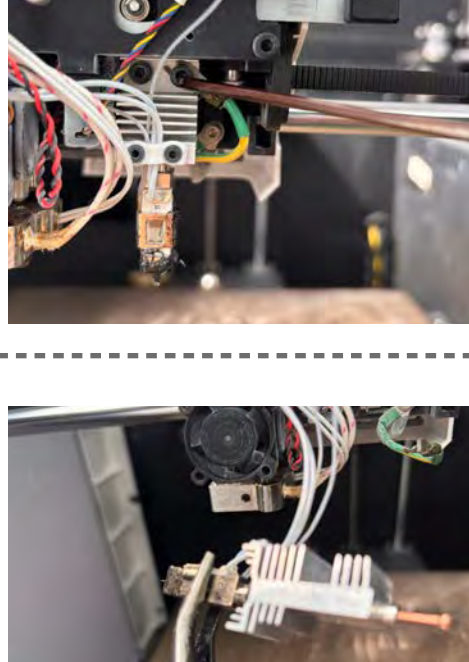
Use an H2.5 hex key to loosen the 2 fixing screws and take them out. Slightly pry the rear cover, wiggle the rocker left and right, and remove it.



Step 7.1.5

Use the H2.5 hex key to loosen the 3 fixing screws and disassemble the extruder.

5



Step 7.1.6

Use a tool to gently pry it apart for easier disassembly.



Step 7.1.7

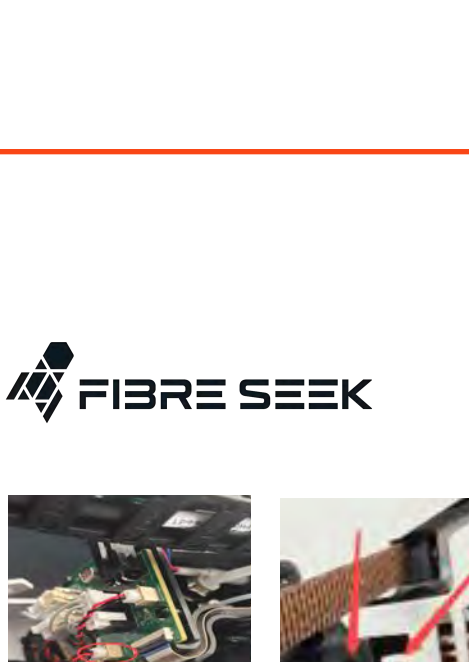
Check if the extruder is jammed or broken due to internal filament.



Step 7.1.8

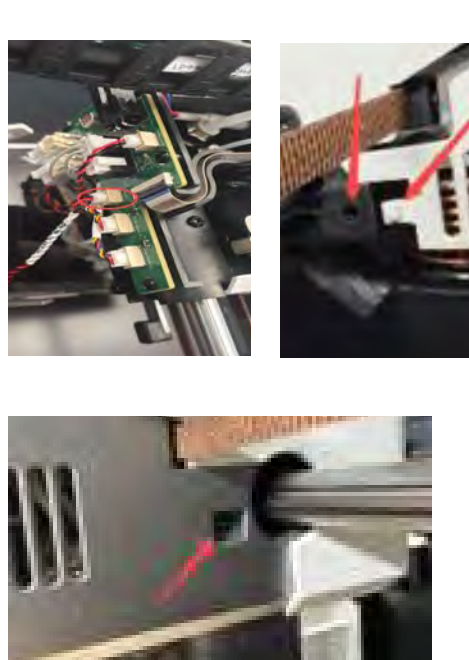
If filament is stuck, clean it thoroughly, then reassemble in order. Align the parts and press firmly until you hear a click to ensure no gaps.

6



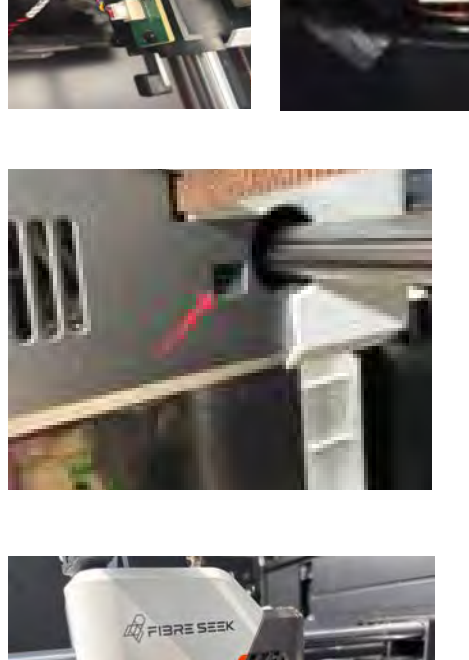
Step 7.1.9

Reinstall and tighten the fixing screws.



Step 7.1.10

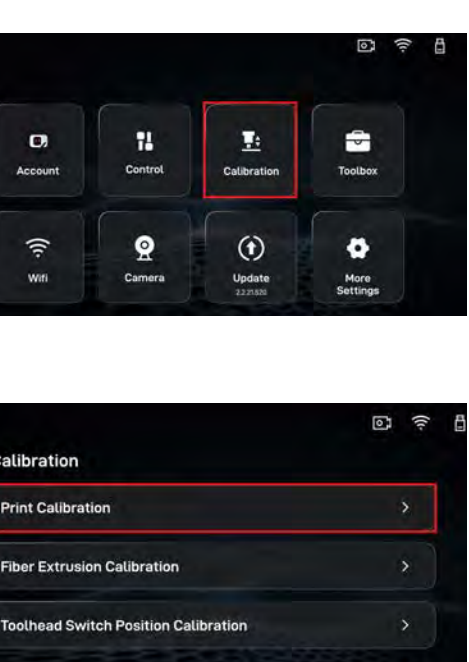
Insert filament and reextrude.



Step 7.1.11

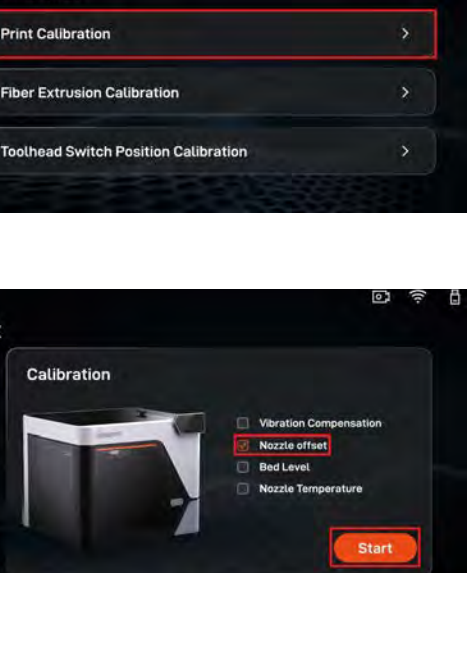
Once normal extrusion is confirmed, reinstall the extruder.

7



Step 7.1.12

After reinstalling the extruder, test extrusion to verify nozzle output. If no material exits, go to Step 7.2. Otherwise, run a normal print test.



Step 7.2

Nozzle Clogging

Step 7.2.1

Power off the printer. Press the bottom clips on both sides of the front cover simultaneously, then pull the cover forward to remove it.



Step 7.2.2

Pinch the latch on the fan at the "FAN-R" position on the mainboard and unplug it to remove the front cover of the toolhead.

8

Step 7.2.3

Heat the nozzle, then use an unclogging pin tool to push through from the bottom (nozzle tip) upward. Repeat several times. If the needle moves smoothly, try extruding material.

Step 7.2.4

If extrusion succeeds, extrude several times to flush out residual debris and clean with tweezers. Once cleaned, run a print test. If no material comes out, proceed to the step below.

9

Step 7.2.5

Use an H2.5 hex key to loosen 2 fixing screws and remove the plastic hotend assembly.

Step 7.2.6

Clean the internal channel with the unclogging pin tool. If that fails, replace the plastic hotend assembly (see the Replacement Guide).

Step 7.2.7

If cleared (needle moves back and forth smoothly), reinstall the plastic hotend and try extruding again.

10

Step 7.2.8

Plug the fan cable of the front cover back into the corresponding port on the mainboard. Make sure the two bottom clips align with their holes and snap into place.

11

Step 7.2.9

After installation, manually calibrate the nozzle offset.

12

Step 7.2.10

After installation, manually calibrate the nozzle offset.

13

Timing Belt Tension Adjustment

Pre-Operation Information

1.1 Purpose

This SOP outlines how to adjust the X/Y-axis timing belts of the FibreSeeker 3 3D printer. It resolves issues including belt slippage caused by insufficient tension, skipped-tooth noise, homing collision, and print position deviation to prevent accelerated wear of motors and bearings arising from excessive belt tension.

1.2 Scope

Applicable to the installation, tension inspection, and adjustment of the X- and Y-axis timing belts of the FibreSeeker 3.

1.3 Notes

It is necessary to inspect, adjust, and test the belt in accordance with the procedure; record the printer status before and after tuning; report any abnormalities that cannot be resolved promptly.

1.4 Required Tools

Tensiometer, hex key, feeler gauge, screwdriver



Troubleshooting Steps

Step 1: Power Off

Disconnect the printer's power cord. Do not perform any adjustment or tightening of fasteners while the printer is powered on.



Step 2: Set Parameters

Configure the tensiometer with the following parameters:

- M=000.9 g/m
- W=006.0 mm/R
- S=0360 mm



Step 3: Check Belt Tension

Move the print head to the far left. Apply vertical pressure at the midpoint of the belt span with a tensiometer, and keep the tension value within the range of 12~20.

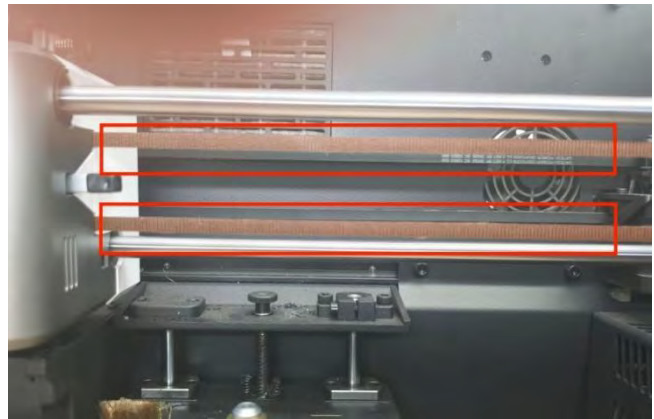


If no tensiometer is available, press the belt manually; obvious slackness indicates insufficient belt tension.



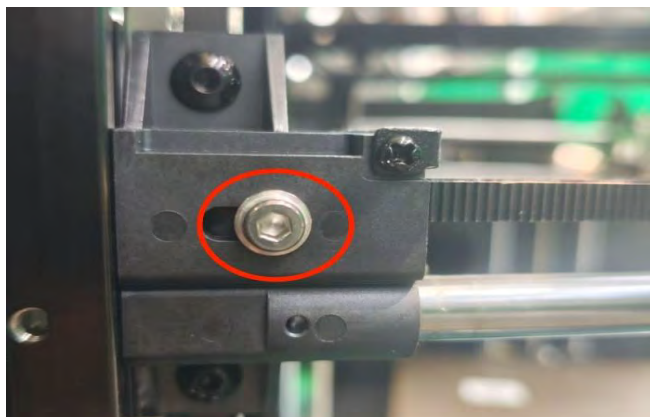
Step 4: Confirm the Adjustment Position

There are two belts vertically: the upper belt's adjustment point is on the left, while the lower belt's adjustment point is on the right side.



Step 5: Locate the Fixing Screws

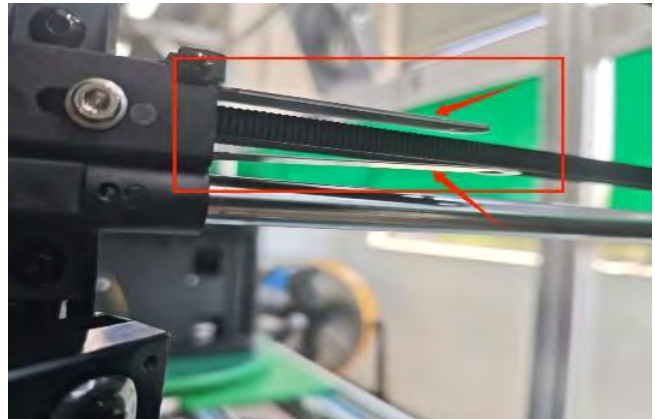
Locate the fixing screws at the outer end of the belt and confirm the target screws to be adjusted.



Step 6: Adjust

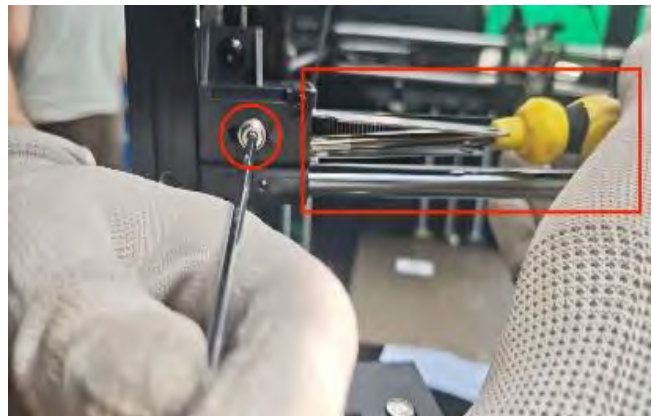
Insert feeler gauges into the top and bottom gaps of the idler pulley:

- Left: 1.00 mm feeler gauge top, 0.4 mm bottom
- Right: 1.00 mm feeler gauge bottom, 0.4 mm top



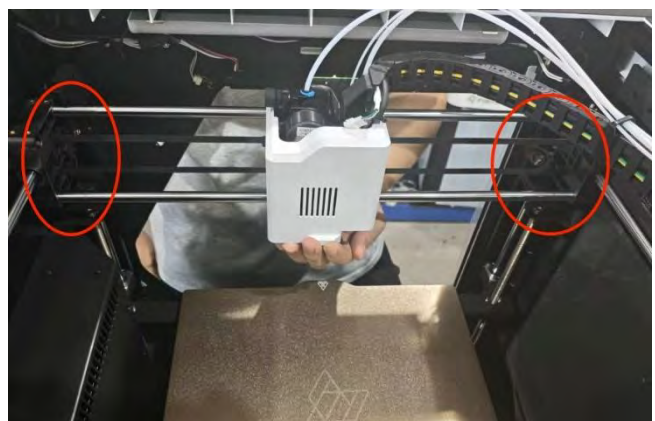
Step 7: Tension Adjustment

Loosen the fixing screws with a hex key; hold the tensioning block fixed with a screwdriver. Push the block inward to increase belt tension, or pry it outward to reduce tension.



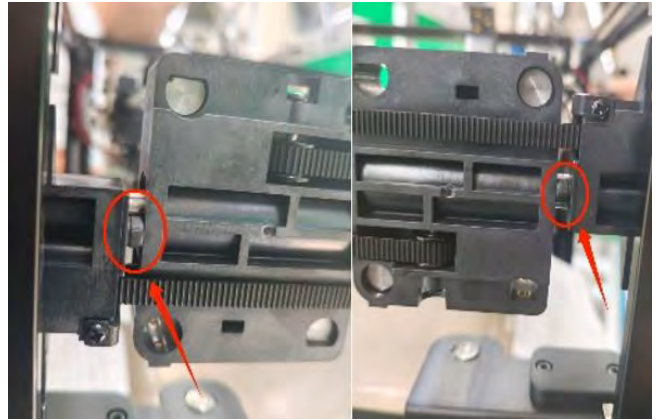
Step 8: Level Assemblies

Move the print head to the middle of the printer. Shift the assemblies to the far left and far right ends of the Y-axis. Manually move the print head to check whether



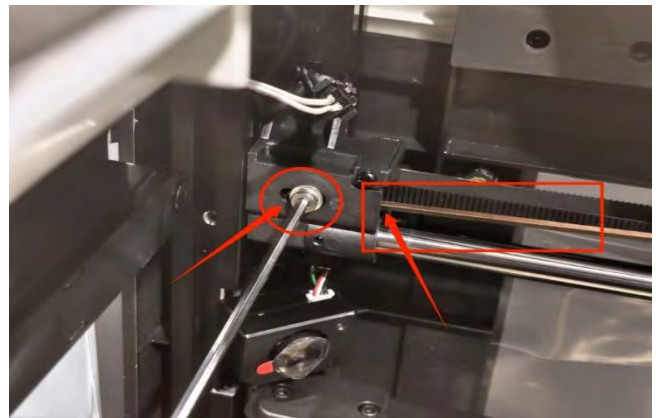
the left and right assemblies are at the same height.

If it is difficult to judge whether the assemblies are level, inspect the balance screws on both sides.



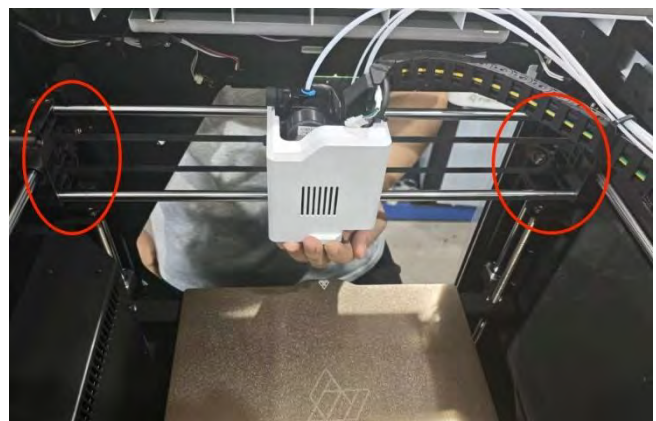
Step 9: Recheck Alignment

If the left and right assemblies are clearly misaligned, carry out secondary adjustment: tighten the screws on the uneven side accordingly as shown in the picture.



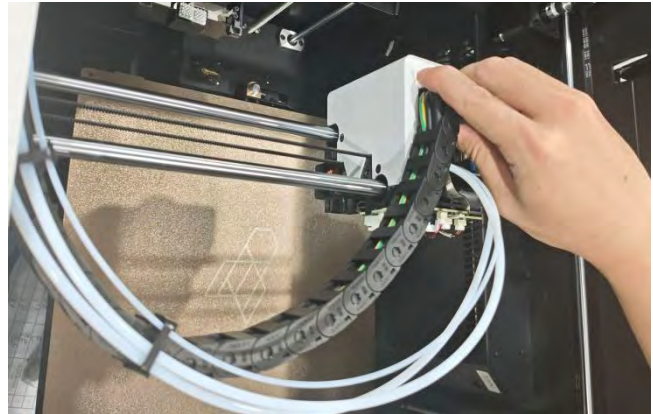
Step 10: Confirm Alignment

Manually move the print head to recheck whether the left and right assemblies are level and confirm the alignment result.



Step 11: Homing Calibration Test

1. Hold the print head and move it forward, backward, leftward and rightward for inspection.
2. Enter 0 for both X and Y coordinates to trigger the printer's automatic X/Y-axis homing procedure. Inspect the following conditions:
 - No tooth skipping, slipping or impact noise occurs when the print head triggers the limit switches.
 - No persistent clicking or slipping noise during printer operation.



Not Just Printing, Evolving.



Troubleshooting Layer Shifting in Plastic-Only or Fiber-Only Printing

This SOP provides adjustment procedures for layer shifting in single-plastic or single-fiber printing on FibreSeeker 3, aiming to safely and effectively resolve the issue.

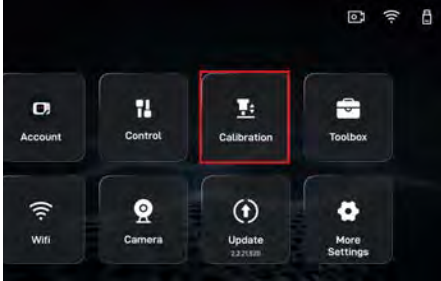
1. Tools

H3, H2.5, and H1.5 hex keys, an unclogging pin tool, and heat-resistant gloves.

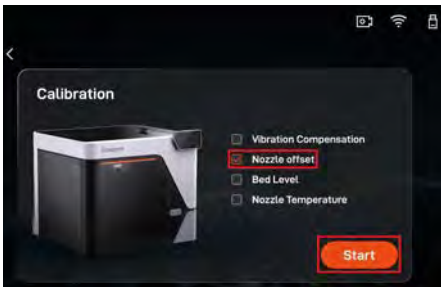
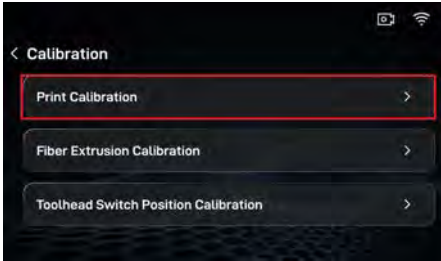
2. Precautions :



1



Step 1
Nozzle Offset
Step 1.1
 Calibrate nozzle offset.



2



Step 1.2
 If the nozzle offset calibration does not resolve the issue, measure the deviation between the plastic and the fiber, and adjust the offset value accordingly. Using the machine zero point as the reference: if the plastic is overall to the right, decrease the X offset value; if the plastic is overall upward, decrease the Y offset value. The specific amount to increase or decrease is equal to the actual model offset, in millimeters. For example, if the plastic is offset to the right by 2 mm, and the current X offset value is -32.55, the adjusted X offset value shall be -34.55.

Step 2
X/Y Belt Wear
 Refer to the Replacement guide for more details.
Step 3
X/Y Motor Step Loss
 Refer to the Replacement guide for more details.

3

Troubleshooting First Layer Print Failure

This SOP describes how to troubleshoot and adjust first layer print failures on FibreSeeker 3. Its purpose is to safely and effectively resolve first layer printing issues.

1. Tools :

None

2. Precautions :

None

1

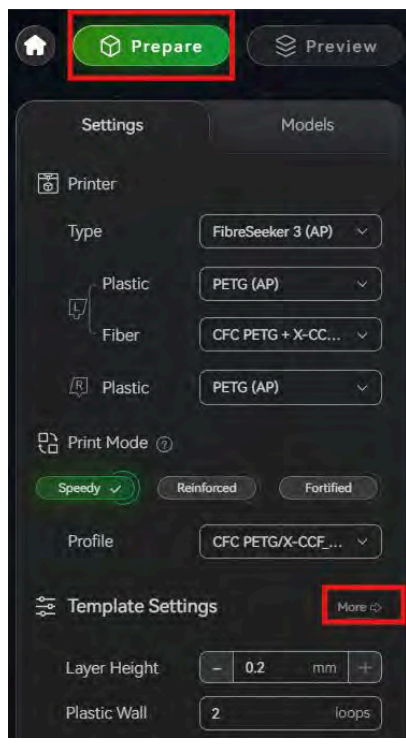


Step 1 Model Warping Step 1.1

No adhesive has been applied to the print bed.



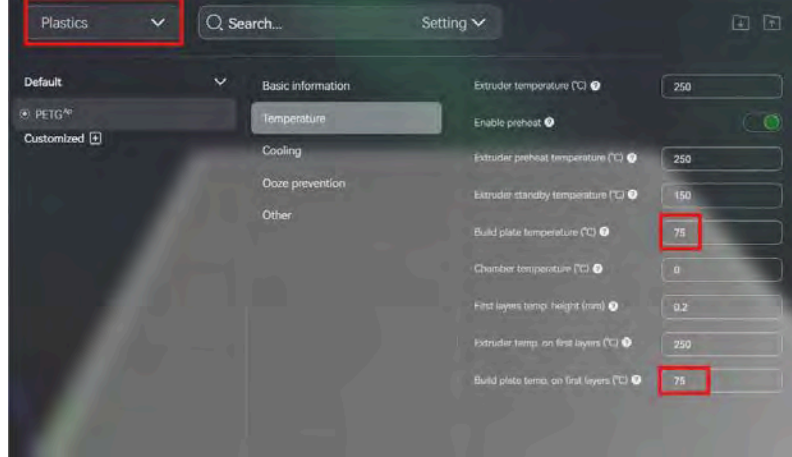
2



Step 1.2

Insufficient heatedbed temperature:

In the slicer, check if the temperature is too low and adjust it.



3



Step 2

If the gap between the platform and nozzle is too large, enable Auto Bed Leveling before printing. If the issue persists, replace the piezoelectric ceramic (see the Replacement Guide).



Step 3

If the gap between the platform and nozzle is too small, enable Auto Bed Leveling before printing. If the issue persists, replace the piezoelectric ceramic (see the Replacement Guide).

4

X-CCF 1K Technical Parameters

As the core reinforcing material for continuous fiber printing systems, X-CCF 1K features finer individual carbon fiber tows. It delivers steadier filament feeding and consistent resin impregnation, which effectively reduces nozzle clogging and boosts the printing success rate.

Mechanical Properties

Standard test specimens of X-CCF 1K are fabricated with a minor dosage of CFC co-extrusion matrix filament. The table below presents the mechanical properties of typical CFC material combinations:

Test Metric	Method (ASTM)	X-CCF/CFC PLA	X-CCF/CFC PETG	X-CCF/CFC PA
Tensile Strength (MPa)	D3039	883 ± 156.3	859 + 71	865 ± 37.7
Tensile Modulus (GPa)	D3039	84 ± 3.6	82 ± 11.7	81 ± 4.7
Tensile Strain at Break (%)	D3039	0.91 ± 0.2	1.05 ± 0.09	1.0 ± 0.09
Compressive Strength (MPa)	D6641	287.3 ± 43.8	202.9 ± 30.6	140.8 ± 22.5
Compressive Modulus (GPa)	D6641	52.3 ± 3.8	32.5 ± 4.6	27.2 ± 4.7
Compressive Strain at Break (%)	D6641	0.5 ± 0.03	0.62 ± 0.02	0.5 ± 0.02
Flexural Strength (MPa)	D790	494.1 ± 41.5	294.2 ± 37.7	319.4 ± 21.3
Flexural Modulus (GPa)	D790	40.1 ± 2.6	33.8 ± 1.8	32.8 ± 1.4

Filament Application

When matched with different co-extrusion materials, X-CCF 1K exhibits unique performance traits suited to different application scenarios, as summarized in the table below:

Filament Combination	Characteristics	Scenarios
X-CCF/CFC PLA	High rigidity, excellent dimensional stability, easy printing, low cost	UAV frames, low-load robotic parts, racing chassis, educational demo prototypes...
X-CCF/CFC PETG	High tensile strength, low moisture absorption, stable dimensions, wide versatility	Load-bearing UAV arms, robotic arm parts, sports equipment, device frame housings, functional jigs or fixtures, lightweight metal substitutes ...
X-CCF/CFC PA	Superior strength, toughness, fatigue resistance, wear & impact resistance, long service life	Robotic joints, transmission brackets, automotive functional parts, industrial fixtures, gear & pulley substrates, cyclic load-bearing components, ...

Disclaimer: Values in this data sheet are for reference and comparison only, and should not be used for design specifications or quality control. Actual values may vary with printing conditions. The final performance of printed parts depends not only on the material but also on part design, environmental conditions, printing conditions, etc.



X-CGF 1K Technical Parameters

X-CGF 1K enables consistent, smooth printing while endowing printed parts with outstanding tensile strength, stiffness and dimensional stability.

Mechanical Properties

Standard test bars of X-CGF 1K are manufactured with a small amount of CFC co-extrusion matrix resin. The table below presents the mechanical properties of typical CFC material combinations:

Test Metric	Method (ASTM)	X-CGF/CFC PLA	X-CGF/CFC PETG
Tensile Strength (MPa)	D3039	714 ± 165	625 ± 48
Tensile Modulus (GPa)	D3039	31 ± 0.4	25 ± 1.7
Tensile Strain at Break (%)	D3039	1.59 ± 0.73	2.36 ± 0.38
Compressive Strength (MPa)	D6641	133 ± 10.5	104 ± 5.5
Compressive Modulus (GPa)	D6641	26 ± 2.0	21 ± 3.8
Compressive Strain at Break (%)	D6641	0.5 ± 0.03	0.49 ± 0.07
Flexural Strength (MPa)	D790	185.7 ± 20.8	144.5 ± 9.4
Flexural Modulus (GPa)	D790	17.7 ± 1.0	11.9 ± 0.7

Filament Application

X-CGF delivers distinct characteristics and applicable scenarios when paired with different co-extrusion materials, as detailed below:

Filament Combination	Characteristics	Scenarios
X-CGF/CFC PLA	Low cost, electrically insulating, moderate stiffness	Teaching test pieces, consumer goods frames, electronic brackets, low-cost jigs or fixtures, beginner prototypes, non-conductive structural parts
X-CGF/CFC PETG	High toughness, excellent impact resistance and insulation	Electrically insulated supports, outdoor equipment frame skeletons, tool grips, sports protective gear, impact-resistant fixtures, general industrial functional parts

Disclaimer: Values in this data sheet are for reference and comparison only, and should not be used for design specifications or quality control. Actual values may vary with printing conditions. The final performance of printed parts depends not only on the material but also on part design, environmental conditions, printing conditions, etc.



Composite Material Performance

Flexural Strength

The figure below compares the bending mechanical properties of neat matrix materials, short-fiber reinforced thermoplastics, continuous fiber reinforced composites and 6061-T6 aluminum alloy.

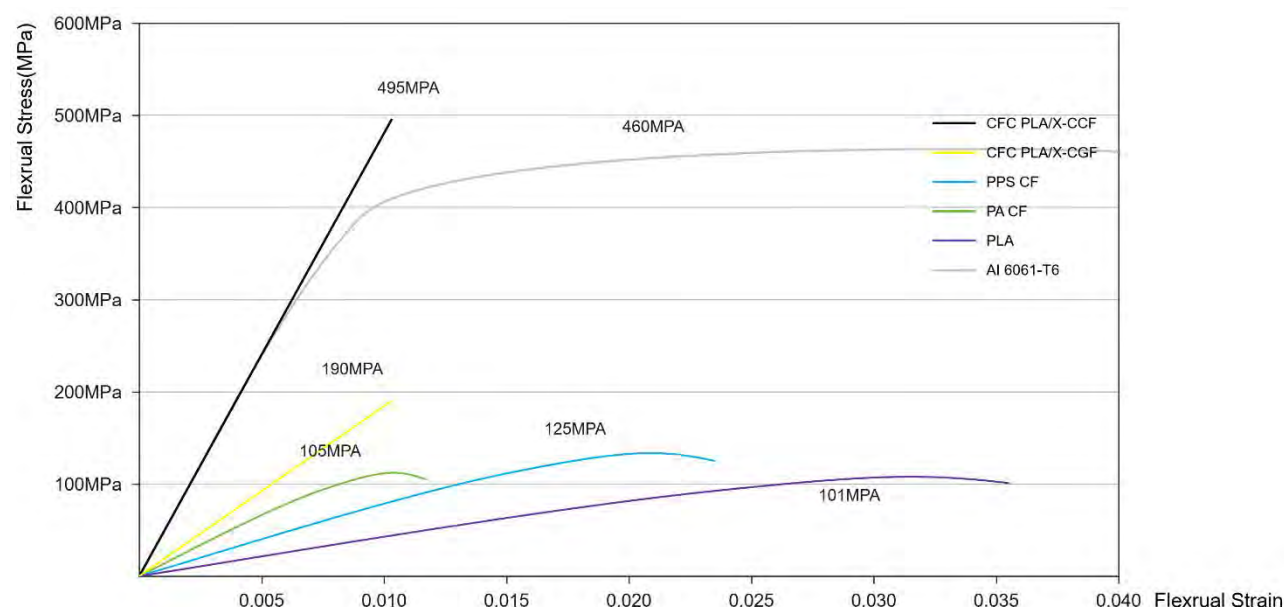


Figure 1 Bending Stress-Strain Comparison of CFC Composites, Engineering Plastics and Al 6061-T6

Flexural stress-strain curves demonstrate that continuous fiber composites possess better flexural performance than conventional printed thermoplastics and aluminum alloys. The X-CCF continuous carbon fiber composite reaches an ultimate flexural stress of 495 MPa and boasts the highest flexural modulus among all test samples. Its stiffness and flexural load-bearing capacity surpass those of 6061-T6 aluminum alloy (460 MPa). Continuous carbon fiber composites can reach or exceed metal-level flexural strength while remaining drastically lighter. Thanks to their high stiffness and low mass, they are perfect structural materials for components bearing large bending loads.

Tensile Strength

Conventional FDM plastics (neat PLA, short-carbon-fiber PA, short-carbon-fiber PPS) range 30–135 MPa in tensile strength, while 6061 aluminum alloy reaches 290 MPa. In contrast, X-CGF glass fiber composite hits 714 MPa and X-CCF carbon fiber composite hits 883 MPa. Their tensile strength is 2–3 times stronger than aluminum and dozens of times stronger than ordinary short-fiber materials. By transferring stress through unbroken continuous fibers, CFC technology overcomes the performance limits of standard short-fiber filaments for lightweight, high-load parts.

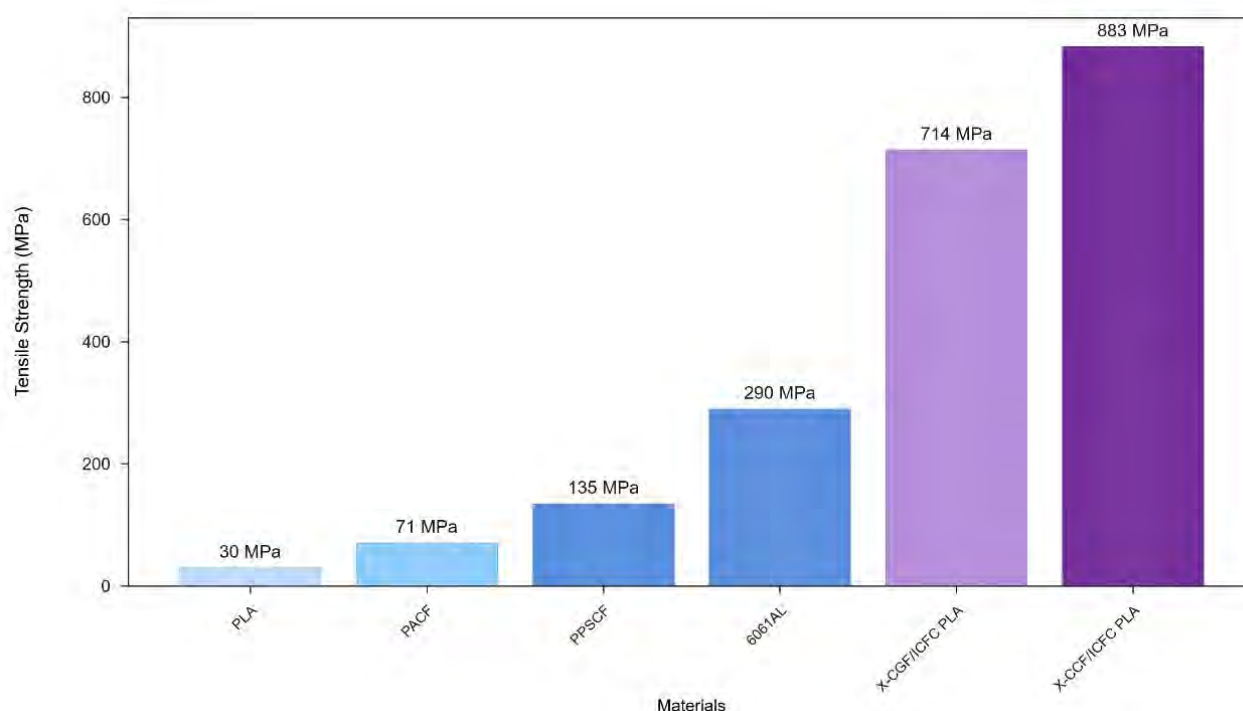


Figure 2 Tensile Strength Comparison of Various Materials



Specimen Testing

Standard

ASTM D3039: Standard Test Method for Tensile Properties of Polymer Matrix Composite Materials, see [Astm D3039 PDF | PDF](#) for details.

ASTM D6641: Standard Test Method for Compressive Properties of Polymer Matrix Composite Materials, see [ASTM D6641 PDF | PDF](#) for details.

ASTM D790: Standard Test Methods for Flexural Properties of Unreinforced and Reinforced Plastics and Electrical Insulating Materials (3-Point Bending), see [ASTM D790 PDF | PDF](#) for details.

Method

Test equipment: UTM5504X-F Universal Testing Machine

Load Sensor: 50 kN

Fixture: Hydraulic fixture for tensile testing

Strain Measurement:

- Extensometer with 50 mm gauge length
- Transverse extensometer with 25 mm gauge length
- Deflectometer with 20 mm gauge length

Tensile loading rate: 2 mm/min;

Compressive loading rate: 1.3 mm/min;

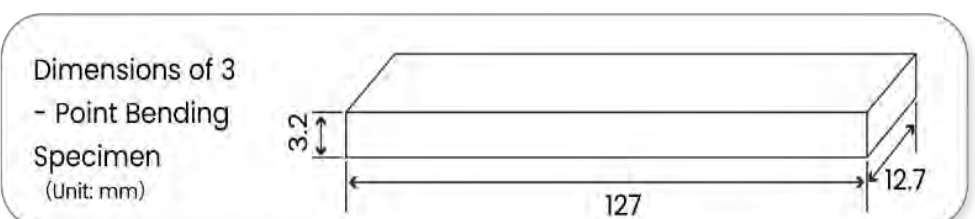
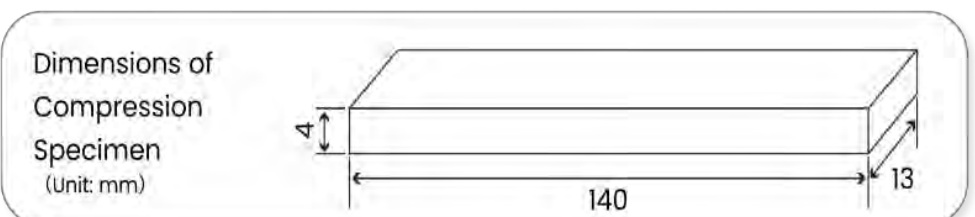
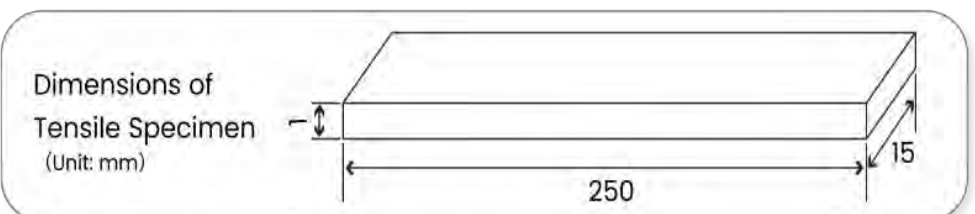
Flexural loading rate: 1.37 mm/min

Specimen Dimensions

Tensile Specimen: 250 mm (Length) × 15 mm (Width) × 1 mm (Thickness)

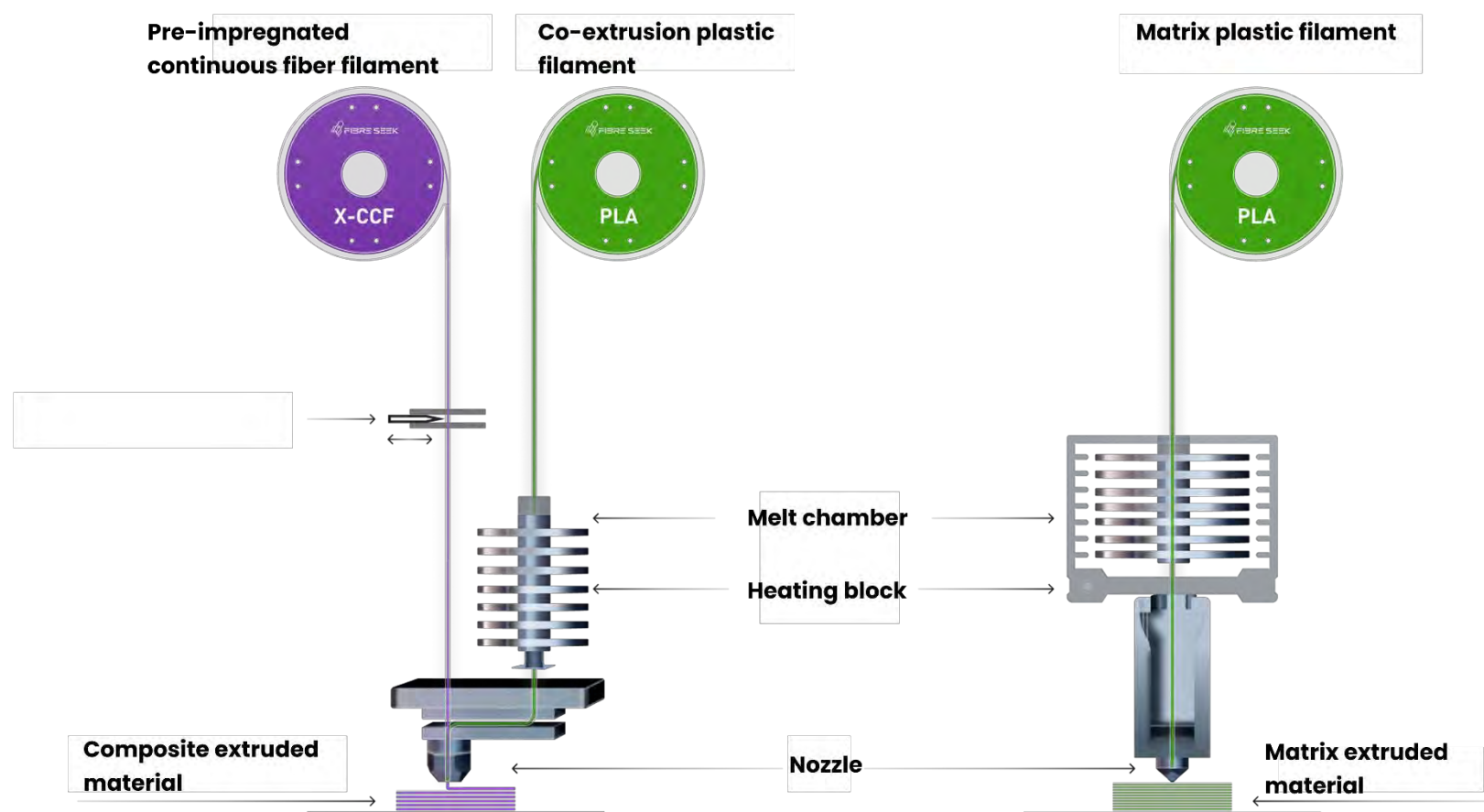
Compression Specimen: 140 mm (Length) × 13 mm (Width) × 4 mm (Thickness)

3-Point Bending Specimen: 127 mm (Length) × 12.7 mm (Width) × 3.2 mm (Thickness)



CFC Technology

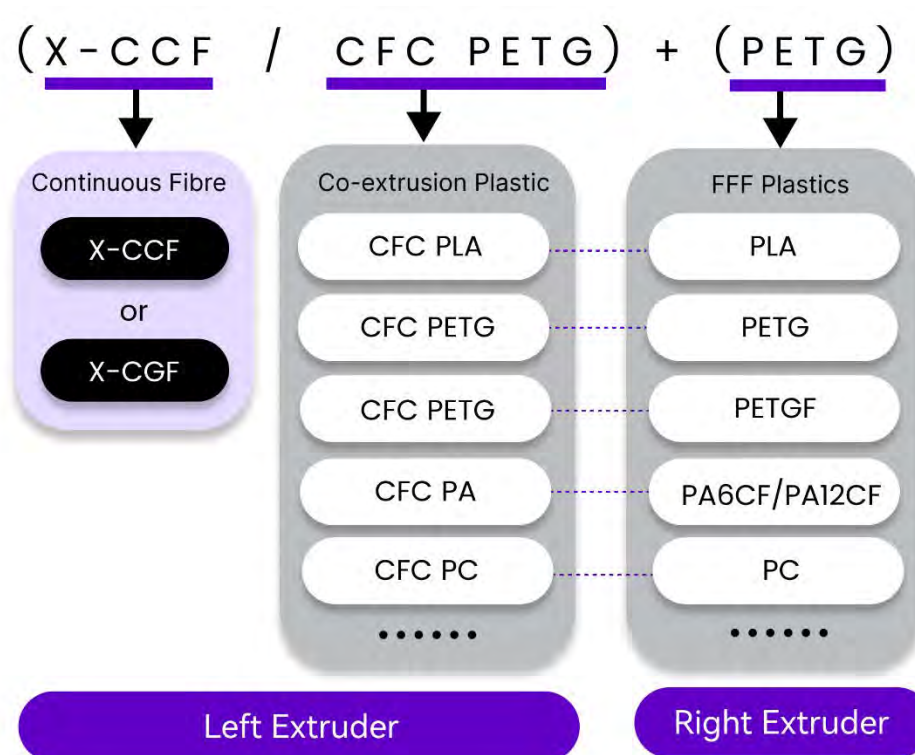
CFC (Composite Fiber Co-extrusion) is Anisoprint's proprietary core additive manufacturing technology. It was originally developed by the founding team in 2015 and specially optimized for fabricating high-performance composite components. It relies on a dual-printhead system. The composite co-extrusion printhead extrudes molten thermoplastic matrix alongside continuous reinforcing filaments (carbon fiber /glass fiber) in one pass to form an internal load-bearing skeleton. The second printhead prints pure base resin simultaneously to form the outer surface, guaranteeing premium surface finish and full structural integrity of finished parts.



Introduction to Composites

X-CCF and X-CGF are continuous fiber-reinforced filaments pre-impregnated with thermosetting resin. They cannot be printed independently and must be co-extruded with matrix plastics such as CFC PLA, CFC PETG and CFC PA.

The complete material system enables composite manufacturing based on the Continuous Fiber Co-extrusion (CFC) technology. The figure on the right elaborates the naming rule and combined molding principle of this material series.



Disclaimer: Values in this data sheet are for reference and comparison only, and should not be used for design specifications or quality control. Actual values may vary with printing conditions. The final performance of printed parts depends not only on the material but also on part design, environmental conditions, printing conditions, etc.



FFF and CFC Polymer System Overview

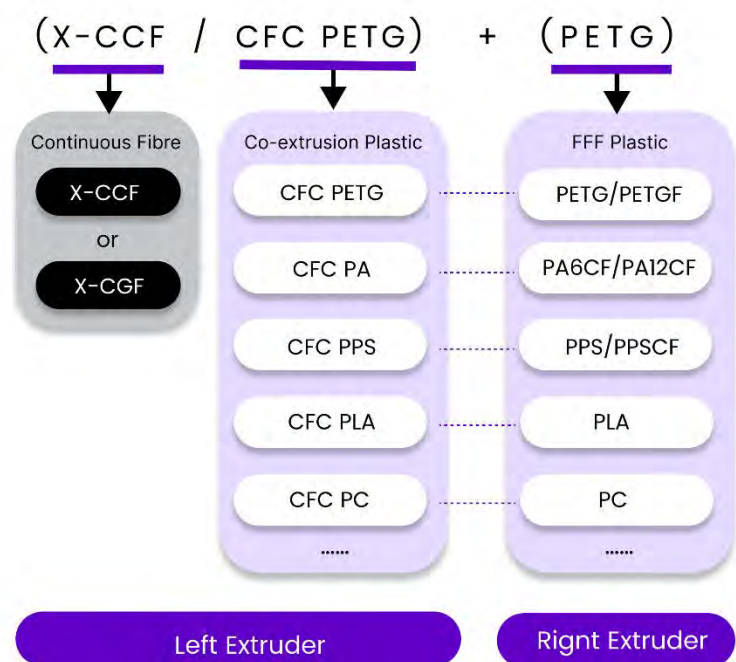
CFC continuous fiber 3D printing requires two types of filaments: FFF matrix filament and CFC composite material. Common FFF filaments include PLA, ABS, PETG, PA (Nylon), PC, etc. These materials deliver differentiated performance in strength, toughness, heat resistance and chemical resistance, and can be selected flexibly according to application scenarios.

CFC stands for **Composite Fiber Co-extrusion**. As a high-performance composite printing material system based on FibreSeeker’s core patented technology, it retains the forming flexibility of thermoplastics. Developed specifically for continuous fiber reinforced printing, this co-extrusion binder system targets applications requiring high strength, high stiffness and lightweight design.

The continuous fiber is pre-impregnated with thermosetting resin. Acting as a high-strength bonding medium, CFC material improves adhesion between fibers and FFF filaments, and ensures stable transfer of interlayer load.

CFC and FFF System

X-CCF and X-CGF are thermosetting resin pre-impregnated continuous fiber filaments. Unable to print alone, they require matching matrix filaments such as CFC PLA, CFC PETG and CFC PA for co-printing. The full material system realizes composite manufacturing through the CFC process. The diagram on the right details the material naming rule and co-forming mechanism.



Why must FFF and CFC filaments be matched in pairs?

Not all polymer combinations produce reliable bonding. At the molecular level, adhesion between thermoplastics relies on the interdiffusion and entanglement of polymer chains at material interfaces. Material pairs with vastly different chemical structures exhibit minimal molecular interdiffusion. Bonding only depends on weak mechanical interlocking, which is prone to delamination and print failure.



PLA-Basic Technical Properties

FFF Polymer

PLA-Basic is an entry-level filament developed for 3D printing enthusiasts and beginners. This low-cost base material modified from PLA features easy printing performance, zero irritating odor, minimal stringing and no warpage with favourable forming results. It stands out as a cost-effective 3D printing filament and must be printed in combination with CFC PLA fiber.

Mechanical Properties

Property	Test Method	Typical Values
Tensile Strength (X-Y)	GB/T 1040	63.95 MPa
Tensile Strength (Z)	GB/T 1040	28.29 MPa
Elongation at Break (X-Y)	GB/T 1040	4.17 %
Elongation at Break (Z)	GB/T 1040	2.41 %
Bending Strength (X-Y)	GB/T9341	101.2 MPa
Bending Strength (Z)	GB/T9341	40.5 MPa
Bending Modulus (X-Y)	GB/T9341	3085.87 MPa
Bending Modulus (Z)	GB/T9341	2832.46 Mpa
Izod Impact Strength of Notched Specimen (X-Y)	GB/T 1843	3.09 KJ/m²
Izod Impact Strength of Notched Specimen (Z)	GB/T 1843	2.15 KJ/m²

Physical Properties

Property	Test Method	Typical Values
Density	GB/T 1033	1.24 g/cm³
Melt Index	GB/T 3682	3.5-4.5 (190 °C /2.16kg)

Thermal Properties

Property	Test Method	Typical Values
Heat Deflection Temperature	GB/T 1634	50-60 °C (0.45Mpa)
Continuous Service Temperature	IEC 60216	N/A

Disclaimer: Values in this data sheet are for reference and comparison only, and should not be used for design specifications or quality control. Actual values may vary with printing conditions. The final performance of printed parts depends not only on the material but also on part design, environmental conditions, printing conditions, etc.



PETG Technical Properties

FFF Polymer

PETG is an easy-to-print transparent 3D printing filament with high flowability, balanced performance and good chemical resistance. Printed components feature high toughness, low warpage, glossy transparent surfaces, weather resistance and anti-yellowing capability. Eco-friendly and BPA-free, it is ideal for aesthetic models, home appliance, electronic and cosmetic parts requiring toughness and impact resistance. It needs co-printing with CFC PETG fiber.

Mechanical Properties

Property	Test Method	Test Condition	Typical Values
Tensile Strength (X-Y)	ISO527/2	50 mm/min	48 ± 10 MPa
Young's Modulus (X-Y)	ISO527/2	1 mm/min	-
Elongation at Break (X-Y)	ISO527/2	50 mm/min	11 ± 5 %
Flexural Strength (X-Y)	ISO 178	2 mm/min	70 ± 10 MPa
Flexural Modulus (X-Y)	ISO 178	2 mm/min	2100 ± 300 MPa
Izod Impact Strength of Notched Specimen (X-Y)	ISO 180	23 °C	3.0 ± 1 KJ/m²
Izod Impact Strength of Notched Specimen (Z-X)	ISO 180	23 °C	5.0±2 KJ/m²
Shore Hardness	ISO 868	23 °C	80D

Thermal Properties

Property	Test Method	Test Condition	Typical Values
Heat Distortion (X-Y)	ISO75	0.45 MPa	70 ± 5 °C
Glass Transition	ISO11357-2	10 °C /min	79.4 °C
Melting Temperature	ISO11357-3	10 °C /min	195 °C



PC Technical Properties

FFF Polymer

PC is produced using a polycarbonate resin specifically engineered for 3D printing. It delivers good stiffness and heat resistance with light diffusing properties. It needs co-printing with CFC PC fiber.

Mechanical Properties

Property	Test Method	Typical Values
Young's Modulus (X-Y)	ISO 527, GB/T 1040	2497 ± 154 MPa
Tensile Strength (X-Y)	ISO 527, GB/T 1040	69.1 ± 3.0 MPa
Elongation at Break (X-Y)	ISO 527, GB/T 1040	4.8 ± 0.9 %
Young's Modulus (Z)	ISO 527, GB/T 1040	2371 ± 55 MPa
Tensile Strength (Z)	ISO 527, GB/T 1040	52.8 ± 1.7 MPa
Elongation at Break (Z)	ISO 527, GB/T 1040	2.7 ± 0.1 %
Bending Modulus (X-Y)	ISO 178, GB/T 9341	2640 ± 47 MPa
Bending Strength (X-Y)	ISO 178, GB/T 9341	106.1 ± 1.6 MPa
Notched Charpy Impact Strength (X-Y)	ISO 179, GB/T 1043	4.1 ± 0.9 kJ/m ²

Physical Properties

Property	Test Method	Typical Values
Density	ISO1183, GB/T1033	11.19 g/cm ³ at 21.5 °C
Melt Index	260 °C, 2.16kg	8-11 g/10 min
Light Transmission	GB/T 2410	89 %

Thermal Properties

Property	Test Method	Typical Values
Glass Transition Temperature	DSC, 10 °C/min	113 °C
Heat Deflection Temperature	ISO 75 1.8 MPa	107 °C
Heat Deflection Temperature	ISO 75 0.45 MPa	113 °C



PET-GF15 Technical Properties

FFF Polymer

PET-GF15 is the bridge of the gap between every day usability and industrial performance, offering a strong, heat-resistant, and user-friendly filament for both professional and hobbyist applications. It needs co-printing with CFC PETGF fiber.

Mechanical Properties

Property	Test Method	Typical Values
Young’s Modulus (X - Y)	ISO 527, GB/T 1040	4144.2 ± 133.3 MPa
Tensile Strength (X - Y)	ISO 527, GB/T 1040	59.9 ± 0.8 MPa
Elongation at Break (X - Y)	ISO 527, GB/T 1040	4.0 ± 0.5 %
Bending Modulus (X - Y)	ISO 178, GB/T 9341	3705.4 ± 84.5 MPa
Bending Strength (X - Y)	ISO 306, GB/T 1633	104.2 ± 2.4 MPa
Charpy Impact Strength (X-Y) Notched	ISO 179, GB/T 1043	8.7 ± 0.6 kJ/m²
Charpy Impact Strength (X-Y) Nn-notched	ISO 179, GB/T 1043	27.2 ± 2.0 kJ/m²

Physical Properties

Property	Test Method	Typical Values
Density	ISO1183, GB/T1033	1.43 g/cm³ at 23 °C
Melt Index	270°C, 2.16 kg	36.9 g/10 min
Surface Resistivity (Ω)	ANSI ESD S11.11	OL, >1012 Ω

Thermal Properties

Property	Test Method	Typical Values
Glass Transition Temperature	DSC, 10°C/min	59.5 °C
Melting Temperature	DSC, 10°C/min	231.6 °C
Crystallization Temperature	DSC, 10 °C /min	201.4 °C
Heat Deflection Temperature	ISO 75 1.8 MPa	71.8 °C



PA6-CF Technical Properties

FFF Polymer

PA6-CF is a carbon fiber reinforced PA6 (Nylon 6) filament. The carbon fiber reinforcement provides significantly improved stiffness, strength and heat resistance with outstanding layer adhesion. It needs co-printing with CFC PACF fiber.

Mechanical Properties

Property	Test Method	Typical Values
Young's Modulus (X-Y)	ISO 527, GB/T 1040	7453 ± 656 MPa
Tensile Strength (X-Y)	ISO 527, GB/T 1040	105. ± 5.0 MPa
Elongation at Break (X-Y)	ISO 527, GB/T 1040	3.0 ± 0.3 %
Bending Modulus (X-Y)	ISO 178, GB/T 9341	8339 ± 369 MPa
Bending Strength (X-Y)	ISO 306, GB/T 1633	169.0 ± 4.7 MPa
Charpy Impact Strength (X-Y)	ISO 179, GB/T 1043	13.34 ± 0.5 kJ/m²

Physical Properties

Property	Test Method	Typical Values
Density	ISO1183, GB/T1033	1.17 g/cm³ at 23°C
Melt Index	300°C, 2.16 kg	20.5 g/10min
Sheet Resistance in Moisture State	ASTM D991	1 – 10 (108 Ω/sq)

Thermal Properties

Property	Test Method	Typical Values
Glass Transition Temperature	DSC, 10 °C/min	74.2 °C
Melting Temperature	DSC, 10 °C/min	218.5 °C
Crystallization Temperature	DSC, 10 °C /min	184.6 °C
Heat Deflection Temperature	ISO 75 1.8 MPa	173 °C



PPS-CF10 Technical Properties

FFF Polymer

PPS-CF10 is a carbon fiber reinforced PPS (Polyphenylene sulfide) filament, with minimal warping during printing and no need for a heated chamber. With exceptional mechanical strength, high heat resistance, chemical resistance, V0 flame retardancy, and moisture insensitivity, it's specifically designed for professionals operating in extreme conditions. It needs co-printing with CFC PPSCF fiber.

Mechanical Properties

Property	Test Method	Typical Values
Young's Modulus (X-Y)	ISO 527, GB/T 1040	5446.7 ± 149.0 MPa
Tensile Strength (X-Y)	ISO 527, GB/T 1040	59.4 ± 1.3 MPa
Elongation at Break (X-Y)	ISO 527, GB/T 1040	1.4 ± 0.1 %
Young's Modulus (Z)	ISO 527, GB/T 1040	2790.0 ± 152.6 MPa
Tensile Strength (Z)	ISO 527, GB/T 1040	32.0 ± 5.1 MPa
Elongation at Break (Z)	ISO 527, GB/T 1040	1.6 ± 0.2 %
Bending Modulus (X-Y)	ISO 178, GB/T 9341	4646.9 ± 136.9 MPa
Bending Strength (X-Y)	ISO 178, GB/T 9341	94.3 ± 1.9 MPa
Charpy Impact Strength (X-Y) Un-notched	ISO 179, GB/T 1043	11.4 ± 0.7 kJ/m²

Physical Properties

Property	Test Method	Typical Values
Density	ISO1183, GB/T1033	1.29 g/cm³ at 23 °C
Melt Index	300°C, 2.16 kg	26.2 g/10 min
Surface Resistivity (Ω)	ANSI ESD S11.11	OL, >1012 Ω

Thermal Properties

Property	Test Method	Typical Values
Glass Transition Temperature	DSC, 10 °C/min	97.7 °C
Melting Temperature	DSC, 10 °C/min	279.7 °C
Crystallization Temperature	DSC, 10 °C /min	218.8 °C
Heat Deflection Temperature	ISO 75 1.8 MPa	71.8 °C

Disclaimer: Values in this data sheet are for reference and comparison only, and should not be used for design specifications or quality control. Actual values may vary with printing conditions. The final performance of printed parts depends not only on the material but also on part design, environmental conditions, printing conditions, etc.



CFC PETG Technical Properties

CFC Polymer

CFC PETG is a highly transparent PETG filament with excellent compatibility with PETG and continuous fibers. It is primarily used for 3D printing via composite nozzles alongside continuous fibers.

Performance Test Data

Property	Test Condition	Test Standard	Value
Basic Physical Properties			
Density	23 °C	GB/T 1033	1.29 g/cm³
Heat Deflection Temperature (HDT)	0.45 MPa	GB/T 1634	76 °C
Mechanical Properties of Printed Specimens			
Tensile Strength (X-Y)	50mm/min	GB/T 1040.2	57 MPa
Tensile Strength (X-Z)	50mm/min	GB/T 1040.2	32 MPa
Flexural Strength	2mm/min	GB/T 9341	82 MPa
Flexural Modulus	2mm/min	GB/T 9341	2200 MPa
Notched Charpy Impact Strength	2.75 J	GB/T 1843	4 kJ/m²



CFC PA Technical Properties

CFC Polymer

CFC PA is high-purity PA12, featuring excellent compatibility with PA6-CF and continuous fibers. It is mainly applied to continuous fiber 3D printing with composite nozzles.

Mechanical Properties

Property	Test Standard (ASTM)	Value
Young's Modulus (X - Y)	D638 (ISO 527, GB/T 1040)	1580± 195 MPa
Tensile Strength (X - Y)	D638 (ISO 527, GB/T 1040)	56.8 ± 0.53 MPa
Elongation at Break (X - Y)	D638 (ISO 527, GB/T 1040)	15.86±2.41 %
Flexural Modulus (X - Y)	D790 (ISO 178, GB/T 9341)	1580 ± 195 MPa
Flexural Strength (X - Y)	D790 (ISO 178, GB/T 9341)	68.7 ± 1.8 MPa
Charpy Impact Strength (X - Y)	D256 (ISO 179, GB/T 1043)	11.42 ± 0.9 kJ/m²

Physical Properties

Property	Test Standard	Value
Density	ASTM D792(ISO 1183, GB/T 1033)	1.43 23 °C, g/cm³
Melt Flow Rate	270 °C, 2.16 kg	36.9 g/10 min
Glass Transition Temperature	DSC, 10 °C/min	59.5 °C
Melting Temperature	DSC, 10 °C/min	231.6 °C
Crystallization Temperature	DSC, 10 °C /min	201.4 °C



Recommended Printing Settings

FFF matrix filaments and CFC composites require different printing parameters and settings cannot be shared. Recommended printing conditions are shown in the three tables below.

FFF Polymer

Parameter	PLA-Basic	PETG	PC
Nozzle Temperature	210-230 °C	240-260 °C	250-270 °C
Bed Temperature	45-60 °C	60-70 °C	90-105 °C
Build Plate Material	PEI	Normal	Texture PEI (Glue when needed)
Printing Speed	< 300 mm/s	≤ 300 mm/s	50-200 mm/s
Retraction Distance	-	0.8-1.2 mm	1-3 mm
Retraction Speed	-	30-40 mm/s	20-40 mm/s
Closure Chamber		—	70-100 °C
Drying Condition	50°C >8H	60-65°C	75°C, 6h
Cooling Fan	ON	ON	OFF

Parameter	PET-GF15	PA6-CF	PPS-CF10
Nozzle Temperature	280-310 °C	200-300 °C	310-350 °C
Bed Temperature	70-80 °C	25-50 °C	80-90 °C
Printing Speed	≤250 mm/s	30-60 mm/s	≤ 300 mm/s
Environmental Temperature	Room Temp.	Room Temp.-50 °C	Room Temp.
Cooling Fan	OFF	OFF	OFF

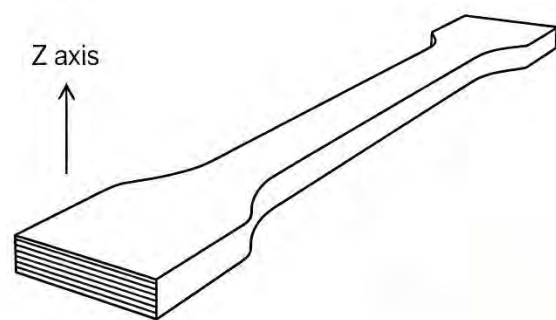
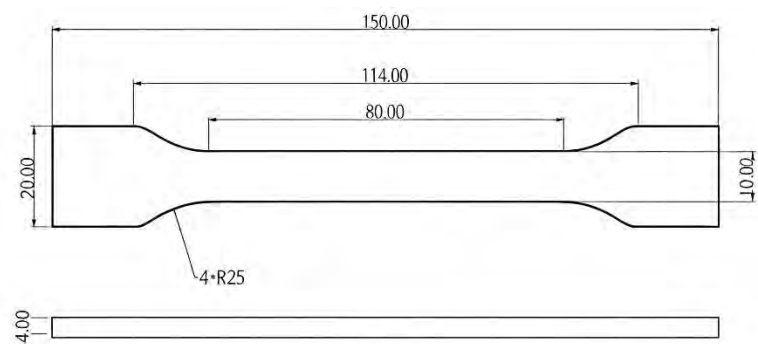


CFC Polymer

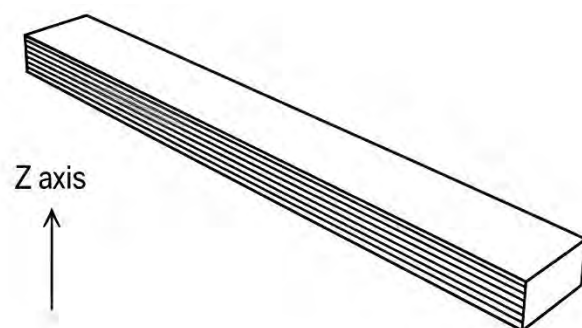
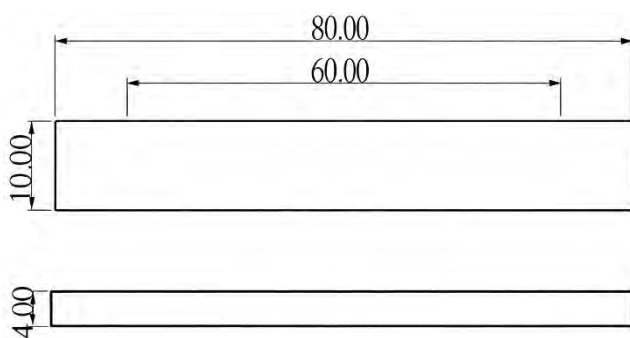
Parameter	CFC PETG	CFC PA
Nozzle Temperature	250 °C	250–260 °C
Bed Temperature	75 °C	75 °C
Build Plate Material	Glass, PEI, Spring Steel Sheet	lass, PEI; apply PVP adhesive if needed
First Layer Temperature	250 °C	250–260 °C
Cooling Fan	—	OFF
Recommended Ambient Temperature	Room Temperature – 50°C	Room Temperature – 50 °C
Drying Condition	60 °C, 4–8 h	70–80°C, 8–12 h



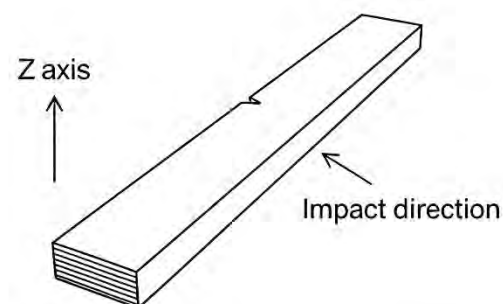
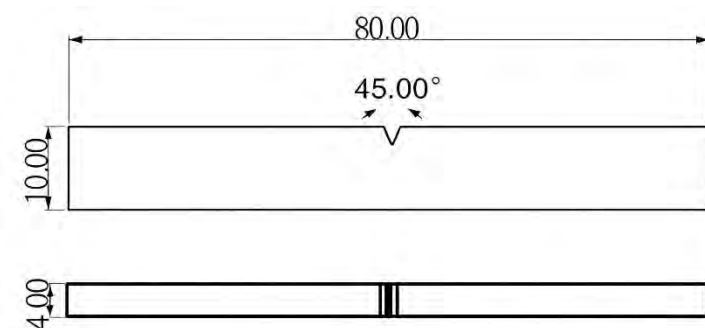
Tensile Testing Specimen



Flexural Testing Specimen



Impact Testing Specimen



Moisture Content & Drying Conditions for FFF Polymer

Different FFF filaments feature varying hygroscopicity. Drying treatment is recommended prior to printing. The two tables below list the maximum allowable moisture content and recommended drying conditions for various FFF filaments.

Moisture Content

Material	Maximum Moisture Content	Moisture Susceptibility
PLA-Basic	250 ppm	Low
PETG/CFC PETG	500 ppm	Medium
PC	200 ppm	Medium-High
PET- GF15	500 ppm	Medium-High
CFC PA	500 ppm	Very High
PA6-CF	500 ppm	Very High
PPS-CF10	100 ppm	Low

Recommended Drying Conditions

Material	Temperature	Time	Instructions
PLA-Basic	50-55 °C	4-6 h	Low-temp hot air drying to avoid spool deformation
PETG/CFC PETG	60-70 °C	6 h	Prone to stringing and bubbling; seal after opening
PC	75-85 °C	6-8 h	Fully dry before printing
PET- GF15	65-70 °C	8 h	store and use in dry environment
CFC PA	70-80 °C	8-12 h	Covers PA6, PA66 and copolyamides; significant grade-to-grade differences.
PA6-CF	70-80 °C	8-12 h	Continuous feeding from heated dryer
PPS-CF10	120 °C	6-8 h	Requires heat-resistant spools and specialized oven

Note: Maximum moisture content refers to the recommended upper limit before feeding into the printer / melt processing.1% = 10,000 ppm. Drying parameters are specified for hot-air circulating or dehumidifying ovens. Consumer household ovens have large temperature fluctuations. Confirm the temperature resistance of filament spools when drying high-temperature materials.